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RESEARCH**

APPLICATION NUMBER:

209531Orig1s000

PHARMACOLOGY REVIEW(S)

Tertiary Pharmacology Review

By: Paul C. Brown, Ph.D., ODE Associate Director for Pharmacology and Toxicology, OND IO

NDA: 209531

Submission date: 8/9/2016, 9/23/2016

Drug: nusinersen

Applicant: Biogen

Indication: spinal muscular atrophy

Reviewing Division: Division of Neurology Products

Discussion:

The pharmacology/toxicology reviewer and supervisor conducted a thorough evaluation of the nonclinical information submitted in support of this NDA. They have determined that the information is adequate to support approval for the indication of spinal muscular atrophy.

Toxicity studies of nusinersen were conducted in mice and juvenile cynomolgus monkeys. Histological evidence of neurotoxicity was observed in monkeys. The NOAEL for this toxicity occurred at a dose that produced exposures relevant to the clinical exposure.

Developmental and reproductive toxicity studies in mice and rabbit showed no fetal effects related to treatment with nusinersen. A pre-/postnatal study is ongoing and the supervisor recommends that completion of this study be a post-marketing requirement.

Carcinogenicity studies have not been conducted with nusinersen at this time. The Division has recommended that a carcinogenicity study be conducted after approval.

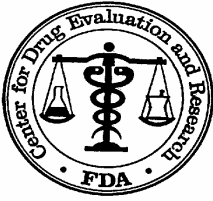
Conclusions:

I agree that the pharmacology/toxicology information is adequate to support approval of this application. The post-marketing requirements of a carcinogenicity study and completion of the pre-/postnatal study are reasonable.

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/s/

PAUL C BROWN
12/22/2016



DEPARTMENT OF HEALTH AND HUMAN SERVICES
PUBLIC HEALTH SERVICE
FOOD AND DRUG ADMINISTRATION
CENTER FOR DRUG EVALUATION AND RESEARCH

PHARMACOLOGY/TOXICOLOGY REVIEW AND EVALUATION

NDA NUMBER(S):	209531
SERIAL NUMBER:	000
DATE RECEIVED BY CENTER:	8/09/16
PRODUCT:	Nusinersen (ISIS 396443)
INDICATION:	Spinal Muscular Atrophy (SMA)
SPONSOR:	Biogen
REVIEW DIVISION:	Division of Neurology Products (DNP)
PHARM/TOX REVIEWER:	Ed Fisher
PHARM/TOX SUPERVISOR:	Lois Freed
DIVISION DIRECTOR:	Billy Dunn
PROJECT MANAGER:	Fannie Choy

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Note: All figures and tables in this review were excerpted from the sponsor's submission or literature.

I. EXECUTIVE SUMMARY

A. Drug

Trade name: Spinraza

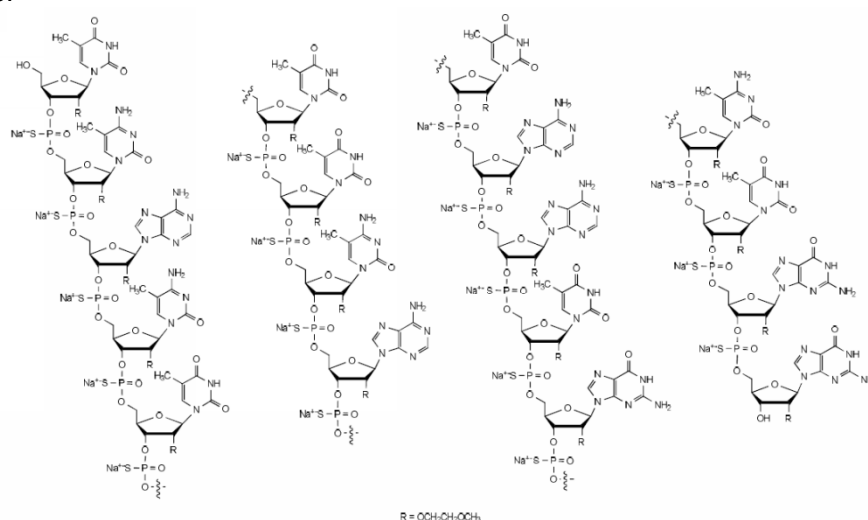
Generic name: Nusinersen

Code names: ISIS 396443

Chemical name: RNA, [2'-O-(2-methoxyethyl)](P-thio)(m5U-m5C-A-m5C-m5U-m5U-m5U-m5C-A-m5U-A-A-m5U-G-m5C-m5U-G-G)

CAS registry number: 1258984-36-9

Structure:



Drug class: 2'-O-(2-methoxyethyl) antisense oligonucleotide (ASO)

Indication: Spinal muscular atrophy

Clinical dose: 12 mg on Days (b) (4) followed by maintenance dosing every 4 months

Dosage forms: Sterile solution for intrathecal (IT) injection

Relevant IND: 110011

B. Background and brief discussion of nonclinical findings

ISIS 396443 is a fully modified 18-base 2'-MOE (2'-O-(2-methoxyethyl)) phosphorothioate oligonucleotide that was designed to be complementary to and hybridize with the pre-mRNA for human SMN2 to increase exon 7 inclusion and full length SMN protein expression. This structure is similar to endogenous RNA except for the substitution of a sulfur for a non-bridging oxygen at each internucleotide linkage and the addition of methoxyethyl groups (i.e., 2'-O-MOE) to the 2'-position of each nucleotide. In pharmacology studies, ISIS 396443 appeared to promote SMN2

exon 7 splicing in SMA mouse models, showing activity at tissue concentrations achieved clinically (~2-10 µg/g tissue).

In safety pharmacology studies, there were no pulmonary or cardiovascular effects in rats using continuous intrathecal (IT) infusion for 25 days, or effects on ECG in the repeat-dose IT toxicology studies in monkeys. The only safety pharmacology effects observed were transient post-dose effects on lower spinal reflex observed at the highest doses tested in monkeys (≥3 mg).

The toxicity and pharmacokinetics (PK) of ISIS 396443 were evaluated following single IT bolus doses (up to 7 mg) in adult monkeys or following repeated IT bolus doses (weekly or every other week) for 14 (0.3, 1, or 3 mg/dose) and 53 (0.3, 1, or 4 mg/dose) weeks in juvenile monkeys. A 13-week toxicity study was conducted in juvenile CD-1 mice using SC administration (1, 10, or 50 mg/kg weekly or biweekly) to assess systemic effects. IT bolus injection of ISIS 396443 produced dose- and time-dependent increases in exposure in the spinal cord and brain. Neurotoxicity (decreased lower spinal reflexes) was observed following single doses ≥ 3 mg in both adult and juvenile monkeys but appeared to be reversible and did not increase in incidence or severity with continued dosing. In the 14- and 53-week juvenile monkey studies, no drug-related mortality or effects on body weight, ECG, CSF chemistry/cell counts, clinical pathology, or complement activation (as measured by concentrations of Bb split product) were observed, and there was no evidence of systemic inflammation or non-CNS histopathology. Possible long-term effects on performance in a learning and memory test were observed at the HD in the 1-year study, which could be related to drug-related hippocampal neurohistopathology. However, due to data variability and small group sizes, this neurobehavioral effect did not reach statistical significance. Neuronal vacuolation (graded slight to minimal) in the inferior region of the hippocampus was observed at the MD and HD in both studies. In some animals at both doses, this vacuolation was associated with low incidences of neuronal and glial cell necrosis and cellular debris. Hippocampal vacuolation and necrotic cells/cellular debris were still present following 12 and 24 weeks of recovery in the 14- and 53-week studies, respectively. The NOAEL for neurohistopathology in monkeys (0.3 mg/dose or 39 mg/year) is similar to the proposed human maintenance dose (36 mg/yr) when calculated on a yearly basis and was associated with tissue levels similar to those measured in patient tissue samples at autopsy (see Tables V.1-3, pp.69-70).

In the 13-week SC mouse study, histopathologic findings attributed to the expected effects of local and systemic accumulations of oligonucleotide and/or their pro-inflammatory effects were observed in a variety of tissues included basophilic granules in the kidney, Kupffer cell hypertrophy, and vacuolated macrophages at the injection sites, lymph nodes and less commonly in multiple other tissues. These findings were generally not associated with evidence of necrosis, degeneration, or inflammation. However, 1 HD animal was found to have renal tubular degeneration characterized by basophilia, vacuolation, and single cell necrosis. Vacuolation and degeneration of renal tubular epithelial cells associated with basophilic granulation have been previously reported in animal studies of other phosphorothioate oligonucleotides.

No clear evidence of reproductive or developmental toxicity was seen in mouse and rabbit studies, and ISIS 396443 was negative for genotoxicity in a standard battery of in vitro and in vivo assays. Carcinogenicity studies were not conducted. The sponsor has requested a waiver based on the infeasibility of conducting lifetime studies in rodents by the IT route, but given the significant systemic exposure documented in clinical studies, it is recommended that a parenteral study in one species be conducted postmarketing.

C. Recommendations

The application is approvable from a pharmacology/toxicology standpoint.

II. PHARMACOLOGY

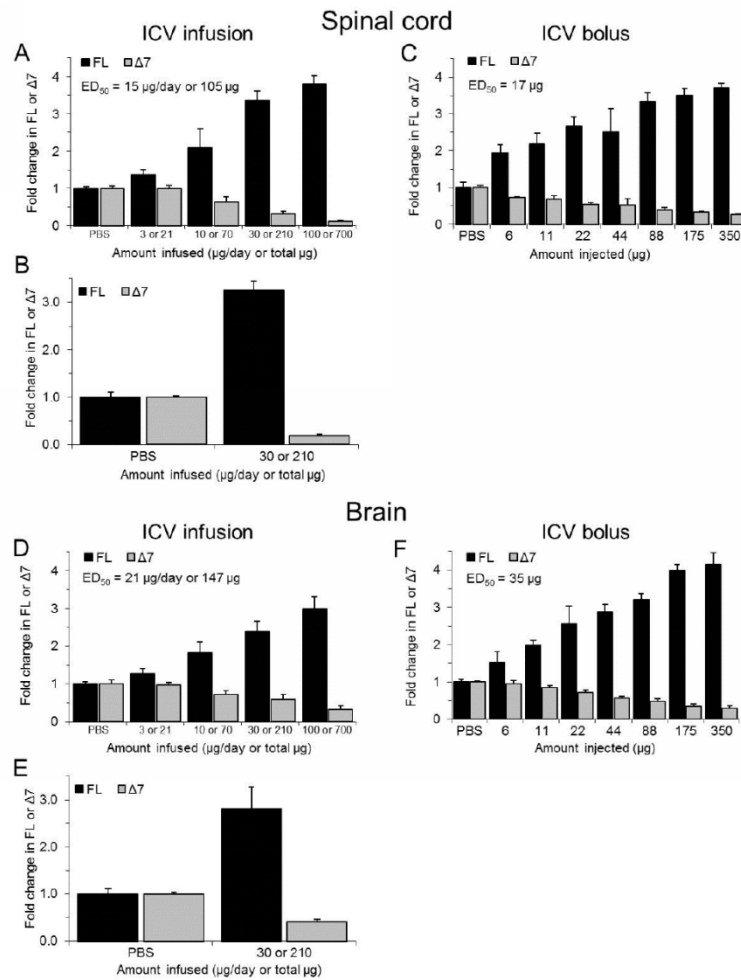
A. Brief summary

In vitro studies

ISIS 396443 was identified in studies in which candidate 2'-MOE antisense oligonucleotides (ASOs) were screened in in vitro splicing assays, reporter gene assays, and SMA patient fibroblasts. Two regions within exon 7 were found that promoted exon 7 inclusion in the splicing studies and SMA patient fibroblasts when bound by an ASO. Further studies identified a novel site approximately 10 nucleotides downstream from the intron/exon junction that promoted inclusion of exon 7 in cell culture assays. This site was shown to be effective in promoting exon 7 inclusion on SMN2 pre-mRNA. This site on the SMN2 pre-mRNA is referred to as intron splicing silencer N1 (ISS-N1). Further screening in SMA patient fibroblasts confirmed that ASOs targeting the ISS-N1 site were the most efficient at promoting SMN2 exon 7 inclusion. In the absence of ASO, approximately 30% of the SMN2 transcripts in patient fibroblasts cells are full length, i.e. contain exon 7. Treatment with the lead candidate increased the amount of full length transcript expressed and decreased the amount of transcripts missing exon 7. The effects of ISIS 396443 are sequence specific as scrambling six nucleotides in the sequence results in loss of activity. Treatment of patient fibroblasts with ISIS 396443 increased exon 7 inclusion in the SMN2 transcript and also increased the amount of SMN protein produced in these cells.

Since humans are the only species known to have the *SMN2* gene, in vivo pharmacology was examined in mouse models of SMA in which the endogenous mouse gene has been removed and the human *SMN2* gene is expressed. Studies were conducted in a mild phenotype model that expresses 4 copies of the human *SMN2* gene (*SMN*^{-/-}; *SMN2*^{2TG/2TG}). ISIS 396443 was either infused into the CSF over 7 days or administered as a single ICV bolus injection. Although both dosing regimens resulted in dose-dependent increases of properly spliced SMN2 and loss of transcripts lacking exon 7 (Figure II.1), the ICV bolus injection decreased the ED50 from 105 µg to 17 µg in the spinal cord. This increased potency was confirmed by quantifying ISIS 396443 levels in the spinal cord and brain and comparing this to SMN2 splicing. Preliminary duration of effect was examined in these studies as well. Comparable levels of splicing were observed up to 71 days post initiation of ICV infusion (Figure II.1 B and E). The prolonged PD effect is consistent with the long half-life of the ASO.

Figure II.1 Administration of ISIS 396443 by ICV infusion or ICV bolus injection



(A) Real-time RT-PCR analysis of SMN2 transcripts including exon 7 (FL) or excluding exon 7 (Δ7) in the lumbar spinal cord, 2 days after administration of ISIS 396443 by ICV infusion for 7 days at the indicated daily doses and total dose administered (e.g., 3 µg/day for 7 days = 21 µg). For each dose level, n = 5. Error bars represent the S.D. The calculated ED₅₀ is shown. (B) SMN2 splicing was assessed 71 days after the end of the 30 µg/day ICV infusion in A. (C) Real-time RT-PCR analysis of FL and Δ7 SMN2 transcripts in the lumbar spinal cord, 9 days after administration of ISIS 396443 by a single ICV bolus injection at the indicated dose. For each dose level n = 4. Error bars represent the S.D. The calculated ED₅₀ is shown. (D-F) Same as in A-C, except that the analysis is for the brain.

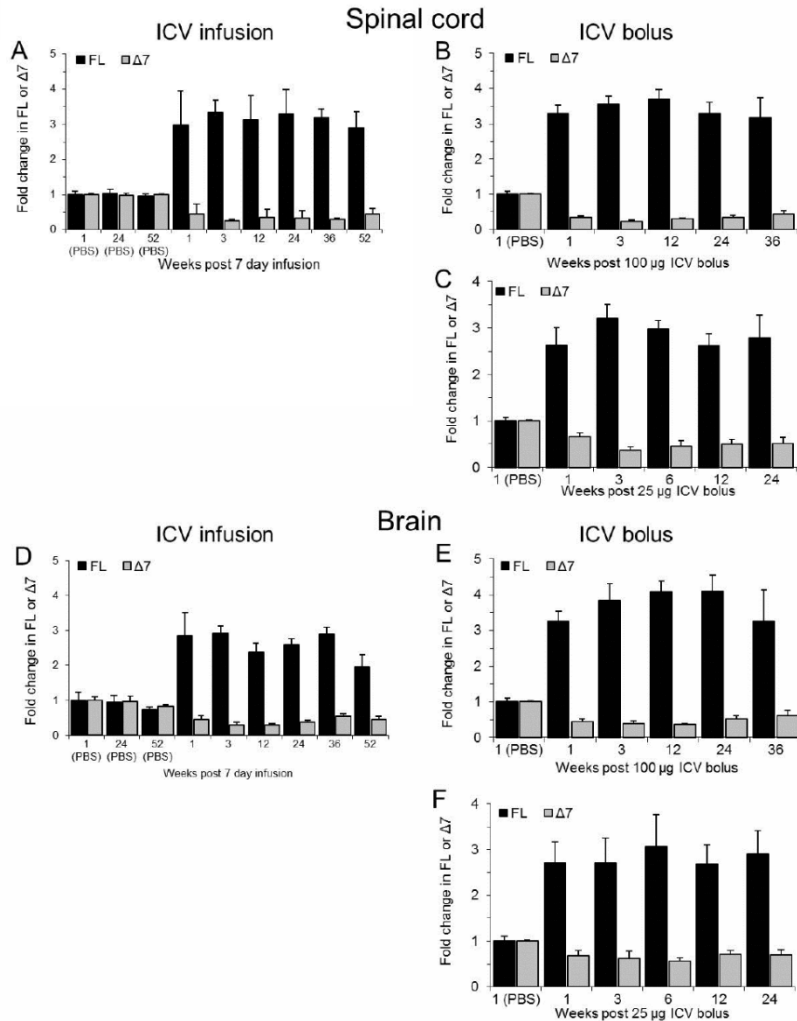
ISIS 396443 distributed broadly throughout the spinal cord and brain, with accumulation in cortical, striatal, and hippocampal regions. Highest concentrations of ISIS 396443 were found in neurons, including motor neurons in the lumbar spinal cord. SMN protein staining showed increased expression in ASO-dosed mice relative to PBS controls.

When induction of allograft inflammatory factor-1 (Aif1), a marker for monocyte/microglia activation, was measured in mice, no increase in Aif1 transcript levels was seen at ISIS 396443 doses of up to 700 µg (ICV infusion) or 350 µg (ICV bolus).

In similar studies using a different mouse model (C/C), in which the mouse *Smn1* allele (C-allele) has the human genomic SMN2 sequence after exon 6 as well as the 42-kb human SMN2 locus, ICV bolus injection of ISIS 396483 increased SMN2 expression in a dose-dependent manner. The ED₅₀ and EC₅₀ values in both the spinal cord and brain were comparable in the two models.

When SMN2 transgenic mice were dosed by ICV infusion for 7 days and SMN2 splicing monitored out to 1 year in brain and spinal cord, SMN2 exon 7 inclusion was maintained for the duration of the experiment in both the brain and spinal cord. Translation of the mRNA was confirmed by immunohistochemistry, showing SMN positive staining in the spinal cord. ICV bolus provided a similar duration, of at least 24 to 36 weeks in brain and spinal cord (Figure II.2). There was no significant increase in *Aif1* over the course of the study.

Figure II.2 Duration of action after ICV infusion or ICV bolus injection of ISIS 396443



(A) Real-time RT-PCR analysis of SMN2 FL and Δ7 transcripts in the lumbar spinal cord at the indicated time points after administration of ISIS 396443 by ICV infusion at 50 μg/day for 7 days. PBS, n = 4; 1 and 3 weeks, n = 5; 12 weeks, n = 6; 24 weeks n = 7; 36 weeks, n = 6; 52 weeks n = 7. Error bars represent the S.D. (B) Real-time RT-PCR analysis of FL and Δ7 SMN2 transcripts in the lumbar spinal cord at the indicated time points after administration of 100 μg of ISIS 396443 by a single ICV bolus injection. For each time point, n = 5, except for the 24 weeks group, for which n = 4. Error bars represent the S.D. (C) Same as in B, except that 25 μg of ISIS 396443 was injected. (D-F) Same as in A-C, except that the analysis is for the brain.

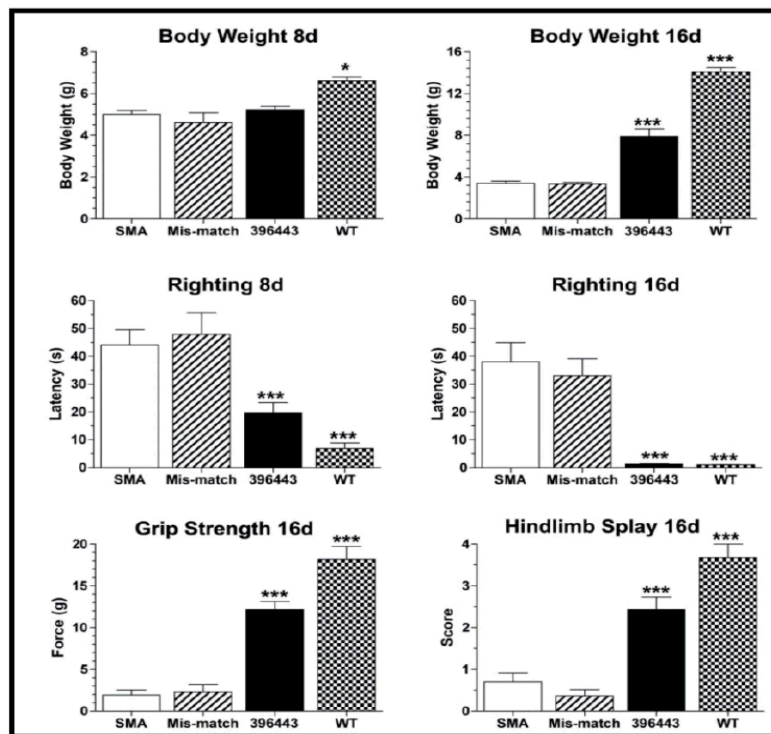
Additional evidence that the prolonged induction of SMN2 is directly caused by ISIS 396443 was provided by an experiment performed with an ASO fully complementary to ISIS 396443 (α443). Injection of α443 three weeks after injection of ISIS 396443 completely reversed SMN2 splicing to baseline within two weeks in both the brain and spinal cord of SMN2 transgenic mice (mild phenotype).

The functional consequences of increased SMN2 were evaluated in SMA models of varying severity. In the mild phenotype SMN2 mouse (loss of the tail and ear necrosis beginning 3 to 4 weeks after birth), which lacks mouse *Smn1* but has four copies of human SMN2, intrauterine ICV bolus injection of ISIS 396443 on GD15 produced a dose dependent effect on exon 7 inclusion in pups 7 days after birth, with maximal effect (92% inclusion) occurring at a dose of 20 μ g. At this dose, ISIS 396443 significantly delayed the onset of tail necrosis and ear necrosis (which are the major symptoms in mouse model but not seen in humans) from 3 weeks in untreated mice to 9 weeks. A dose of 10 μ g had an intermediate effect on tissue necrosis. ICV dosing on GD15 had no effect on SMN2 splicing in tail tissue measured on PND 14, suggesting that the beneficial activity was not due to a direct effect of ISIS 396443 in tail tissue. Dosing of pups with 20 μ g on PND1 or 2 delayed tail necrosis by approximately 2 weeks indicating that early treatment was more effective.

In a severe SMA mouse model (SMN Δ 7) that better replicates some of the features of prenatal or severe type I SMA, a single ICV injection of 4 μ g on PND 0 resulted in a significant increase in body weight at PND16 compared to untreated or control oligonucleotide treated animals. Motor functions, including righting reflex, grip strength, and hindlimb splay, were also improved.

A single ICV bolus injection produced a dose-dependent effect on survival of severe SMA mice. Median survival for untreated mice and control oligonucleotide treated mice was 16 days after birth. A dose of 0.5 μ g increased survival to 20 days; maximal effects on survival were achieved at the 4 μ g dose (median survival 26 days). Some animals survived longer than 50 days in groups treated with 2 and 4 μ g (Figure II.3). It was also reported that ISIS 396443 administration preserved the morphology of neuromuscular junctions, compared to untreated animals, and increased the size of myofibers in Δ 7 mice.

Figure II.3 ISIS 396443 Improves Body Weight and Motor Function in Severe SMA Mice



ISIS 396443 (4 μ g) was administered as a single ICV bolus on PND0 pups of SMA Δ 7 mice. Body weight and various motor function tests were performed 8 and 16 days after birth and show that ISIS 396443 produces a significant increase in weight, measure of strength (i.e. grip strength), and motor function (i.e., righting reflex and hindlimb splay) that approach that of wild type animals. * $p < 0.05$; *** $p < 0.001$

The duration of action of ISIS 396443 was determined following a single 4 µg ICV bolus injection on PND0 in SMA Δ7 mice by measuring the concentration of ISIS 396443 and level of SMN2 exon 7 inclusion in the spinal cord tissue at 3, 16, and 30 days after the single injection. At three days after injection, ISIS 396443 increased exon 7 inclusion in spinal cord tissue and reached a maximal effect 16 days after injection. The effect was lost by 30 days after the single injection. The same dose produced a median survival of 26 days, consistent with the duration of effect. A single 4 µg ICV bolus injection produced spinal cord concentrations of approximately 8-10 µg/g at 3 days and 2 µg/g at 16 days post-dosing, when maximal efficacy was observed. Tissue concentration at 30 days post dosing was below 2 µg/g. Based upon these results, the sponsor predicted that tissue concentrations between 2 and 10 µg/g are required to produce efficacy in this mouse model. The duration of action following ICV bolus injection in mice on PND0 differ from the results following ICV infusion in adult mice, in which the duration of action was greater than 6 months, although the effective tissue concentrations were similar in ranges. This was thought to possibly be due to more rapid clearance of ASO from the tissue in the developing and growing neonate and to a dilution effect as the tissue increases in size.

ISIS 396443 was shown to be active in peripheral tissues in studies in SMN2 transgenic mice. IP bolus injection every 2 days for a total of 4 doses resulted in a dose-dependent increase in *SMN2* splicing correction in the liver, with an ED50 of 31 mg/kg. The EC50 value obtained in the liver (135 µg/g) was higher than had been observed in the spinal cord (1.6 µg/g) and brain (5.7 µg/g). In contrast to the long-duration of action observed with ISIS 396443 in the CNS, the duration of action of ISIS 396443 in the liver after IP bolus injection was much shorter, with minimal SMN2 splicing correction observed 8 weeks after dosing. In agreement with the shorter duration of SMN2 splicing correction, the half-life of ISIS 396443 in liver was about 20 days. The longer duration of effect in neurons relative to liver has been observed for other ASOs.

CSF SMN protein level was evaluated as a possible biomarker using another ASO (ISIS 449323) with the same sequence as ISIS 396443 but with mixed phosphorothioate and phosphodiester modifications in the backbone. ICV bolus injection of ISIS 449323 into the CSF of adult mice transgenic for human SMN2 resulted in almost complete SMN2 exon 7 inclusion in CNS tissues and a 3.5-fold increase in the amount of SMN protein in CSF. Similar results were found in a second study in which a control ASO did not increase SMN protein in the CSF while ISIS 449323 did.

B. Safety Pharmacology

1. A pulmonary and cardiovascular safety evaluation of ISIS 396443 administered as a continuous intrathecal infusion in albino rats (Study #396443-AS04 (727-038), conducted by (b) (4), report dated 12/14/12, GLP)

a. Methods

Male S-D rats (8/gr) were administered ISIS 396443 (0, 0.02, 0.06, or 0.2 mg/day; Lot #RP396443-001) by continuous intrathecal infusion (0.25ul/hour) via implanted minipumps for 28 days.

Group	Dose ^b (mg/day)	Study Design Concentration (mg/mL)	Number of Animals	
			Male	Female
1 ^a	0	0 (Vehicle)	8	0
2	0.02	3.3	8	0
3	0.06	10	8	0
4	0.2	33	8	0

^a Group 1 received the vehicle, aCSF

^b ISIS 396443 or the vehicle was administered as a continuous intrathecal lumbar infusion using osmotic infusion pumps at a flow rate of approximately 0.25 µL/hour set to deliver material for approximately 28 days.

b. Results

i. Mortality, clinical signs, body weights

One MD (#318) and 3 HD (#s 327, 331, and 332) animals died or were sacrificed, but these deaths were attributed to surgical procedure or accident and not drug. No drug-related signs or effects on body weight were observed.

ii. Pulmonary monitoring (monitoring in plethysmograph chamber on Day 11, 12, or 22, for 60 seconds at 15 minute intervals for at least 4 hours)

No drug-related effects on pulmonary function were observed.

iii. Cardiovascular monitoring (systolic, diastolic, MAP, and HR monitored on SD19, 22, 23, or 25)

No significant drug-related effects on heart rate or blood pressure were observed.

iv. Postmortem evaluations (lumbar spinal cord samples collected after sacrifice following completion of cardiovascular monitoring)

ISIS 396443 concentrations in the lumbar spinal cord were 70, 170, and 389 µg/g at the end of the infusion at the LD, MD, and HD, respectively.

III. PHARMACOKINETICS

PK in monkeys was evaluated in a dedicated PK study in adults (Study 396443-APK01) and in the 3 toxicology studies: 1) single bolus IT study in adult monkeys (Study 396443-AS01), 2) a 14-week IT study in juvenile monkeys (Study 396443-AS03), and 3) a 53-week IT study in juvenile monkeys (Study 396443-AS06). The results of the latter 2 studies are described in the Toxicology section of this review.

A. Absorption

In the single dose adult monkey study in which animals received either the vehicle (artificial CSF) or ISIS 396443 (1, 3, or 7 mg given as 3 min IT bolus injection via lumbar catheter), peak CSF concentrations were observed within 15 minutes of dosing and then rapidly declined (up to 24–48 hours post injection). Following the initial rapid distribution phase, ISIS 396443 elimination from the CSF was much slower with an observed elimination half-life of approximately 11 days. MRT_{0-48 hr} values ranged from 4.3 to 7.7 hrs at the 3 mg dose (Table III.1). AUC values indicated that the majority of CSF exposure was during the first 12 hrs. Plasma concentrations peaked at 2-5 hrs and CSF-to-plasma ratios ranged from 95-524 within 2 to 48 hr (Table III.2).

Table III.1. ISIS 396443 CSF PK parameters in adult monkeys after single IT bolus dose

Dose Level	Variable	Units	N	Mean ± SD	Median (Range)
3 mg	AUC _{0-12hr}	hr*µg/mL	8	1468 ± 700	1215 (877 - 2824)
3 mg	AUC _{0-24hr}	hr*µg/mL	8	1641 ± 758	1425 (977 - 3159)
3 mg	AUC _{0-48hr}	hr*µg/mL	8	1734 ± 768	1552 (1033 - 3250)
3 mg	CL _{csf,0-48hr} ^a	mL/hr	8	1.99 ± 0.720	1.93 (0.920 - 2.90)
3 mg	C _{max}	µg/mL	8	592 ± 374	420 (322 - 1304)
3 mg	MRT _{0-48hr}	hr	8	5.40 ± 1.15	5.15 (4.28 - 7.70)
3 mg	T _{max}	hr	8	0.310 ± 0.120	0.250 (0.250 - 0.500)
3 mg	t _{1/2} ^b	days	-	11.35	-

^a CL_{csf,0-48hr} = Dose (mg)/AUC_{0-48hr} (hr*µg/mL) x 1000

^b Half-life was determined using the sparse sampling function in Phoenix WinNonlin, therefore only a single value is reported

Table III.2. ISIS 396443 Plasma PK parameters in adult monkeys after single IT bolus dose

Dose (mg)	C _{max} (µg/mL)		T _{max} (hr)		AUC _{0-24hr} (hr*µg/mL)		CL/F (mL/hr) ^a	
	N	Mean ± SD	N	Median (Range)	N	Mean ± SD	N	Mean ± SD
1	6	0.222 ± 0.131	6	5 (1.0 - 10)	6	2.03 ± 0.408	6	513 ± 124
3	13	0.455 ± 0.213	13	4 (2 - 8)	13	4.50 ± 1.83	13	772 ± 304
7	6	2.61 ± 1.93	6	2 (1 - 4)	6	14.6 ± 4.32	6	521 ± 163

^a CL/F = Dose (mg)/(AUC_{0-24hr}/1000)

In the dedicated PK study, male cynomolgus monkeys received either 4 IT bolus doses or 4 IV bolus doses at 1 mg/dose on Days 1, 8, 15, and 22. After the IT bolus injection, peak CSF concentrations occurred 1 hour (the first evaluated time point), and plasma concentrations peaked 4 hours post dose. CSF concentrations (collected prior to the 2nd through 4th dose)

increased approximately 2.5-fold from the first to third IT dose (Table III.3). CSF concentrations declined rapidly over the first 48 to 168 hrs after the fourth IT injection, but terminal half-life was estimated at 102 days. This was considered the most representative estimate of the elimination half-life in CSF since sampling continued for up to a year.

Table III.3 ISIS 396443 CSF concentrations after IT and IV bolus administration of 1 mg dose

Group	Route ^a	No. of Doses	Day of Collection	Time Since Previous Dose (hr)	ISIS 396443 Concentration (µg/mL)		
					N	Mean ± SD	Median (range)
2	IT	0	1	0	16	BLQ	BLQ
		1	8	0	15	0.116 ± 0.146	0.0507 (0.00557 - 0.465)
		2	15	0	15	0.105 ± 0.0902	0.0687 (0.0186 - 0.347)
		3	22	0	15	0.134 ± 0.0879	0.128 (0.0309 - 0.330)
		4	22	1	4	119 ± 30.4	109 (94.8 - 163)
		4	22	4	4	47.0 ± 16.5	43.2 (32.5 - 69.1)
		4	22	12	4	17.9 ± 7.57	15.4 (12.2 - 28.4)
		4	24	48	4	2.65 ± 1.15	2.77 (1.39 - 3.67)
		4	29	168	8	0.179 ± 0.0801	0.164 (0.0662 - 0.316)
		4	85	1512	7	0.0326 ± 0.0131	0.0367 (0.0153 - 0.0542)
		4	183	3864	7	0.0183 ± 0.0183	0.0111 (0.00151 - 0.0498)
		4	253	5544	7	0.0118 ± 0.0117	0.0124 (0.000149 - 0.0310)
1	IV	4	365	8232	4	0.00534 ± 0.00571	0.00367 (0.000868 - 0.0131)
		4	29	168	3	BLQ	BLQ

Plasma AUC values after IT and IV administration are shown in Table III.4. Plasma concentrations measured 168 hours after 1, 2, and 3 IT doses accumulated up to ~3-fold with multiple dosing. The calculated AUC values for the IV group were thought to be underestimated due to lack of sampling prior to 1 hr on Day 22, but the higher AUC with IT administration (IT:IV ratio >1.0) was thought to support a tissue distribution advantage after IT administration.

Table III.4 Plasma PK after 1 mg IT and IV bolus injections

PK Parameter	N	IT - Lumbar	N	IV
C _{max} (µg/mL)	17	0.154 ± 0.114	3	0.367 ± 0.0440
T _{max} (hr) ^a	17	4 (1 - 8)	3	1 (1 - 1)
AUC _{0-24hr} (µg*hr/mL)	17	1.53 ± 0.441	3	0.720 ± 0.0662
AUC _{0-168hr} (µg*hr/mL)	17	2.07 ± 0.532	3	1.02 ± 0.163

Values reported as mean ± standard deviation

^a Tmax values reported as median (range)

Comparison of adult and juvenile monkey studies indicated that plasma exposures in juveniles were 2-4-fold higher at the same mg doses compared to adults (Table III.5), consistent with body weight differences and supporting a direct relationship between plasma clearance and body weight, as observed previously for similar classes of ASOs.

Table III.5 Plasma exposures in juvenile and adult monkeys after single 1 mg or 3 mg IT dose

Exposure Measure	IT Dose (mg)	Juvenile Monkeys	Adult Monkeys
$C_{\max}$ ($\mu\text{g/mL}$)	1	0.569 to 0.683 ^{a,b}	0.222 ^c
	3	1.97 ^b	0.455 ^c
$AUC_{0-24\text{hr}}$ ($\mu\text{g}\cdot\text{hr/mL}$)	1	3.64 to 4.02 ^{a,b}	2.03 ^c
	3	12 ^b	4.50 ^c
Body Weight (kg)	NA	1.0	3.0

B. Distribution

Tissue distribution in adult monkeys following 3 weeks of IT or IV bolus injection of a 1 mg dose (Study 396443-APK01) is shown in Tables III.6-7. After IT dosing, the highest CNS tissue concentrations were measured in the lumbar spinal cord followed by the thoracic and cervical spinal cord regions and the frontal and temporal cortex brain regions. Lower concentrations were observed in the brain stem, cerebellum, hippocampus, and putamen. (The high cerebellum concentration on Day 85 was considered anomalous.)

Table III.6 CNS tissue concentrations in monkeys after multiple 1 mg IT bolus doses

Tissue	Day 29		Day 85		Day 183		Day 253		Day 365	
	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD
Brain Stem	4	10.2 $\pm$ 8.65	3	6.6 $\pm$ 0.797	3	0.931 $\pm$ 0.402	3	1.86 $\pm$ 2.27	4	0.472 $\pm$ 0.343
Cerebellum	4	3.74 $\pm$ 2.49	3	13.5 $\pm$ 6.86	3	0.575 $\pm$ 0.437	3	1.35 $\pm$ 1.33	4	1.09 $\pm$ 0.800
Cervical Spinal Cord	4	19.9 $\pm$ 4.69	3	9.58 $\pm$ 0.602	3	3.64 $\pm$ 2.44	3	3.50 $\pm$ 3.25	4	1.86 $\pm$ 1.76
Frontal Cortex	4	13.6 $\pm$ 4.98	3	7.39 $\pm$ 5.36	3	5.00 $\pm$ 3.61	3	7.30 $\pm$ 8.61	4	1.96 $\pm$ 1.10
Hippocampus	4	3.47 $\pm$ 3.24	3	1.75 $\pm$ 0.745	3	2.54 $\pm$ 1.85	3	2.19 $\pm$ 2.96	4	1.61 $\pm$ 0.953
Lumbar Spinal Cord	4	91.4 $\pm$ 37.9	3	37.5 $\pm$ 15.9	3	24.2 $\pm$ 18.0	3	10.8 $\pm$ 9.14	4	4.34 $\pm$ 4.78
Putamen	4	0.493 $\pm$ 0.352	3	0.658 $\pm$ 0.317	3	0.230 $\pm$ 0.106	3	0.437 $\pm$ 0.392	4	0.306 $\pm$ 0.250
Temporal Cortex	4	16.6 $\pm$ 5.93	3	15.3 $\pm$ 6.7	3	4.91 $\pm$ 1.86	3	4.64 $\pm$ 4.86	4	2.07 $\pm$ 0.948
Thoracic Spinal Cord	4	35.8 $\pm$ 9.51	3	15.7 $\pm$ 4.62	3	5.93 $\pm$ 3.79	3	3.53 $\pm$ 2.27	4	1.96 $\pm$ 1.20

Values reported as $\mu\text{g/g}$

SD = Standard Deviation

Table III.7 Systemic tissue concentrations in monkeys after multiple 1 mg IT or IV bolus doses

Tissue	Route	Day 29		Day 85		Day 183		Day 253		Day 365	
		N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD
Kidney Cortex	IT	4	61.3 $\pm$ 15.8	3	14.3 $\pm$ 11.6	3	0.962 $\pm$ 0.797	3	0.362 $\pm$ 0.197	4	0.652 $\pm$ 0.363
	IV	3	112 $\pm$ 73.5	-	NSC	-	NSC	-	NSC	-	NSC
Liver	IT	4	4.74 $\pm$ 2.02	3	0.113 $\pm$ 0.0556	3	0.0598 $\pm$ 0.0425	3	BLQ	4	BLQ
	IV	3	3.47 $\pm$ 1.32	-	NSC	-	NSC	-	NSC	-	NSC
Skeletal Muscle	IT	4	0.0140 $\pm$ 0.0280	3	BLQ	3	BLQ	3	BLQ	4	BLQ

Values reported as $\mu\text{g/g}$

BLQ = Below limit of quantitation ($<0.015 \mu\text{g/g}$); SD = Standard deviation; NSC = No sample collected

CNS tissue concentrations in adult and juvenile monkeys were generally similar while liver concentrations were higher in juveniles than in adults. Lumbar spinal cord concentrations 7 days after 4 weekly 1 mg doses in adult monkeys (total dose = 4 mg over 3 weeks [1.33 mg/week])

were 91.4 ± 37.9 were consistent with those measured in juveniles 7 days after last dose in the 14-week toxicity study (total dose = $10 \times 1 \text{ mg} = 10 \text{ mg}$ [0.77 mg/week]) ($78.6 \pm 53.3 \text{ } \mu\text{g/g}$).

In mice, monkeys, and humans, ISIS 396443 was highly bound to plasma proteins (>94% bound) at clinically relevant or higher plasma concentrations (100 ng/mL and $5 \text{ } \mu\text{g/mL}$), which would tend to limit glomerular filtration and urinary excretion. Protein binding in monkey and human CSF, however, was much lower (<25%) at $5 \text{ } \mu\text{g/mL}$, with no measurable binding observed at $150 \text{ } \mu\text{g/mL}$, consistent with lower protein concentration existing in CSF (Study 396443-IS05 and Study 396443-IS06).

Tissue distribution and placenta transfer were evaluated in a combined fertility and developmental toxicity study in mice following multiple SC bolus injections (Study 396443-AS08) and in an embryofetal development toxicity study in rabbits following multiple SC bolus injections (Study 396443-AS09) as described in the toxicology section.

C. Metabolism

Metabolism was evaluated in selected CSF, plasma, and tissue samples from the 14-week and 53-week monkey toxicity studies. Following IT administration, intact ISIS 396443 was the only oligonucleotide detected in monkey CSF extracts. In monkey plasma samples, intact ISIS 396443 was the most abundant oligonucleotide detected and accounted for >95% of the total oligonucleotides. The only drug-related moiety detected in plasma sample extracts was a 17-mer oligonucleotide (N-1 from 3'-end only; Table III.8). Intact ISIS 396443 was also the most abundant oligonucleotide detected in all evaluated monkey tissues, including brain cortex, lumbar spinal cord, liver, and kidney, and accounted for 79% to 81% of the total oligonucleotides (Table III.9). The metabolites detected in these tissues were the chain-shortened oligonucleotides N-1, N-2, N-3, and N-4 from 3'-end, as well as the chain shortened oligonucleotides N-1, N-2, and N-3 from 5'-end. The N-1 metabolite from the 3'-end was the most abundant metabolite in all evaluated tissues (up to 20% of the total oligonucleotide present when using peak area to measure abundance), while each individual remaining metabolite accounting for less than 5%. These findings were thought to be consistent with a stable backbone structure, with metabolism appearing to occur mainly via slow exonuclease-mediated hydrolysis at both the 3'- and 5'-ends.

Table III.8: Relative abundance (%) of ISIS 396443 and associated metabolite in monkey plasma

Oligonucleotide	Day 1 2 hrs Post-dose	Day 36 7 days Post-dose	Day 99 2 hrs Post-dose
ISIS 396443 (18-mer, Full-Length)	97.74 ± 1.34	ND	96.93 ± 0.52
ISIS 524402 (17-mer, 3'-deletion)	2.26 ± 1.34	ND	3.07 ± 0.52

Table III.9 (%) ISIS 396443 and metabolites in monkey liver, brain, and lumbar spinal cord

Oligonucleotide	Brain	Liver	LSC
ISIS 396443 (Full Length)	80.5 ± 2.05	78.77 ± 4.05	78.61 ± 1.91
17-mer (N-1, 3'-Deletion)	17.16 ± 1.56	17.05 ± 3.76	18.44 ± 1.64
16-mer (N-2, 3'-Deletion)	2.33 ± 0.75	2.20 ± 0.51	2.96 ± 0.32
15-mer (N-3, 3'-Deletion)	ND	1.98 ± 0.58	ND

Data presented are mean $\pm$ standard deviation (n=7 for Brain and Liver, n=6 for LSC).

ND = not detected

LSC = Lumbar Spinal Cord

D. Excretion

Urinary excretion was not evaluated. The chain shortened oligonucleotides are expected to be excreted in urine following slow metabolism in tissues, which represents the major pathway for whole body clearance of ASO compounds reported in the literature.

IV. TOXICOLOGY

A. SINGLE DOSE TOXICITY

1. A single dose toxicity and pharmacokinetic study of ISIS 396443 administered by intrathecal lumbar slow bolus in cynomolgus monkeys ^{(b) (4)} Study #004-016, ISIS Study #396443-AS01, conducted by ^{(b) (4)}, report dated 1/1/11, GLP)

a. Methods

Cynomolgus monkeys (2-3/sex/group (groups 2-4), 5/sex HD TK; 3.6-4.2 years old) were given single IT bolus injections (0, 1, 3, or 7 mg; Lot RP396443-001) via surgically implanted lumbar catheters at least 6 days prior to initiation of dosing. The dose was administered over an approximately 3 minute period after which the catheter was flushed with 0.5 ml aCSF. Observations included BW, clinical observations, and neurological and physical examinations. Blood and CSF samples for clinical pathology, complement activation, and toxicokinetic (TK) analysis were taken at various time points. Seven days after dosing 2-3 animals per sex per group were euthanized. TK animals were euthanized on SD3, 21, 57 and 71.

Study Design: Single IT Lumbar Bolus								
Group	Treatment Regimen ^a			Total Number of Animals	Sacrifice			
	Dose (mg)	Concentration (mg/mL)	Volume (mL/dose)		Day 8		TK ^c	
					M	F	M	F
1 ^b	0	0	1	4	2	2	0	0
2	1	1	1	6	3	3	0	0
3	3	3	1	16	3	3	5	5
4	7	7	1	6	3	3	0	0

^a Single IT-L Bolus on Day 1 administered over 3 minutes

^b Group 1 received vehicle (aCSF)

^c TK animals (1 or 2/sex) were sacrificed 2 (2 M/1 F), 20 (1 M/1 F), 56 (1 M/1 F) or 70 (1 M/2 F) days following dose administration

b. Results

i. Mortality (assessed twice daily)

All animals survived to the scheduled euthanasia.

ii. Clinical observations: (data collected at least once daily)

Clinical signs related to drug were reported in 1 HD male (animal 11) that was observed lying on the bottom of its cage at the 2-hour post dose observation. Subsequent neurological and physical examinations found slightly decreased body temperature and neurological deficits (2 hour: cutaneous reflex on the right side and tail reflex; 4 hour – proprioception in the right hindlimb) in this animal. No other drug-related clinical signs were observed.

iii. Body weight: (data collected prior to surgery, on the day of dosing, then weekly, and prior to necropsy)

No significant dose-related effects on body weight or food consumption were observed.

iv. Physical and neurological exams: (completed prior to catheter placement, once prior to initiation of dosing, 4 hours after dosing, prior to necropsy)

Male animal 11 (above) showed physical and neurological impairment. At 2 hrs post-dose, the animal appeared drowsy with hindlimb weakness. Body temperature (99.8 °F) and respiratory rate (32 breaths/minute) were below the normal range. By 4 hours post-dose, respiratory rate was within normal limits, but body temperature remained low (98.9°F). Since a slight decrease in body temperature was observed in most animals, including controls, after dosing, the relationship of this finding to drug was considered questionable. The neurological exam for this animal indicated negative responses for the cutaneous reflex on the right side and the tail reflex at 2 hours following dose administration. At the 4-hour evaluation, a negative response for the proprioceptive reflex for the right hindlimb was observed but the drowsiness and hindlimb weakness had reversed. No abnormal findings on neurological exam were observed prior to necropsy on SD8 or prior to the TK necropsies on SD 3, 21, 57 or 71. TK animal (# 025) (male, 3 mg group) showed negative cutaneous reflex on the left side prior to necropsy on SD 21, but due to the long interval after dosing, this finding was not considered related to the test article.

v. Clinical Pathology

No significant dose-related effects on clinical pathology (hematology, clinical chemistry, coagulation) were observed. No apparent dose-related effects on CSF chemistry or cellularity were observed. No clear dose-related changes in complement split product Bb were observed. At 6 hours post-dose, Bb levels increased ~2X over baseline levels in all dose groups including controls.

vi. Histopathology

Cellular infiltrate (rated slight to minimal) was observed in the meninges of 1/3 LD and 1/3 HD males, 1/3 MD and 2/3 HD females, and 4/5 MD TK males. No similar cellular infiltrates were observed in controls. An inflammatory mass was observed at the catheter tip of 1 MD female. Such masses have been observed in other intrathecal studies and the study pathologist considered this finding a result of the presence of the catheter. No drug-related histologic findings were observed in any non-CNS tissues.

c. Conclusions:

Single IT bolus injections of ISIS 396443 (0, 1, 3, or 7 mg) to adult monkeys produced apparent acute neurological impairment after dosing at the HD (single male).

B. SUBCHRONIC TOXICITY

1. A 14 week multiple dose toxicity study of ISIS 396443 administered as an intrathecal lumbar bolus injection in juvenile cynomolgus monkeys with a 4 week interim necropsy and 12 week recovery (b) (4) Study Number: 004-018, Isis Study Number: 396443-AS03, conducted by (b) (4), report dated 6/1/12, GLP (except CSF cell counts, complement Bb analysis, metabolite analysis, and sponsor performed bioanalysis of hippocampus)

a. Methods

Young cynomolgus monkeys (3/sex/group interim, 3-4/sex/group terminal, and 2/sex/group C and HD recovery; 7-8 months of age, 0.9-1.6 kg at start of dosing) were administered ISIS 396443 (Lot RP396443-001) at doses of 0 (artificial CSF vehicle), 0.3, 1, or 3 mg/dose (0.4, 1.3, or 4 mg/mL, 0.75 mL/dose) IT by lumbar bolus injection (slow bolus injection over 3 minutes via implanted lumbar catheter). Doses were administered to all animals on Days 1, 8, 15, 22, and 29. An interim sacrifice was performed on 3/sex/group ~7 days after the 5th dose (Day 36). The

remaining C, LD, and MD monkeys received additional doses once every other week on Days 43, 57, 71, 85, and 99. The remaining HD monkeys received additional doses once weekly on Days 36, 43, 50, 57, 64, 71, 78, 85, 92, and 99. 3-4/sex/group were sacrificed on Day 106 (approximately 7 days following the final dose administration). 2/sex from the C and HD groups were sacrificed following a 12-week recovery (Day 183).

Study Design: Multiple IT Lumbar Bolus											
Treatment Regimen ^a				Total Dose (mg)	Total Number of Animals	Interim (Day 36)		Sacrifice Terminal (Day 106)		Recovery (Day 183)	
Group	Dose (mg)	Conc. (mg/mL)	Volume (mL/dose)			M	F	M	F	M	F
1 ^b	0	0	0.75	0	17	3	3	3	4	2	2
2	0.3	0.4	0.75	3	13	3	3	3	4	0	0
3	1	1.3	0.75	10	13	3	3	3	4 ^d	0	0
4	3	4	0.75	45	18	3	3	4 ^c	4	2	2

^a Groups 1-3: IT Bolus on Days 1, 8, 15, 22, 29, 43, 57, 71, 85 and 99 administered over 3 minutes. Animals assigned to the interim sacrifice received the final dose on Day 29. Group 4: IT Bolus on Days 1, 8, 15, 22, 29, 36, 43, 50, 57, 64, 71, 78, 85, 92 and 99 administered over 3 minutes. Animals assigned to the interim sacrifice received the final dose on Day 29.

^b Group 1 received vehicle (aCSF).

^c Group 4 Animal 024 was found dead on Day 8. Animal 024A was added to the study to preserve the number of animals at each dose level.

^d Group 3 Animal 049 was sacrificed on Day 64 (~ 7 days after 7th dose) due to a non-functioning catheter system.

Observations included body weight and food consumption; clinical observations; neurological (general sensory and motor functions) and physical examinations; TK in plasma, CSF, and tissue; ophthalmology; ECG; clinical pathology; complement activation; immunohistochemistry; gross pathology; microscopic examination of a full panel of non-CNS tissues in C and HD groups (10% neutral buffered formalin-fixed); and an expanded examination of brain and spinal cord from all animals. Histopathologic evaluations were completed by Tox Path Specialists. Brain sections from all animals (8 coronal sections from the left side of the brain) were paraffin embedded and serial sections were stained with H&E and examined microscopically. In addition, two sections of the brains were stained with Fluoro-jade B for evaluation of neuronal degeneration. For selected slides (or a single slide) from the brain of all animals, GFAP, Fluoro-jade B, and PGP 9.5 staining was performed on serial sections from the tissue in the existing paraffin block to investigate the cellular location (if intracellular) of the vacuoles (noted at the original microscopic evaluation) and assist with characterizing other possible changes not apparent with H&E staining. In those cases a single section in the area noted to be affected during the original microscopic evaluation or the section closest to that area was stained from all animals.

Doses were based on the results of the single dose study in adult monkeys and two pilot PK studies of a similar uniform 2'-MOE-modified oligonucleotide. The HD (3 mg/dose) was said to be limited by the potential for acute and transient changes thought to be associated with the IT bolus administration. In the initial pilot study in which adult monkeys received escalating lumbar IT bolus injections of 1, 3, and 10 mg, the HD, which used a 10 mg/mL concentration, produced transient clinical signs, including hindlimb ataxia, hindlimb neurological deficits, and decreased heart rate, body temperature, and respiration rate. In the single dose adult monkey study with ISIS 396443 (396443-AS013 above), transient hindlimb weakness and deficits in spinal reflexes were observed in a single monkey at an IT dose of 7 mg (7 mg/mL). In the second pilot study in juvenile monkeys, lumbar bolus IT injections of 0.3 and 1.5 mg appeared to be well-tolerated after 4 weeks of dosing. Based on these data, the HD of 3 mg (4 mg/mL) in the current juvenile monkey study was expected to be on the threshold for acute toxicity.

The HD of 3 mg administered weekly throughout the study was also intended to produce high spinal cord concentrations. This dose was said to be within the dose/exposure range associated with signs of drug-related histologic changes in the cerebellum with a related oligonucleotide (SOD-1 inhibitor for ALS indication). Due to the chemical modification of ISIS 396443, which included 2'-MOE on each of the nucleotides, the tissue residence time was anticipated to be longer than observed in the previous program and thus result in higher tissue concentrations at comparable doses. Dosing at the LD and MD (weekly for first 4 weeks followed by every other week) was intended to more closely reflect the anticipated clinical dosing regimen.

The LD of 0.3 mg produced spinal cord concentrations in juvenile monkeys that were similar to those associated with the desired pharmacologic activity in SMN2 transgenic mice, i.e., increased exon 7 inclusion and SMN protein (~2 to 8 µg/g tissue concentration).

b. Results

i. Mortality and Clinical Observations

There was no mortality considered drug-related. A single HD male (024) was found dead on Day 8 approx 8 hrs after the second dose, but this death was considered unrelated to drug based on the pathology findings. The cause of death was determined to be hemorrhagic meningoencephalitis (inflammation of the meninges and brain) and was thought likely secondary to a bacterial infection resulting from the placement and/or presence of the catheter.

No drug-related clinical signs were observed.

ii. Body Weight

There were no drug-related effects on BW or food consumption.

iii. Physical and Neurological Examinations

No ISIS 396443-related changes in physical examination parameters (heart rate, respiration, temperature, auscultation (respiratory and cardiovascular), gait, disposition, abdominal palpation, lymph nodes palpation, and general appearance of the eyes, ears, oral cavity, skin, and nails) were observed.

ISIS 396443-related neurological changes (deficits in spinal reflexes) were seen in HD animals ~4 hr after dosing on Days 1 and 29 (Table IVB.1). Deficits remained in 2 HD females (059 and 060), at the Day 99 evaluation (approximately 4 hours after dosing). Neurological examinations were normal for all other animals at the Day 99 evaluation and for all animals at the Day 183 (recovery) evaluation.

Table IVB.1 Neurological examinations

Males

Time Interval		Day 1 (~4 hours)				Day 29 (~4 hours)			
Group Number		1	2	3	4	1	2	3	4
IT-L Dose		0 mg	0.3 mg	1 mg	3 mg	0 mg	0.3 mg	1 mg	3 mg
General Sensory and Motor Function	Tracking		8/8	6/6	6/6	8/8	8/8	6/6	6/6
	Level of Consciousness		8/8	6/6	6/6	8/8	8/8	6/6	6/6
	Motor Function		8/8	6/6	6/6	8/8	8/8	6/6	6/6
Cerebral Reflexes	Pupillary Reflex	Constriction	Left	8/8	6/6	6/6	8/8	8/8	6/6
			Right	8/8	6/6	6/6	8/8	8/8	6/6
		Dilation	Left	8/8	6/6	6/6	8/8	8/8	6/6
			Right	8/8	6/6	6/6	8/8	8/8	6/6
	Orbicularis oculi Reflex	Cross	Left	8/8	6/6	6/6	8/8	8/8	6/6
			Right	8/8	6/6	6/6	8/8	8/8	6/6
		Reaction	Left	8/8	6/6	6/6	8/8	8/8	6/6
			Right	8/8	6/6	6/6	8/8	8/8	6/6
	Corneal Reflex	Left	8/8	6/6	6/6	8/8	8/8	6/6	6/6
		Right	8/8	6/6	6/6	8/8	8/8	6/6	6/6
Spinal Reflexes	Knee Jerk	Left Hind Limb	8/8	6/6	6/6	8/8	7/8	5/6	5/6
		Right Hind Limb	8/8	6/6	6/6	8/8	8/8	6/6	5/6
	Proprioceptive Reflex	Left Hind Limb	8/8	6/6	6/6	8/8	8/8	6/6	6/6
		Right Hind Limb	8/8	6/6	6/6	8/8	8/8	6/6	6/6
	Cutaneous Reflex	Left Side	8/8	6/6	6/6	7/8	8/8	6/6	6/6
		Right Side	8/8	6/6	6/6	7/8	8/8	6/6	6/6
	Sensory Foot Reflex	Left Hind Limb	8/8	6/6	6/6	7/8	8/8	6/6	6/6
		Right Hind Limb	8/8	6/6	6/6	7/8	8/8	6/6	6/6
	Tail Reflex		8/8	6/6	6/6	7/8	8/8	6/6	4/8

Females

Time Interval		Day 1 (~4 hours)				Day 29 (~4 hours)			
Group Number		1	2	3	4	1	2	3	4
IT-L Dose		0 mg	0.3 mg	1 mg	3 mg	0 mg	0.3 mg	1 mg	3 mg
General Sensory and Motor Function	Tracking		9/9	7/7	7/7	9/9	9/9	7/7	7/7
	Level of Consciousness		9/9	7/7	7/7	9/9	9/9	7/7	7/7
	Motor Function		9/9	7/7	7/7	9/9	9/9	7/7	7/7
Cerebral Reflexes	Pupillary Reflex	Constriction	Left	9/9	7/7	7/7	9/9	9/9	7/7
			Right	9/9	7/7	7/7	9/9	9/9	7/7
		Dilation	Left	9/9	7/7	7/7	9/9	9/9	7/7
			Right	9/9	7/7	7/7	9/9	9/9	7/7
	Orbicularis oculi Reflex	Cross	Left	9/9	7/7	7/7	9/9	9/9	7/7
			Right	9/9	7/7	7/7	9/9	9/9	7/7
		Reaction	Left	9/9	7/7	7/7	9/9	9/9	7/7
			Right	9/9	7/7	7/7	9/9	9/9	7/7
	Corneal Reflex	Left	9/9	7/7	7/7	9/9	9/9	7/7	7/7
		Right	9/9	7/7	7/7	9/9	9/9	7/7	7/7
Spinal Reflexes	Knee Jerk	Left Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
		Right Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
	Proprioceptive Reflex	Left Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
		Right Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
	Cutaneous Reflex	Left Side	9/9	7/7	7/7	8/9	9/9	7/7	7/9
		Right Side	9/9	7/7	7/7	9/9	9/9	7/7	8/9
	Sensory Foot Reflex	Left Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
		Right Hind Limb	9/9	7/7	7/7	9/9	9/9	7/7	7/7
	Tail Reflex		9/9	7/7	7/7	9/9	9/9	7/7	8/9

iv. Clinical Pathology

There were no drug-related changes in hematology, serum and CSF chemistry, or urinalysis parameters and no effect on CSF cell counts.

v. Ophthalmoscopy

There were no drug-related effects.

vi. Cardiovascular

There were no clearly drug-related effects on ECG or blood pressure parameters (measured prior to initiation of dosing and prior to necropsy). Several animals (2 C, 3 LD, 2 MD, and 3

HD) had mild (up to ~20%) prolongation of QT and/or QTc intervals with and without changes in heart rate but due to the lack of a dose-response these changes were not considered drug-related.

vii. TK

Plasma, CSF, and tissue exposure data are shown below (Tables IVB.2-4). The highest concentrations were measured in the lumbar spinal cord, brain cortex, and liver (169, 166, and 115 µg/g, respectively, at the HD) at the terminal necropsy on Day 106 following once-weekly administration. Tissue concentrations at all doses generally increased from Day 36 to Day 106. Clearance was slow following the end of dosing in the HD group. The decrease in concentrations from tissues during the recovery period was somewhat slower than the observed clearance from plasma and CSF. Levels in the hippocampus (measured separately) were relatively high (Table IVB.5), but not higher than in the cortex.

Table IVB.2. Plasma TK parameters after single or multiple IT bolus doses

Dose (mg/dose)	Number of Doses	Profile Day	C_{max} (µg/mL)	T_{max} (hr)	AUC_{0-24} (hr*µg/mL)	$C_{168hr\ postdose}$ (ng/mL)
			Mean ± SD	Median (range)	Mean ± SD	Median (range)
0.3	1	1	0.16 ± 0.11	4 (2 – 8)	1.06 ± 0.42	0 (0 – 0)
	5	29	0.13 ± 0.05	5 (2 – 5)	1.14 ± 0.27	0 (0 – 1.7)
	10	99	0.12 ± 0.06	2 (2 – 5)	1.05 ± 0.40	0 (0 – 0)
1	1	1	0.68 ± 0.50	2 (0.5 – 8)	3.64 ± 1.09	0 (0 – 6.6)
	5	29	0.61 ± 0.25	2 (2 – 5)	3.93 ± 1.38	2.5 (0 – 4.3)
	10	99	0.56 ± 0.14	2 (2 – 2)	3.51 ± 0.67	2.3 (0 – 3.9)
3	1	1	1.97 ± 0.97	2 (2 – 6)	12.0 ± 2.42	2.2 (0 – 4.7)
	5	29	1.92 ± 1.13	2 (2 – 5)	12.7 ± 4.21	7.7 (3.3 – 20.3)
	15	99	1.56 ± 0.26	2 (2 – 2)	11.1 ± 2.26	9.8 (7.1 – 13.1)

^a Animals at 0.3 and 1 mg/dose received 5 weekly doses followed by biweekly dosing to Day 99 (10 total doses). Animals at the 3 mg/dose received weekly dosing throughout the study (15 total doses)

Table IV.3 CSF concentrations pre-dose, 7 days after the 5th (Day 29) and last (Day 99) IT bolus dose, and after recovery (HD)

Dose (mg/dose)	Study Day	CSF Sampling Site			
		Cisterna Magna		IT-Lumbar	
		N	ISIS 396443 ^a (µg/mL)	N	ISIS 396443 ^a (µg/mL)
0.3	1	0	d	3	0 ± 0
	36	6	0.0047 ± 0.0032	5	0.010 ± 0.013
	106	6	0.016 ± 0.0064	4	0.015 ± 0.010
1	1	0	d	2	0 ± 0
	36	6	0.050 ± 0.045	5	0.023 ± 0.027
	106	6	0.022 ± 0.011	3	0.011 ± 0.0085
3	1	0	d	3	0 ± 0
	36	5	0.066 ± 0.049	4	0.057 ± 0.066
	106	7	0.139 ± 0.083	3	0.078 ± 0.073
	183	3	0.031 ± 0.029	2	0.016 (0.0034-0.029) ^b

^a Reported as mean ± SD

^b Reported as median (range)

^c Animals at 0.3 and 1 mg/dose group received 5 weekly doses followed by biweekly dosing to Day 99 (10 total doses). Animals at 3 mg/dose received weekly dosing throughout the study (15 total doses)

^d No predose sample collected via cisterna magna

Table IV.4 Tissue concentrations 7 days after the 5th (Day 29) and last (Day 99) IT bolus dose, and after recovery (HD)

Organ	Study Day	0.3 mg/dose ^a		1 mg/dose ^a		3 mg/dose ^a	
		N	ISIS 396443 (µg/g)	N	ISIS 396443 (µg/g)	N	ISIS 396443 (µg/g)
Cervical	36	6	7.15 ± 4.43	6	16.3 ± 7.83	6	26.7 ± 13
Spinal	106	7	10.6 ± 6.21	6	26.8 ± 6.4	7	82.8 ± 27.8
Cord	183					4	69.1 ± 4.91
Thoracic	36	6	8.43 ± 3.65	6	12.2 ± 4.07	6	23.3 ± 13.2
Spinal	106	7	20.2 ± 4.59	6	38.4 ± 7.06	7	101 ± 46.5
Cord	183					4	107 ± 35.2
Lumbar	36	6	12.9 ± 4.22	6	33.8 ± 7.77	6	62.9 ± 32.4
Spinal	106	7	31.9 ± 15.5	6	78.6 ± 53.3	7	169 ± 54.2
Cord	183					4	242 ± 246
Brain	36	5	3.9 ± 2.3	6	10 ± 6.08	6	17.6 ± 19.8
Cerebellum	106	7	15 ± 10.2	6	23.7 ± 20.1	7	51.2 ± 15.9
	183					4	65.3 ± 21.5
Brain	36	6	9.98 ± 1.56	6	26.3 ± 3.68	6	68 ± 39.3
Cortex	106	7	15.2 ± 3.9	6	35.4 ± 13.7	7	166 ± 45.8
	183					4	122 ± 26.3
Brain Pons	36	6	6.79 ± 5.13	5	15.2 ± 12	5	27.1 ± 14.9
	106	7	7 ± 2.39	6	13 ± 3.67	7	45.9 ± 18.4
	183					4	32.4 ± 18.5
Liver	36	6	1.59 ± 0.414	6	16.4 ± 21.9	6	45.2 ± 23.8
	106	7	1.88 ± 0.377	6	42.5 ± 40.9	7	115 ± 65.1
	183					3 ^b	8.52 ± 6.19

^a Animals at 0.3 and 1 mg/dose received 5 weekly doses followed by biweekly dosing to Day 99 (10 total doses). Animals at the 3 mg/dose received weekly dosing throughout the study (15 total doses)

^b Animal 059 removed from analysis due to abnormal values. Refer to [Section 8.1](#) of the Toxicokinetic report.

Table IVB.5. Concentrations in hippocampus 7 days after the last (Day 99) IT bolus HD

Group	Animal No.	Study Day	Time from Last Dose	Dose Level (mg)	Route	# of Doses	ISIS 396443 (µg/g)	Group Mean ± SD (µg/g)
4	024A	106	7 days	3	IT	15	83.16	148.79 ± 48.98
	025	106	7 days	3	IT	15	119.11	
	026	106	7 days	3	IT	15	153.75	
	055	106	7 days	3	IT	15	183.2	
	056	106	7 days	3	IT	15	182.04	
	057	106	7 days	3	IT	15	102.54	
	058	106	7 days	3	IT	15	217.71	

viii. Complement analysis

No drug-related increases in plasma or CSF complement split product Bb were observed (Figures IV.1-2). While Bb values were elevated in a few individual animals, there were no consistent dose-related patterns that would indicate an effect on complement activation.

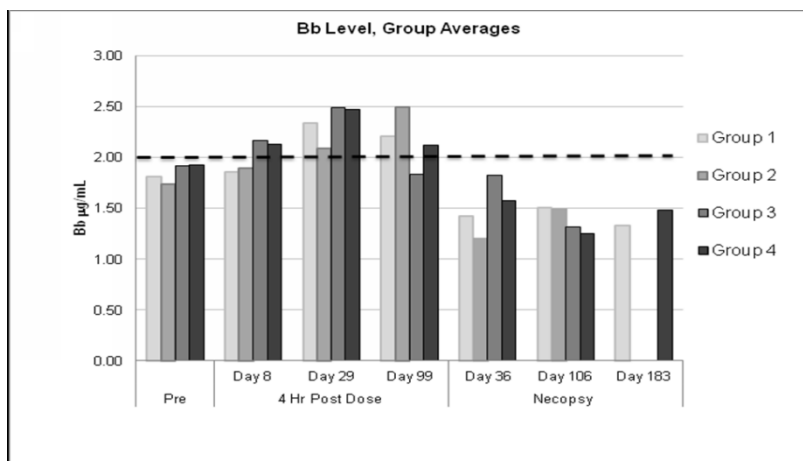


Figure IV.1: Average Bb in plasma specimens from each group at the pre, 4 hr post/dose and necropsy time points. Dotted line represents top of normal range for Bb (2 SD above mean).

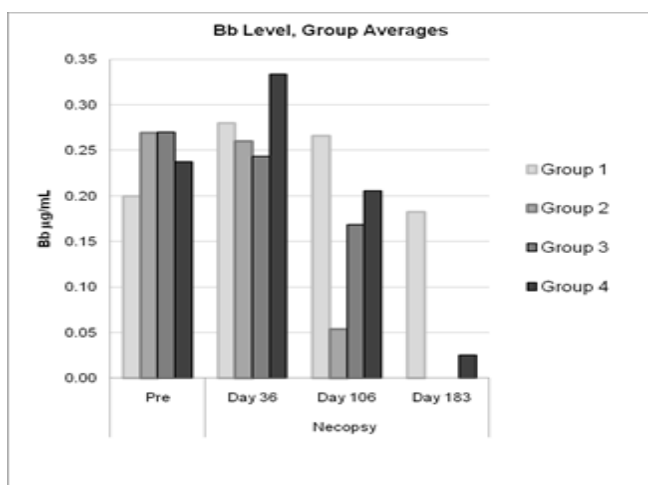


Figure IV.2: Average Bb level in CSF at pre and necropsy time points

ix. Necropsy

There were no gross lesions and no changes in organ weights considered attributable to administration of ISIS 396443.

Histopathology

Day 36

The only findings considered possibly drug-related (relationship to drug considered “equivocal” in report) were “scattered, slight microglial foci and slight perivascular infiltrates” in the brain and spinal cord in a single MD male which were thought by the study pathologist to represent a focal reaction caused by penetration of the oligo into the brain and spinal cord parenchyma.

Day 106

Vacuolation, considered by the pathologist to be neuronal vacuolation based on general morphology and immunohistochemical staining (“PGP 9.5 and GFAP stains indicated that the

vacuoles were most likely within neurons”), was present in the inferior region of the hippocampus in MD and HD animals (Table IVB.6; pathology summary below).

- Brain, Hippocampus, Hippocampal (Inferior) Area, Vacuolation (slight to minimal severity)
 - 1 mg males (1/3 animals) (slight severity) and 3 mg males (3/3) and females (4/4)(slight to minimal severity) (Day 106)
 - 3 mg males (2/2) and females (2/2) (slight to minimal severity) (Day 183)
- Brain, Hippocampus, Hippocampal (Inferior) Area, Neuron, Necrosis
 - 3 mg females (1/4) (slight severity) (Day 106)
 - A single necrotic neuron was observed in the neuronal layer. This was verified with the Fluoro-Jade B stain.
- Brain, Hippocampus, Hippocampal (Inferior) Area, Glial Cell, Necrosis
 - These cells were not Fluoro-Jade B positive and were interpreted to be necrotic glial cells.
 - 3 mg males (1/4) (slight severity) (Day 106)
 - 3 mg recovery females (1/2) (slight severity) (Day 183)

In 2 HD animals (026 male; 058 female) there were “one or two necrotic cells within or adjacent to the zone of vacuolation.” Based on Fluoro-Jade B staining, these necrotic cells were determined to be a neuron in animal 058 and glial cells in animals 026. According to the pathology report, “the biologic significance of these vacuoles, and the apparently associated low level of neuronal or glial cell necrosis, was not apparent. Although extremely slight, the loss of neurons, present only in the 3 mg (high dose) animals, was considered an adverse effect.” The report further stated that “the very specific and limited anatomic area affected by this change was considered to be a very unusual event. What unique feature of these particular neurons (assuming the vacuoles were within neurons) made them susceptible to the formation of these vacuoles was not apparent.” According to the report, the finding of a small number of necrotic cells “indicated that “the vacuolation seldom (if ever) progressed to neuronal necrosis. As a very rare necrotic neuron can be encountered in control animals (especially when brain sections are stained with Fluoro-Jade B as was performed for selected brain sections in this study), this single necrotic neuron may have been unrelated to test article administration.”

A slight accumulation of macrophages near the central canal of the spinal cord was found in a single HD female and slight increases (compared to C) of infiltrates in the meninges overlying the brain was present in MD and HD males and females. The infiltrates were graded slight to mild, and the average severity was only marginally increased compared to C.

Day 183

Vacuolation was observed with a similar incidence and severity in all (2/sex) HD recovery animals. This was thought to be due to the persistence of high concentrations of oligonucleotide. Necrotic glial cells were also reported in one of the HD animals (060 female).

Table IVB.6. Brain histopathology

STUDY ID: 004-018				Microscopic Evaluation Data - TERMINAL DAY 106																		
Group Title		GROUP 1 MALES		Summary Data		GROUP 2 MALES		Summary Data		GROUP 3 MALES		Summary Data		GROUP 4 MALES		Summary Data						
		GROUP 1 MALES		GROUP 1 MALES		GROUP 2 MALES		GROUP 2 MALES		GROUP 3 MALES		GROUP 3 MALES		GROUP 4 MALES		GROUP 4 MALES						
Group Desc		VEHICLE		Group Size	3	ID# 106443 (0.1 MG)		Group Size	3	ID# 106443 (0.1 MG)		Group Size	3	ID# 106443 (0.1 MG)		Group Size	3					
		Animals		INC	AVG SEV	Animals		INC	AVG SEV	Animals		INC	AVG SEV	Animals		INC	AVG SEV					
Test		SS	SS			SS	SS			SS	SS			SS	SS							
Days on Test		106	106			106	106			106	106			106	106							
		Sex	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M					
TISSUE	DIAGNOSES (Shading indicates test article related findings)	0	0	0		0	1	0	0		0	0	0		0	0	2					
		4	5	6		A	3	4		5	9	0		2	2	4	A					
BRAIN, TEMPORAL CORTEX	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	CEREBRAL CORTEX INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	CEREBRAL CORTEX HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, THALAMUS/HYPOTHALAMUS	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	DILTRATES, PERIVASCULAR, EOSINOPHIL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, HIPPOCAMPUS	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	GFAP EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	PGP 9.5 EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	FB EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, VASCULATURE	-	-	-	0	0.00	-	-	-	0	0.00	1	-	-	1	0.33	2	2	1	3	1.67	
-	HIPPOCAMPAL (SUPERIOR) AREA, NEURONS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, GLIAL CELL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	1	-	1	0.50	
-	NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, PRESENT	P	P	P	3	0.00	P	P	P	3	0.00	P	P	P	3	0.00	P	P	P	3	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, PRESENT SLIDE NUMBER	3	3	4	3	3.33	4	4	4	3	4.00	4	4	4	3	4.00	4	4	4	3	4.00	
BRAIN, MIDBRAIN	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	DILTRATES, PERIVASCULAR, MONONUCLEAR CELL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	1	1	0.33	
-	DILTRATES, PERIVASCULAR, EOSINOPHIL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NECROSIAL FOCUS FOC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, MIDBRAIN, SUBSTANTIA NIGRA	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
BRAIN, PARIENTAL CORTEX	H&E EXAMINED	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	NORMAL	Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		Y	Y	Y	3		
-	CEREBRAL CORTEX INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	CEREBRAL CORTEX HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	PERIVASCULAR MACROPHAGES	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
Group Title		GROUP 1 FEMALES		Summary Data		GROUP 2 FEMALES		Summary Data		GROUP 3 FEMALES		Summary Data		GROUP 4 FEMALES		Summary Data						
		GROUP 1 FEMALES		GROUP 1 FEMALES		GROUP 2 FEMALES		GROUP 2 FEMALES		GROUP 3 FEMALES		GROUP 3 FEMALES		GROUP 4 FEMALES		GROUP 4 FEMALES						
Group Desc		VEHICLE		Group Size	4	ID# 106443 (0.1 MG)		Group Size	4	ID# 106443 (0.1 MG)		Group Size	4	ID# 106443 (0.1 MG)		Group Size	4					
		Animals		INC	AVG SEV	Animals		INC	AVG SEV	Animals		INC	AVG SEV	Animals		INC	AVG SEV					
Test		SS	SS			SS	SS			SS	SS			SS	SS							
Days on Test		106	106			106	106			106	106			106	106							
		Sex	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F					
TISSUE	DIAGNOSES (Shading indicates test article related findings)	0	0	0	4		0	0	0	4		0	0	0	4		0	0	0	4		
		2	3	3	2		4	4	4	4		4	4	5	5		5	5	5	5		
-		2	3	4	2		1	2	3	4		5	9	0	1		5	6	7	8		
BRAIN, TEMPORAL CORTEX	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	CEREBRAL CORTEX INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	CEREBRAL CORTEX HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	CEREBRAL CORTEX HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, THALAMUS/HYPOTHALAMUS	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	DILTRATES, PERIVASCULAR, EOSINOPHIL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, HIPPOCAMPUS	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	GFAP EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	PGP 9.5 EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	FB EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, VASCULATURE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	1	1	1	1	1.25	
-	HIPPOCAMPAL (SUPERIOR) AREA, NEURONS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	1	1	0.25
-	NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, GLIAL CELL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, PRESENT	P	P	P	4	0.00	P	P	P	4	0.00	P	P	P	4	0.00	P	P	P	4	0.00	
-	HIPPOCAMPAL (SUPERIOR) AREA, PRESENT SLIDE NUMBER	3	3	4	4	4.00	4	4	4	4	4.00	4	4	4	4	4.00	4	4	4	4	4.25	
BRAIN, MIDBRAIN	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	DILTRATES, PERIVASCULAR, MONONUCLEAR CELL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	DILTRATES, PERIVASCULAR, EOSINOPHIL	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NECROSIAL FOCUS FOC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
BRAIN, MIDBRAIN, SUBSTANTIA NIGRA	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
BRAIN, PARIENTAL CORTEX	H&E EXAMINED	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	NORMAL	Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		Y	Y	Y	4		
-	CEREBRAL CORTEX INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	CEREBRAL CORTEX HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	
-	PERIVASCULAR MACROPHAGES	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	

STUDY ID: 004-018				Microscopic Evaluation Data-RECOVERY DAY 183																	
Group Title		GROUP 1 MALES		Summary Data		GROUP 4 MALES		Summary Data		GROUP 1 FEMALES		Summary Data		GROUP 4 FEMALES		Summary Data					
		GROUP 1 MALES		GROUP 1 MALES		GROUP 4 MALES		GROUP 4 MALES		GROUP 1 FEMALES		GROUP 1 FEMALES		GROUP 4 FEMALES		GROUP 4 FEMALES					
Group Dose		VEHICLE	Group Size=	2	ISIS 396443 (3 MG)	Group Size=	2	VEHICLE	Group Size=	2	ISIS 396443 (3 MG)	Group Size=	2	ISIS 396443 (3 MG)	Group Size=	2					
		Animals:			Animals:			Animals:			Animals:			Animals:							
		SS	SS	-	SS	SS	-	SS	SS	-	SS	SS	-	SS	SS	-					
		Days on Test	183	183	183	183	-	183	183	-	183	183	-	183	183	-					
		Sex	M	M	M	M	-	F	F	-	F	F	-	F	F	-					
TISSUE	DIAGNOSIS (Shading indicates test article related findings)	0	0		0	0		0	0		0	0		0	0						
		0	0		2	2		3	3		3	3		5	6						
		7	8		7	8		6	7		6	7		9	9						
BRAIN, TEMPORAL CORTEX	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	NORMAL	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	CEREBRAL CORTEX, INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
-	CEREBRAL CORTEX, HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
BRAIN, THALAMUS/HYPOTHALAMUS	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	NORMAL	Y	Y	-	2			-	Y	-	1			Y	Y	-	2				
	INFILTRATES, PERIVASCULAR, EOSINOPHILS	-	-	-	0	0.00	-	-	-	0	0.00	1	-	-	1	0.50	-	-	-	0	0.00
BRAIN, HIPPOCAMPUS	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	GFAP EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	PGP9.5 EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	FJB EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	NORMAL	Y	Y	-	2			Y	Y	-	2			-	-	-	0				
	HIPPOCAMPAL (INFERIOR) AREA, VACUOLATION	-	-	-	0	0.00	2	1	-	2	1.50	-	-	-	0	0.00	1	1	-	2	1.00
	HIPPOCAMPAL (INFERIOR) AREA, NEURON NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
	HIPPOCAMPAL (INFERIOR) AREA, GLIAL CELL NECROSIS	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	1	-	1	0.25
	HIPPOCAMPAL (INFERIOR) AREA, PRESENT	-	P	-	1	0.00	P	P	-	2	0.00	P	P	-	2	0.00	P	P	-	2	0.00
	HIPPOCAMPAL (INFERIOR) AREA, PRESENT SLIDE NUMBER	-	3	-	1	1.50	3	3	-	2	3.00	3	3	-	2	3.00	3	3	-	2	3.00
	BRAIN, MIDBRAIN	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2			
		NORMAL	Y	Y	-	2			-	Y	-	1			Y	Y	-	2			
INFILTRATES, PERIVASCULAR, MONONUCLEAR CELL		-	-	-	0	0.00	-	-	-	0	0.00	2	-	-	1	1.00	-	-	-	0	0.00
INFILTRATES, PERIVASCULAR, EOSINOPHILS		-	-	-	0	0.00	-	-	-	0	0.00	1	-	-	1	0.50	-	-	-	0	0.00
-	MICROGLIAL FOCUS/FOCI	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
BRAIN, MIDBRAIN, SUBSTANTIA NIGRA	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	NORMAL	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
BRAIN, PARIETAL CORTEX	H&E EXAMINED	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	NORMAL	Y	Y	-	2			Y	Y	-	2			Y	Y	-	2				
	CEREBRAL CORTEX, INFLAMMATION	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
	NEUTROPHILIC	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
	CEREBRAL CORTEX, HEMORRHAGE	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00
	PERIVASCULAR, MACROPHAGES	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00	-	-	-	0	0.00

c. Conclusions:

IT bolus injections of ISIS 396443 (0, 0.3, 1, or 3 mg/dose) weekly or biweekly to juvenile monkeys for 14-weeks (total doses of 3, 10, or 45 mg) produced acute neurological deficits (lower spinal reflexes) at the HD and brain histopathology at the MD (neuronal vacuolation in hippocampus) and HD (neuronal vacuolation and rare cell necrosis in hippocampus). The neurohistopathology findings were seen at the end of dosing and recovery but not at the 1-month interim sacrifice.

2. A 13-Week Subcutaneous Injection Toxicity Study of ISIS 396443 in Juvenile CD-1 Mice (Study Number: (b) (4)-192015, Isis Study Number: 396443-AS07, conducted by (b) (4), report dated 12/15/15, GLP (except liver metabolite analysis))

a. Methods

ISIS 396443 (Lot RP396443-001) was administered by subcutaneous injection (10 mL/kg) to juvenile mice (Crl:CD1(ICR); 10/sex/grp; 1 pup/sex/litter assigned to each group) once weekly from PND 4 to 25 and once every other week through PND 95 at doses of 0 (vehicle:0.9% sodium chloride for injection), 1, 10, or 50 mg/kg/dose. Pups were observed twice daily for mortality and moribundity and clinical observations, body weights, ophthalmic examinations (1 week prior to termination), clinical pathology (hematology and serum chemistry evaluations performed for 5/sex/group at termination), and macroscopic and microscopic pathology (all tissues from all C and HD, selected tissues from all LD and MD, on PND 97; 10% neutral buffered formalin fixation). Liver tissue was collected from 6/sex/grp at necropsy for TK.

b. Results

i. Mortality and Clinical Observations

There were no drug-related effects on survival or clinical signs. One LD female (no.8772-09) was found dead on PND 84, with malignant lymphoma in multiple tissues considered the cause of death. Lymphoproliferation and multiorgan lymphohistiocytic infiltrates have previously been reported with oligonucleotide administration to mice. However, since these pro-inflammatory effects were not seen microscopically in the current study and there was no dose-response, the sponsor considered the lymphoma an incidental finding unrelated to drug administration.

ii. Body Weight

There were no drug-related effects on BW or food consumption.

iii. Clinical Pathology

There were no clear drug-related changes in hematology parameters. Platelet values were decreased slightly (<10%) in males in a dose-related manner, but SS was not reached and there was no comparable change in females.

Decreases (up to 10%, SS) in total protein and globulin were seen in HD females only. There were no other apparent drug-related changes in serum chemistry.

iv. Ophthalmic examination

There were no drug-related effects.

v. Necropsy

Increased liver and mesenteric lymph node weights at the HD (21 and 32% compared to C, respectively; NS) were attributed to Kupffer cell hypertrophy (HD males) and vacuolated macrophages in the mesenteric lymph node (HD both sexes). Increased (>12%) spleen weights in MD (males) and HD (males and females) groups considered to represent known pro-inflammatory effects of oligonucleotide administration in mice were without microscopic correlates. There were no drug-related macroscopic findings.

Histopathologic findings considered related to tissue accumulation of oligonucleotide and/or their pro-inflammatory effects were observed at the injection sites at ≥ 1 mg/kg/dose and systemically at ≥ 10 mg/kg/dose (Table IVB.7). These included basophilic granules in the kidney, Kupffer cell hypertrophy in liver, and vacuolated macrophages at the injection sites, lymph nodes, and less commonly in multiple other tissues. Additional drug-related findings were noted in the mandibular salivary gland in MD and HD males and sternal bone marrow in HD males.

The kidney findings at the MD and HD were described as small round basophilic granules, which were said to represent oligonucleotide stained with hematoxylin, either within the tubular epithelium or in the lumen of the proximal convoluted tubules of the cortex. In the absence of morphological changes in the tubules (i.e., vacuolation) they were not considered toxicologically significant. However, 1 HD male (no. 8766-05) had multifocal renal tubular degeneration characterized by basophilia, vacuolation, and single cell necrosis; according to the report, it was unclear if this was an incidental finding or associated with the basophilic granules. Vacuolation and degeneration of renal tubular epithelial cells associated with basophilic granulation have been reported previously with oral ASO administration (e.g., with ISIS 301012, another 2'-MOE ASO, in dogs; Henry *et al.*, 2008).

In the liver, Kupffer cells hypertrophy at the MD and HD was due to "abundant basophilic fibrinous to foamy to cytoplasm." This again was thought to represent accumulation of oligonucleotide or possibly secretory substances such as cytokines. However, the distinctive basophilic granules seen in the kidney were not seen in the liver or any other tissue. Basophilic granules in the cytoplasm of Kupffer cells and Kupffer cell hypertrophy have been reported previously with other oligonucleotides (Henry *et al.*, 2008), but according to the pathology report, these granules were not seen with ISIS 396443. No necrosis, degeneration, or inflammation was associated with the enlarged Kupffer cells in this study.

Vacuolated macrophages containing abundant cytoplasm similar to that seen in the Kupffer cells were also observed locally at injection sites and systemically in multiple tissues. These macrophages were noted in greatest number and prominence in the dermis and subcutis at the injection sites (interscapular and dorsal thoracic) and all examined lymph nodes (sinus histiocytes of the mandibular, mesenteric, and inguinal nodes). Vacuolated macrophages were seen at injection sites at all doses and in the lymph nodes in the MD and HD groups. Fewer numbers of vacuolated macrophages were seen in the GI tract, male and female reproductive tracts, joint, salivary and lacrimal glands, urinary bladder, and skin distant from the injection sites (inguinal and associated with the rectum and vagina). While it was thought likely that the actual number of individual affected tissues was larger than was observed and reported, the apparent absence of vacuolated macrophages in muscle (skeletal or cardiac), bone marrow, pulmonary, and CNS tissues was consistent with a list of tissues/systems previously reported to be unaffected by systemic ASO administration. The granular and/or microvacuolated cytoplasm of macrophages have been related to cellular activation and cytokine production resulting from the pro-inflammatory properties of the ASO in some studies (Frazier, 2015); however, no necrosis, degeneration, or inflammation was associated with the enlarged and vacuolated macrophages in this study.

An increase in myeloid cells (myeloid predominance) relative to other cell types was seen in the sternal bone marrow of HD males. This was thought to possibly represent a pro-inflammatory effect of oligonucleotide administration. However, since there were no hematological alterations or similar effects in femoral bone marrow it was unclear whether this was a true drug effect.

An increased incidence of "minimal decreased secretory granules within the granular ducts" was seen in the mandibular salivary glands of MD and HD males. This change was said to be a common finding in anorectic or moribund animals and was noted in the found dead LD

female. However, there was no adverse body weight effect in treated males (BW increased). Decreased androgen levels were considered another possible mechanism in the study report, but there were no apparent changes in the male reproductive organs. This finding was considered “a nonadverse effect of test article administration of uncertain but likely limited toxicological importance” in the MD and HD groups.

Table IVB.7 Selected histopathologic findings, PND 97 scheduled necropsy

Dose (mg/kg):	Males				Females			
	0	1	10	50	0	1	10	50
Liver ^a	10	10	10	10	10	10	10	10
Hypertrophy, Kupffer cell	0	0	1	10	0	0	1	10
Minimal	-	-	1	8	-	-	1	9
Mild	-	-	0	2	-	-	0	1
Kidney ^a	10	10	10	10	10	10	10	10
Basophilic granules	0	0	5	10	0	0	9	10
Minimal	-	-	5	8	-	-	9	6
Mild	-	-	0	2	-	-	0	4
Lymph node, mandibular ^a	10	10	10	10	10	10	10	10
Macrophages, vacuolated	0	0	2	9	0	0	5	9
Minimal	-	-	2	5	-	-	5	6
Mild	-	-	0	4	-	-	0	3
Lymph node, mesenteric ^a	10	10	10	10	10	10	10	10
Macrophages, vacuolated	0	0	1	10	0	0	3	9
Minimal	-	-	1	7	-	-	3	8
Mild	-	-	0	3	-	-	0	1
Lymph node, inguinal ^a	10	10	8	10	10	7	10	10
Macrophages, vacuolated	0	0	3	9	0	0	6	10
Minimal	-	-	3	8	-	-	6	10
Mild	-	-	0	1	-	-	0	0
Injection site, interscapular^a	10	10	10	10	10	10	10	10
Macrophages, vacuolated	0	1	10	10	0	7	10	10
Minimal	-	1	10	5	-	7	7	1
Mild	-	0	0	5	-	0	3	6
Moderate	-	0	0	0	-	0	0	3
Marrow, sternum ^a	10	10	10	10	10	10	10	10
Myeloid predominance (present)	0	1	1	3	0	0	1	1
Salivary gland, mandibular^a	10	10	10	10	10	1^b	NA	10
Decreased secretion, granular duct	0	1	4	3	0	1	-	0
Minimal	-	1	1	1	-	0	-	-
Mild	-	0	3	2	-	0	-	-
Severe	-	0	0	0	-	1 ^b	-	-

^a - Number of tissues examined from each group. The N=10 in the 1 mg/kg/dose group females included the animal found dead on PND 84.

^b - FD on PND 84.

NA - Not applicable.

ANIMAL NO. 8766-05	GROUP 4: 50 MG/KG/DOSE	MALE	SCHEDULED EUTH 07/02/15	DATE OF DEATH: 07/02/15	GRADE
ORGAN WEIGHT	ABS. (G)	REL.	GENERAL COMMENT	GROSS: MASS NOT CONFIRMED	P
LIVER	2.4131	5.201		MASS #1	
KIDNEYS	0.8198	1.767	^==>	GROSS UNCONFIRMED	
BRAIN	0.5046	1.087	MARROW, STERN	MICRO: MYELOID PREDOMINANCE	P
HEART	0.2221	0.479	JOINT- TIBIOFEM	MICRO: MACROPHAGES, VACUOLATED	2
SPLEEN	0.1696	0.366		JOINT CAPSULE, SUBJACENT TO SYNOVIAL EPITHELIUM.	
LUNGS	0.3111	0.670	CECUM	MICRO: MACROPHAGES, VACUOLATED	1
SAL. GLAND MAND	0.3199	0.689	EPIDIDYIMIDES	MICRO: MACROPHAGES, VACUOLATED	1
PROSTATE	0.0250	0.054		BILATERAL.	
SV/CG/ACC FLUID	0.4193	0.904	KIDNEYS	MICRO: DEGENERATION, TUBULAR	1
TESTES	0.2918	0.629		TUBULAR BASOPHILIA, VACUOLATION AND SINGLE CELL NECROSIS,	
THYMUS	0.0250	0.054		MULTIFOCAL, CORTEX, UNILATERAL. ONE OF PAIR PRESENT.	
LN, MESENTERIC	0.0192	0.041		BASOPHILIC GRANULES	1
ADRENAL GLANDS	0.0057	0.012		FINE TO COARSE GRANULES, INTRACYTOPLASMIC AND LUMINAL, CORTEX,	
PITUITARY	0.0019	0.004		PROXIMAL CONVOLUTED TUBULES, UNILATERAL. ONE OF PAIR PRESENT.	

vi. Toxicokinetics

Dose-dependent exposure to ISIS 396443 was observed in mouse liver tissue (Table IVB.8). In addition to the parent compound, putative oligonucleotide metabolites consistent with nuclease-mediated cleavage were evident in liver samples which were collected from Group

4 on Day 97 (Table IVB.9). ISIS 396443 was the most abundant species of oligonucleotide and accounted for >74% of the total present (based on % total peak area). The metabolite of N-1 from the 3'-end was the most abundant metabolite in all evaluated tissues (up to ~23% of the total oligonucleotide present based on % total peak area); the relative abundance of each individual remaining metabolite was <1%. These data were considered consistent with nuclease metabolism.

Table IVB.8 Liver exposure to ISIS 396443 on PND 97

Group	Dose Level	Postnatal Day	Gender	No. of Doses	N	Liver ISIS 396443 Concentration (µg/g)
2	1 mg/kg	97	Female	9	6	BLQ
			Male	9	6	BLQ
			Combined	9	12	BLQ
3	10 mg/kg	97	Female	9	6	44.0 ± 10.4
			Male	9	6	45.2 ± 4.85
			Combined	9	12	44.6 ± 7.77
4	50 mg/kg	97	Female	9	6	220 ± 13.1
			Male	9	6	214 ± 35.8
			Combined	9	12	217 ± 25.9

Note: The lower limit of quantitation (LLOQ) for the assay was 10.00 µg/g for liver. All values below the lower limit of quantitation (BLQ) were treated as zero (0) for the statistical summary analysis. Values are presented as mean ± standard deviation

Table IVB.9 Relative abundance (% of total peak area) of ISIS 396443 and putative chain-shortened metabolites in liver

Group	Dose Level	Postnatal Day	Gender	Animal No.	No. of Doses	Percent of Total Peak Area (PTA) (%)									
						ISIS 396443 (N)	Shortner Metabolites								
							N-1	N-2	N-3	N-4	N-5	N-6	N-7	N-8	N-9
4	50 mg/kg	97	M	8770-07	9	75.32	22.80	0.6834	0.3561	0.05597	0.04609	0.06490	0.1994	0.05376	0.4199
4	50 mg/kg	97	M	8767-07	9	75.90	22.42	0.6132	0.3425	0.04314	0.04585	0.06759	0.1759	0.03460	0.3597
4	50 mg/kg	97	M	8766-05	9	75.58	23.03	0.4919	0.2663	0.05885	0.04173	0.04128	0.1479	0.02277	0.3183
4	50 mg/kg	97	M	8773-05	9	76.04	22.39	0.5823	0.2969	0.04871	0.04430	0.06379	0.1804	0.02720	0.3217
4	50 mg/kg	97	M	8769-02	9	75.29	23.12	0.5649	0.2951	0.04848	0.03484	0.08175	0.1630	0.05081	0.3543
4	50 mg/kg	97	M	8778-08	9	75.08	22.99	0.6328	0.3837	0.06203	0.04700	0.07644	0.2255	0.06577	0.4471
4	50 mg/kg	97	F	8770-08	9	76.20	21.83	0.7149	0.4174	0.04078	0.05868	0.08553	0.1545	0.05506	0.4425
4	50 mg/kg	97	F	8768-09	9	76.43	21.99	0.6052	0.3364	0.03155	0.04936	0.06768	0.1196	0.04536	0.3267
4	50 mg/kg	97	F	8767-08	9	74.08	23.58	0.8517	0.4547	0.05345	0.07268	0.09329	0.2620	0.04404	0.5003
4	50 mg/kg	97	F	8776-09	9	76.17	22.13	0.6224	0.3286	0.05920	0.06073	0.01393	0.2060	0.01862	0.3814
4	50 mg/kg	97	F	8769-13	9	75.18	23.10	0.6642	0.3353	0.04787	0.05731	0.06503	0.1637	0.04827	0.3391
4	50 mg/kg	97	F	8766-06	9	75.90	22.45	0.5961	0.3026	0.04968	0.04426	0.05712	0.1907	0.02826	0.3874

Note: The metabolites were identified based on the comparison of their relative retention times with the retention times of putative metabolites [i.e., ISIS 522402 [N-1 from 3'-end (17-mer Metabolite of ISIS 396443)] established during the validation of an HPLC-UV method for the determination of ISIS 396443 in mouse liver (CMIC Study No. 1050-041; Isis MV Study No. 396443-MV11).

c. Conclusions

Subcutaneous administration (once weekly PND 4 though PND 25 and once every other week thereafter until PND 95) of ISIS 396443 to juvenile CD-1 mice at doses of 0, 1, 10, and 50 mg/kg/dose resulted in basophilic granules in the kidney (MD and HD, both sexes), Kupffer cell hypertrophy (HD males), and vacuolated macrophages at the injection sites, lymph nodes, and multiple other tissues (both sexes at all doses). The basophilic granulation was associated with a low incidence (single HD male) of renal tubular degeneration characterized by vacuolation and cell necrosis. Decreased secretory granules within the granular ducts of the mandibular salivary gland (MD and HD) and an increase in myeloid cells in the sternal bone marrow (HD) were also observed in treated males. Increased spleen weights (MD males and HD males and females) without microscopic correlates were considered to represent known pro-inflammatory effects of oligonucleotide administration in mice. The spectrum and severity of effects observed in juvenile animals were consistent with those documented in adult animals.

C. CHRONIC TOXICITY

1. ISIS 396443: A one year (53 weeks) repeat dose toxicity study in juvenile cynomolgus monkeys with a 26-week recovery phase (Study Number: 8264391, ISIS Study Number: 396443-AS06, conducted by (b) (4), report dated 3/31/15, GLP)

a. Methods

Young cynomolgus monkeys (5/sex/group main, 2/sex/group C and HD recovery; approximately 10-11 months of age, 1-2 kg at start of dosing) were administered ISIS 396443 (Lot No. CA396443 001) at doses of 0 (artificial CSF vehicle), 0.3, 1, or 4 mg/dose (0.4, 1.33, or 5.33 mg/mL, 0.75 mL/dose) by IT lumbar bolus injection (slow bolus injection over 1 minute via lumbar puncture at L3-L6 level, anesthesia induced with ketamine and medetomidine) once weekly for the first 5 doses and once every 6 weeks thereafter for a total of 53 weeks (total doses: 0, 3.9, 13, or 52 mg).

Group Number	Group Description	Dose Level (mg/dose)	Dose Volume (mL)	Dose Concentration (mg/mL)	Animals/Group		Necropsy after ...	
					Male	Female	53 weeks (Day 372)	79 weeks (Day 554)
1	Control ¹	0	0.75	0.00	7	7	5 M / 5 F	2 M / 2 F
2	Low	0.3	0.75	0.40	5	5	5 M / 5 F	
3	Intermediate	1.0	0.75	1.33	5	5	5 M / 5 F	
4	High	4.0	0.75	5.33	7	7	5 M / 5 F	2 M / 2 F

Doses administered by IT bolus injection once weekly for 5 doses (on Days 1, 8, 15, 22 and 29) and then once every 6 weeks thereafter (on Days 71, 113, 155, 197, 239, 281, 323 and 365)

¹ Group 1 received vehicle (aCSF)

For TK analysis, blood was collected from all animals pre-dose and 1, 2, 4, 8, 12, 24, 48, and 168 hours postdose on Day 1 and pre-dose and at 0.5, 1, 2, 4, 6, 8, 12, 24, 48, and 168 hours postdose on Day 365. In addition, blood samples were collected pre-dose on all dosing days and during the recovery period (Days 379, 393, 407, 421, 435, 449, 479, 518, and 554). CSF for TK analysis was collected pre-dose on all dosing days, 168 hours after the last dose on Day 365 and during the recovery period (Days 379, 407, 449, and 554). Tissue (brain, spinal cord, liver kidney) was collected for TK analysis at scheduled necropsies.

Blood for anti-drug antibody analysis was collected prior to start of dosing and pre-dose on Days 29, 113, 197, and 323 and at the end of the recovery period. CSF was collected for CSF chemistry, total cell counts, and complement analysis pre-dosing and at the end of the dosing and recovery periods.

Endpoints consisted of clinical observations (twice daily; pre-dose, 4, and 8 hrs after dosing on dosing days), body weight and food consumption, and physical (abdominal palpation, eye, ear and nose examination, body temperature, heart and lung auscultation) and neurologic (general sensorimotor, cerebral and spinal reflexes) examinations (in non-sedated animals during the pre-dose, dosing, and recovery phases), neurobehavioral observations (FOB pre-dose, Weeks 29 and 53 of dosing, and last week of recovery), learning and memory testing (conducted pre-dose, during and at the end of the dosing period, and at the end of the recovery phase, using the Wisconsin General Testing Apparatus), ophthalmic examinations, cardiovascular (BP and ECG pre-dose, Week 53, and after recovery), determination of bone mineral density and content and femur length (in anesthetized animals pre-dosing, during Week 53, and at the end of recovery), clinical pathology evaluations including urine analysis, and T cell-dependent antibody response (TDAR).

Complete necropsies were performed on all animals (main and recovery groups), with macroscopic examination of all animals and microscopic examination of a full panel of tissues from C and HD animals (fixation in 10% neutral buffered formalin). Brain sections (8 left hemisphere sections) from all animals were paraffin embedded, H&E stained, and examined microscopically at Tox Path Specialists. In addition, two sections of the brain (at the level of the hippocampus and cerebellum) were stained with Fluoro-jade B for evaluation of neuronal degeneration.

Dose Justification: The doses and dosing regimen for this chronic study were based on the results of the 14-week study in juvenile monkeys (0.3, 1, or 3 mg administered IT weekly for the first 4 weeks then every other week at the LD and MD or weekly at HD during weeks 5-14; Study No. 396443-AS03) in which acute neurological changes (absence of the posterior spinal reflexes associated with slow-bolus [3 minutes] IT administration of the 3 mg HD) were considered dose-limiting. These changes were thought to be related to a local effect on spinal cord function specifically at the site of the dosing catheter placement at the thoraco-lumbar junction. However, the sponsor also stated that because the incidence of this neurological sign was low and did not progress in incidence or severity over time, the HD was considered well-tolerated. According to the sponsor, the somewhat higher HD selected for the chronic toxicology study is “intended to ensure sufficiently high tissue concentrations to identify target organ toxicities.”

The every-other week or weekly dosing regimen used in the 14-week study was intended to exaggerate exposure and identify potential toxicities produced by exposure that exceed the expected clinical exposure. The tissue half-life (elimination half-lives estimated to range from 150 to 170 days in monkeys) was thought to support infrequent dose administration. At the HD, weekly administration for 14 weeks produced lumbar spinal cord and brain cortex concentrations that were approximately 30-fold higher than the targeted therapeutic exposure levels in humans (tissue concentration of 2 - 8 µg/g). At the LD, administration every other week produced exposure that was approximately 3- to 5-fold higher than the targeted therapeutic exposure levels. The slow clearance from tissues was illustrated by concentrations of ASO in tissue that increased from 4- to 14-week time points, and concentrations after the 13-week recovery period that were similar or only slightly lower than at the end of the dosing period. The chronic monkey study used a modified dosing regimen, with weekly administration for the first 4 weeks, followed by once every 6 weeks administration during the maintenance period, which, according to the sponsor, would “more appropriately reflect the intended clinical dose regimen.” The dosing regimen being used in patients is currently no more than 3 doses in the first 3 months of treatment, then every 3 to 6 months thereafter. Therefore, the sponsor considered total administered dose to be the most useful parameter for making exposure comparisons.

In the 14-week study, IT administration of 3 mg/week for 14 weeks (45 mg total dose) was associated with hippocampal neuronal vacuolation in all examined animals at the end of dosing and recovery. According to the sponsor, the location of this lesion was thought to reflect the distribution and cellular uptake of oligonucleotide. There were also observations of single necrotic neurons in a single HD animal at the terminal necropsy, and necrotic glial cells (1 or 2 cells per animal) in 2 additional HD animals, 1 at terminal necropsy and 1 at recovery. Hippocampal vacuolation without evidence of necrosis was also seen in 1/7 MD monkeys (1 mg every other week or 10 mg total dose). These histologic changes were not present at any dose at the 4 week interim sacrifice and were not present in the LD group (0.3 mg every other week or 3 mg total dose) after 14 weeks of dosing. Thus, the finding correlated with dose and duration of administration. While not associated with any apparent long-term neurological changes (only general sensory and motor functions were evaluated in 14-week study), this effect was considered to reflect the high tissue concentration of oligonucleotide and was considered dose limiting for chronic administration.

The doses and regimen in the chronic study was expected to maintain tissue concentrations throughout the 1-year study duration. This regimen resulted in the administration of total doses of

3.9, 13, and 52 mg over 53 weeks, which is slightly higher than the total doses delivered over 14 weeks in the previous study in juvenile monkey. The HD was again said to be limited by the IT bolus associated clinical signs. According to the sponsor, the dose regimens employed were intended to produce exposure levels that range from therapeutically relevant tissue concentrations at the 0.3 mg dose to those associated with target organ effects at 4 mg and exceed the exposure observed at 1 mg dose level in 14-week study. The regimen was also said to be determined by tissue PK and the likely clinical dose interval of every 3 - 6 months in patients.

The sponsor gave the following justification for the starting age of the monkeys for this study: "it was recommended that the animals should be as young as possible at initiation of dosing, since studies in juvenile animals via the intrathecal route have been requested by regulatory authorities to support pediatric clinical trials. However, based on the published article by Morford et al. (Morford LL et al. Birth Defect Res Pt B 92:359-380, 2011), it appeared that initiation of dosing up to 12 months of age should be acceptable."

b. Results

i. Mortality and Clinical Observations

There were no unscheduled deaths during the study. While, the study report stated that there were no clinical observations clearly attributable to drug, the observation of limited or no use of the right leg in 1 HD male (17348M) beginning 4 hours post-dose on Days 1, 8, and 15, with staggered movements observed for this animal on Days 8 and 15, which was associated with the neurological findings in this animal (see vi. below), fits the clinical picture of acute neurological changes seen at the HD in the 14-week study. According to the report, "the relationship of this change to treatment is unclear because it was limited to a single monkey and was fully reversible despite continued treatment with ISIS 396443."

ii. Body Weight

Increased BW gain was seen in the MD and HD females (both SS; Table IVC.1) and BW was increased (SS) in HD females beginning on Day 282 through the end of the dosing period. No BW effects were seen males.

Table IVC.1.

Summary of Body Weight Change				
Test Item	(dosage)	1	2	3
ISIS 396443	mg/dose	0	0.3	1.0
		4.0		
Group/ Sex	Phase Day	Data Presented in "kg"		Interval X through X
		PRED 52 -	RECO	
		DSNG 372	2 - 183	
1/F	Mean	0.7	0.4	
	SD	0.14	0.07	
	N	7	2	
2/F	Mean	0.8	-	
	SD	0.13	-	
	N	5	-	
3/F	Mean	0.9*	-	
	SD	0.19	-	
	N	5	-	
4/F	Mean	0.9*	0.4	
	SD	0.17	0.35	
	N	7	2	
	Statistics	A	X7	

* P<=0.05
 ** P<=0.01
 *** P<=0.001
 A = ANOVA and Dunnett's
 X7 = Not analyzed (mean of actual group sizes too small)

iii. Ophthalmoscopy

There were no drug-related effects.

iv. Cardiovascular

There were no clearly drug-related effects on ECG or blood pressure parameters (measured 24 hr post dose). A small but SS increase in QT interval in LD females was considered spurious due to lack of dose-response.

v. Bone Parameters and Femur length

There were no consistent group differences in bone area, bone mineral content (BMC), bone mineral density (BMD), or femur length.

Table IVC.2

Table IV.C.2

		Summary of Bone Measurement Data (Dexa)				
Test Item		(dosage)	1	2	3	4
ISIS 396443		mg/dose	0	0.3	1.0	4.0
		BMD (g/cm ²) - L2 - Measured in Lumbar Spine				
Group/ Sex	Phase Day	Predose	Dosing		Recovery	
		369				
1/F	Mean	0.356	0.408		0.442	
	SD	0.031	0.040		-	
	N	7	7		2	
2/F	Mean	0.346	0.398		-	
	SD	0.021	0.038		-	
	N	5	5		-	
3/F	Mean	0.354	0.416		-	
	SD	0.018	0.007		-	
	N	5	5		-	
4/F	Mean	0.375	0.458 *		0.451	
	SD	0.021	0.041		-	
	N	7	7		2	
Statistics		A	A		X	
Standard deviation not calculated for less than three values						
Statistical analysis was performed using SAS release 8.2						
A = ANOVA and Dunnett's						
X = No analysis performed						
* P≤ 0.05						
** P≤ 0.01						
*** P≤ 0.001						

vi. Physical and Neurological Examinations

Drug-related effects on neurological exam parameters were observed at the HD in both sexes. These consisted of reduced or absent patellar, anal, and grip reflexes, which are lower spinal reflexes. The effects were most pronounced after the first dose, but were observed in individual animals on single days throughout the dosing period. Following the first dose, these neurological deficits were observed in 4/7 HD females and 1/7 HD males (Table IVC.3). While 4/5 affected HD animals had recovered by 48 h, the grip reflex of 1 HD male (17348M) was still reduced before dosing on Day 8. Decreased reflex responses continued to be observed in this animal before dosing on Days 29 and 113.

Decreased or absent spinal (patellar tendon, anal, or grip) reflexes were observed at 4 to 8 hr after dosing on Days 29 (17388F, 17377F), 71 (17342M), 197 (17388F), and 365 (17342M). Physical examinations did not reveal any drug-related findings.

Table IVC.3

Changes in Neurological Examination Parameters on Day 1				
Day 1	4 hours	8 hours	24 hours	48 hours
17377F	Decreased patellar tendon and grip reflex right leg Missing patellar tendon and grip reflex left leg	Decreased grip reflex both legs		
17382F	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	
17387F	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	
17388F	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	Decreased patellar tendon and grip reflex both legs	
17348M	Missing patellar tendon and grip reflex right leg	Missing patellar tendon and grip reflex right leg	Decreased patellar tendon and grip reflex right leg	Decreased grip reflex right leg

vii. Neurobehavioral Observations

There were no drug effects on neurobehavioral observations in a modified Irwin FOB conducted prior to the dosing period, during Weeks 29 and 53 of dosing, or during the last week of the recovery period.

viii. Learning and Memory

Possible drug-related differences in performance in a double choice object discrimination operant learning test (using the Wisconsin General Testing Apparatus) were observed, although the data were variable and N's small. The initial learning phase was conducted in the pre-dosing period of the study while reversal learning and memory were tested during the dosing and recovery periods.

All (24) males and 18/24 females passed the pre-dose learning test. Females 17366F, 17376F, 17380F, 17382F, 17383F, and 17387F did not pass. Animals were randomized into treatment groups according to the results of the pre-dose learning test to minimize bias.

At the first dosing period testing (Weeks 25-28), single MD (17349M) and HD (17361M) males and LD (17380F), MD (17374F), and HD (17388F) females failed to pass the learning test. This was attributed to the failure of these animals to habituate to the test apparatus. For those animals that did pass, there were no dose-related or SS effects on number of days required to complete the tests; however, means for MD males and females were increased compared to C (Table IVC.4). There were no drug-related decreases in number of correct responses observed during the memory test.

When testing was conducted during Weeks 49-52 (Figure IVC.1), a single HD female (17388F) failed to pass the learning test. There was an increase (NS) in number of days required to complete the learning tests at the HD in both sexes. In males this was primarily due to 1 animal 17340M that required 18 days. This result was attributed by the sponsor to the animal passing the memory test with 100% accuracy, which was thought to have made it more difficult to learn the new object introduced at this testing period. (5/7 HD males [17339M, 17340M, 17342M, 17355M, and 17361M] and 3/7 HD females [17367F, 17368F, and 17387F] were found to have hippocampal histopathology at necropsy)

During the recovery period, all (2/sex) HD animals passed the learning test, including the female (17388F) that failed to pass at both times during the dosing phase, but the average number of days required was increased in both sexes compared to C. (1/2 HD females [17367F] was found to have hippocampal vacuolation at the recovery sacrifice.) The HD male (17340M) that took the longest to learn at the end of the dosing phase, performed similarly to C (passed in 5 days), but another HD male (17361M) needed 15 days. (Both HD males were found to have hippocampal vacuolation and necrotic cells at the recovery sacrifice.)

Table IVC.4

8264391

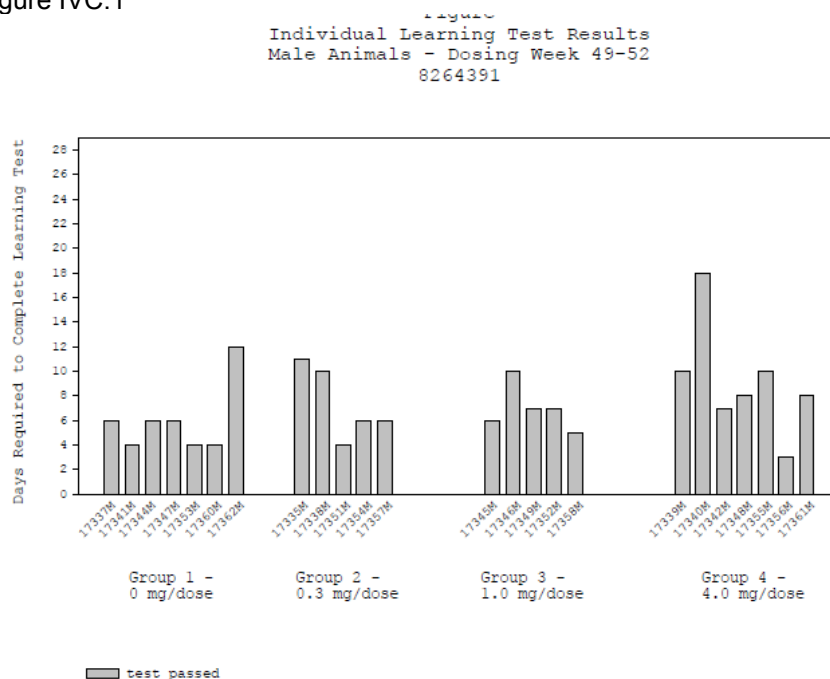
Test Item		(dosage)			
ISIS 396443		mg/dose			
		0	0.3	1.0	4.0

Timepoint		Days $\pm$ SD Required to Complete Learning Test u															
Group	Dosage mg/ dose	Predose				Dosing Week 25-28				Dosing Week 49-52				Recovery Week 22-25			
		Males		Females		Males		Females		Males		Females		Males		Females	
		Days	SD	Days	SD	Days	SD	Days	SD	Days	SD	Days	SD	Days	SD	Days	SD
1	0	9.7 $\pm$ 6.4		12.0 $\pm$ 4.7		10.4 $\pm$ 5.6		12.1 $\pm$ 3.3		6.0 $\pm$ 2.8		5.3 $\pm$ 2.6		5.0 $\pm$ -		3.5 $\pm$ -	
2	0.3	9.8 $\pm$ 3.7		16.8 $\pm$ 6.0		12.2 $\pm$ 6.8		11.0 $\pm$ 1.4		7.4 $\pm$ 3.0		5.0 $\pm$ 1.0		- $\pm$ -		- $\pm$ -	
3	1.0	10.6 $\pm$ 4.8		16.0 $\pm$ 4.4		16.3 $\pm$ 6.8		16.8 $\pm$ 3.2		7.0 $\pm$ 1.9		4.6 $\pm$ 1.5		- $\pm$ -		- $\pm$ -	
4	4.0	7.6 $\pm$ 3.3		11.0 $\pm$ 4.2		7.3 $\pm$ 2.7		12.5 $\pm$ 5.6		9.1 $\pm$ 4.6		7.2 $\pm$ 3.4		10.0 $\pm$ -		6.0 $\pm$ -	
Statistics		AT		AT		AT		AT		AT		AT		X		X	

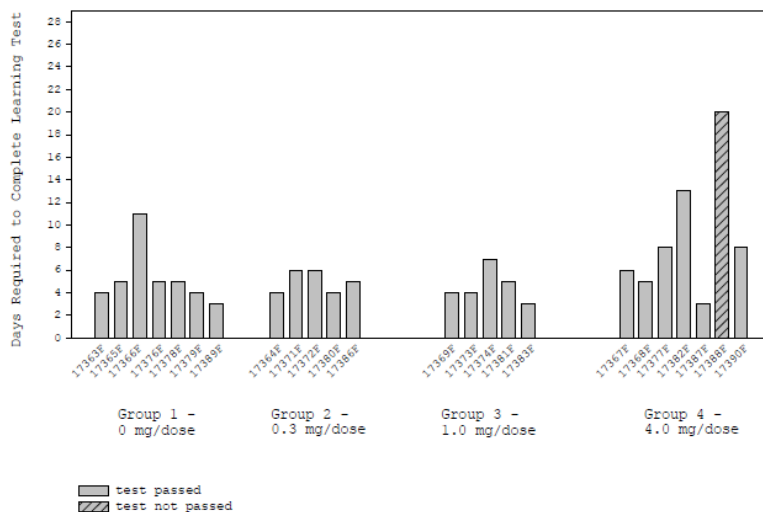
u = excludes animals not passing the learning test

Statistical analysis was performed using SAS release 8.2
A = ANOVA and Dunnett's
T = Rank-transformed data
X = No analysis performed

Figure IVC.1



Individual Learning Test Results
Female Animals - Dosing Week 49-52
8264391



ix. Clinical Pathology

There were no clearly drug-related changes in hematological, clinical chemistry, or urinalysis parameters. Apparent dose-related decreases in APTT times were seen in treated females (Table IVC.5), reaching SS for APTT on day 372 and remaining decreased (NS) compared to C during recovery. The change in APTT is opposite that reported for other oligonucleotides with a phosphorothioate backbone, which have been associated with anticoagulant effects.

CSF total protein (CTP) and albumin (CMIA) were dose-dependently increased during the treatment period in males, reaching SS for CMIA at the HD (Table IVC.6). However, apparent dose-related increases in these parameters were seen in treated females both during the predosing and dosing periods. No drug-related increases in complement split product, Bb, were observed in CSF during the study.

No drug-related changes in the T-cell dependent antibody reaction (TDAR) were observed. Anti-KLH IgM and IgG development to primary and secondary vaccination at the end of the dosing period was similar between C and HD animals, although there was high inter-individual variability; and anti-TT IgM and IgG development to primary and secondary vaccination at the end of the recovery period was comparable for C and HD groups.

Table IVC.5

VC.3

Summary of Hematology							
Test Item		(dosage)					
ISIS 396443		mg/dose					
		0 0.3 1.0 4.0					

Group/ Sex	Phase	Predose		APTT. s			
	Day	41	36	120	204	288	372
1/F	Mean	19.2	19.4	19.8	21.0	20.2	21.6
	SD	2.79	1.54	2.32	2.16	2.09	2.37
	N	7	7	7	7	7	7
2/F	Mean	18.6	18.4	18.7	19.6	19.9	20.1
	SD	0.94	0.76	0.92	0.97	1.17	1.45
	N	5	5	5	5	5	5
3/F	Mean	19.4	19.6	19.3	20.2	19.8	19.8
	SD	1.04	1.88	1.27	1.63	1.19	0.54
	N	5	5	5	5	5	5
4/F	Mean	19.0	18.2	18.5	20.5	19.2	19.2*
	SD	1.48	1.09	0.90	1.49	1.14	1.03
	N	7	7	7	7	7	7
Statistics		A	A	A	A	A	A

* P<=0.05							
** P<=0.01							
*** P<=0.001							
A = ANOVA and Dunnett's							

Table IVC.6

Summary of CSF Chemistry							
Test Item		(dosage)					
ISIS 396443		mg/dose					
		0 0.3 1.0 4.0					
Group/ Sex	Phase	Predose	CTP. mg/dL Dosing	Recovery	Predose	CMIA. mg/dL Dosing	Recovery
	Day	1	372	183	1	372	183
1/M	Mean	22.95	18.02	22.62	9.68	8.39	10.58
	SD	6.382	6.685	7.750	4.100	4.112	5.310
	N	7	7	2	7	7	2
2/M	Mean	19.17	18.33	-	7.16	8.21	-
	SD	4.258	4.570	-	2.087	2.404	-
	N	5	5	-	5	5	-
3/M	Mean	22.34	24.06	-	9.26	12.13	-
	SD	6.231	11.452	-	3.172	6.577	-
	N	5	5	-	5	5	-
4/M	Mean	22.35	32.14	19.50	9.34	16.62*	8.69
	SD	8.440	19.031	1.075	4.831	10.579	0.106
	N	7	7	2	7	7	2
Statistics		A	A	X1	A	AT	X1
1/F	Mean	14.86	14.17	15.73	5.67	6.28	6.59
	SD	2.660	4.920	0.983	1.131	1.638	0.940
	N	7	7	2	7	7	2
2/F	Mean	14.19	15.77	-	4.97	7.45	-
	SD	2.782	6.152	-	1.123	3.448	-
	N	5	5	-	5	5	-
3/F	Mean	17.36	24.42	-	7.11	13.06*	-
	SD	4.141	12.551	-	1.966	7.682	-
	N	5	5	-	5	5	-
4/F	Mean	30.75*	26.56*	26.82	13.68*	13.79**	12.38
	SD	22.415	5.241	4.412	11.503	3.202	2.454
	N	7	7	2	7	7	2
Statistics		AT	A	X1	AT	A	X1

* P<=0.05
 ** P<=0.01
 *** P<=0.001
 A = ANOVA and Dunnett's
 T = Rank-transformed data
 X1 = No analysis required

x. Necropsy

Adrenal gland weights were increased in the MD (SS for left only relative to brain wt) and HD (SS for left and total relative to brain wt) females (Table IVC.7). There were no other SS differences in organ weights at the end of the dosing or recovery periods.

Table IVC.7

Table IV-5.7

Summary of Organ Weights and Organ Weight Ratios

		Terminal Sacrifice						
		Test Item	(dosage)					
		ISIS 396443	mg/dose	0	0.3	1.0	4.0	
		Adrenal, Left				Adrenal, Right		
Group/ Sex		Terminal Body Weight (g)	Unadjusted (g)	Body Weight (%)	Brain Weight (%)	Unadjusted (g)	Body Weight (%)	Brain Weight (%)
1/F	Mean	2158	0.200	0.0093	0.2933	0.150	0.0070	0.2210
	SD	179.7	0.0361	0.00180	0.04010	0.0190	0.00061	0.02605
	N	5	5	5	5	5	5	5
2/F	Mean	2219	0.216	0.0097	0.3274	0.178	0.0080	0.2694
	SD	111.3	0.0482	0.00203	0.06300	0.0403	0.00185	0.05872
	N	5	5	5	5	5	5	5
3/F	Mean	2334	0.260	0.0111	0.3782*	0.182	0.0077	0.2587
	SD	191.6	0.0373	0.00125	0.03895	0.0634	0.00238	0.07527
	N	5	5	5	5	5	5	5
4/F	Mean	2446	0.257	0.0107	0.3809*	0.212	0.0088	0.3143
	SD	268.1	0.0318	0.00228	0.05315	0.0322	0.00216	0.05343
	N	5	5	5	5	5	5	5
	Statistics	A	A	A	A	A	A	A
1/F	Mean	2158	0.350	0.0163	0.5140	68.085	3.1624	100.0000
	SD	179.7	0.0476	0.00219	0.05144	6.3933	0.27591	0.00000
	N	5	5	5	5	5	5	5
2/F	Mean	2219	0.395	0.0178	0.5991	65.819	2.9641	100.0000
	SD	111.3	0.0848	0.00373	0.11563	4.9688	0.11849	0.00000
	N	5	5	5	5	5	5	5
3/F	Mean	2334	0.443	0.0188	0.6387	68.831	2.9426	100.0000
	SD	191.6	0.0950	0.00314	0.08960	8.4795	0.18179	0.00000
	N	5	5	5	5	5	5	5
4/F	Mean	2446	0.469	0.0196	0.6964*	67.554	2.7916	100.0000
	SD	268.1	0.0612	0.00439	0.10293	2.9114	0.36775	0.00000
	N	5	5	5	5	5	5	5
	Statistics	A	A	A	A	A	A	X2

* P<=0.05

** P<=0.01

*** P<=0.001

A = ANOVA and Dunnett's

X2 = Not analyzed (too few distinct values)

* P<=0.05

** P<=0.01

*** P<=0.001

A = ANOVA and Dunnett's

X2 = Not analyzed (too few distinct values)

Macroscopic

There were no gross observations considered drug-related at the main study or recovery necropsies. Bilaterally moderate enlarged lumbar lymph nodes were observed in 1 HD male (17348M), 3 LD males (17335M, 17338M and 17354M), and 1 C male (17341M), but this finding was thought (by the sponsor) to be procedure-related (intrathecal puncture) and was not seen at the end of the recovery phase.

Microscopic

Drug-related microscopic changes in the hippocampus were observed at the terminal and recovery sacrifices (Table IVC.8). At the terminal sacrifice, lesions were present in the hippocampus region (close to or partially within the amygdaloid hippocampal area) of 2/5 MD males and 3/5 HD males and 2/5 HD females. The region affected was the most inferior portion of the hippocampus, at or near the amygdaloid hippocampal area. In 1 HD female, a more medial/superior portion of the hippocampus was also vacuolated. One or more of these findings were still present (same region of the hippocampus) in 2/2 recovery sacrifice males and 1/2 recovery sacrifice females. (Affected animals: MD males 17345M and 17358M; HD males 17339M, 17340M, 17342M, 17355M, and 17361M; HD females 17367F, 17368F, and 17387F.)

The most pronounced lesion was described in the pathology report as "slight to mild vacuolation in the hippocampus, characterized by large vacuoles that appeared to be within the cell body (cytoplasm) of neurons in the hippocampus, close to or within the amygdaloid/hippocampal area (the inferior hippocampal region just lateral to and slightly below the optic tract) and in one high dose terminal sacrifice female in the more medial

CA3 hippocampal region... The vacuoles filled the cytoplasm of individual neurons, marginalizing the nucleus.”

In several animals at both sacrifice times (terminal: 2 HD males, 1 HD female; recovery: 1 HD male), “these large vacuoles were associated with less well-defined vacuoles containing a basophilic (stained blue), flocculent, unidentified material.” According to the report, the cellular origin of these less well-defined vacuoles was not apparent. In addition, “very rare necrotic cells and/or cellular debris” were observed in the hippocampus of MD (1/5) and HD (3/5) males at termination and in a (1/2) HD recovery male. Necrotic cells or necrotic cellular debris were not observed in controls. Fluoro-Jade B staining conducted on two slides/four full coronal (left) sections (two sections at the level of the hippocampus; up to two sections at the level of the cerebellum/brain stem) of the brain from each animal did not reveal any necrotic neurons in any of the study animals (terminal or recovery sacrifice), indicating that the necrotic cells were “most likely not neurons,” but the cell type was not determined. GFAP staining indicated no increases in astrocyte reactivity in the hippocampus or adjacent brain regions, which was thought to suggest that the vacuolation (presumably of neurons) “was not associated with any tissue damage sufficient to elicit an astrocyte response.” Similar morphologic changes (vacuolation and rare necrotic cell in the hippocampus region) were noted at the HD in both terminal sacrifice and recovery animals (recovery not evaluated in LD and MD groups).

According to the study pathologist, “the vacuolation and very rare necrotic cells noted in the region of the hippocampus/amygdaloid hippocampal area, especially given there was no detectable astrocytic response, were unlikely to result in any clinical signs or any alteration that would in any way interfere with the animals ability to function normally in its environment.” However, given the apparent neurobehavioral (L & M) changes seen in HD animals, this statement is questionable.

There were no drug-related changes in non-nervous system tissues.

Table IVC.8 Histopathology findings in 1-year monkey study

Tissue, Finding*	Group 1M 0 mg/dose		Group 2M 0.3 mg/dose		Group 3M 1.0 mg/dose		Group 4M 4.0 mg/dose	
	INC	AVG	INC	AVG	INC	AVG	INC	AVG
Brain, Limbic System/Hippocampus, Inferior Hippocampus/Amygdaloid, Hippocampal Area, Vacuolation, Large Vacuoles, Neurons	0/5	0.00	0/5	0.00	2/5	0.40	3/5	1.00
	0/2	0.00	-	-	-	-	2/2	1.50
Brain, Limbic System/Hippocampus, Inferior Hippocampus, Medial CA3 Region, Vacuolation, Large Vacuoles, Neurons	0/5	0.00	0/5	0.00	0/5	0.00	0/5	0.00
	0/2	0.00	-	-	-	-	0/2	0.00
Brain, Limbic System/Hippocampus, Inferior Hippocampus/Amygdaloid, Hippocampal Area, Vacuoles/Flocculent Material	0/5	0.00	0/5	0.00	0/5	0.00	2/5	0.60
	0/2	0.00	-	-	-	-	1/2	0.50
Brain, Limbic System/Hippocampus, Inferior Hippocampus/Amygdaloid, Hippocampal Area, Necrotic Cells/Cellular Debris	0/5	0.00	0/5	0.00	1/5	0.20	3/5	0.60
	0/2	0.00	-	-	-	-	1/2	0.50

Tissue, Finding*	Group 1F 0 mg/dose		Group 2F 0.3 mg/dose		Group 3F 1.0 mg/dose		Group 4F 4.0 mg/dose	
	INC	AVG	INC	AVG	INC	AVG	INC	AVG
Brain, Limbic System/Hippocampus, <i>Inferior Hippocampus/Amygdaloid, Hippocampal Area, Vacuolation, Large Vacuoles, Neurons</i>	0/5	0.00	0/5	0.00	0/5	0.00	2/5	1.00
	0/2	0.00	-	-	-	-	1/2	0.50
Brain, Limbic System/Hippocampus, <i>Inferior Hippocampus, Medial CA3 Region, Vacuolation, Large Vacuoles, Neurons</i>	0/5	0.00	0/5	0.00	0/5	0.00	1/5	0.40
	0/2	0.00	-	-	-	-	0/2	0.00
Brain, Limbic System/Hippocampus, <i>Inferior Hippocampus/Amygdaloid, Hippocampal Area, Vacuoles/Flocculent Material</i>	0/5	0.00	0/5	0.00	0/5	0.00	1/5	0.20
	0/2	0.00	-	-	-	-	0/2	0.00
Brain, Limbic System/Hippocampus, <i>Inferior Hippocampus/Amygdaloid, Hippocampal Area, Necrotic Cells/Cellular Debris</i>	0/5	0.00	0/5	0.00	0/5	0.00	0/5	0.00
	0/2	0.00	-	-	-	-	0/2	0.00

INC=Incidence; AVG=Average Severity; - = INC or AVG not applicable or not calculated

* For each INC and AVG column there are two cells for each findings: the top cells correspond to the terminal sacrifice, and the bottom cells correspond to the recovery sacrifice.

Shading indicates a test article finding in the corresponding dose group.

For the recovery sacrifice, only the Group 1 and Group 4 animals were processed/evaluated.

xi. Toxicokinetics

TK data indicated continuous and dose-dependent plasma, CSF, and tissue exposure (Tables IVC.9-12; Figure IVC.2). CNS tissue concentrations increased dose dependently in CNS tissue on Day 372, with highest concentrations in the lumbar spinal cord, hippocampus, brain cortex, and thoracic spinal cord. Kidney cortex and liver were the only systemic tissues evaluated in this study. Kidney concentrations were generally higher than CNS tissue concentrations, while liver concentrations were generally lower than all other evaluated tissue concentrations.

Following the end of dosing on Day 365, the estimated terminal half-life value in CSF (111 days) was comparable to the CNS tissue half-life values (117 to 195 days), suggesting an equilibrium between post-distribution CSF and CNS tissue concentrations. Kidney and liver concentrations declined more rapidly (estimated half-life 23 to 34.5 days). Four animals were found to have much higher plasma trough concentrations than the group means (95X in MD animal 17345, 20-35X in HD animals 17340, 17348, and 17367) and were considered outliers. However, the higher plasma trough values did not appear to be associated with differences in CSF concentrations or tissue concentrations. There was no clear relationship between individual brain levels and neurobehavioral or neurohistopathological findings.

Table IVC.9 Pre-dose CSF concentrations

Group	Dose Level (mg/dose)	Study Day	Days After Previous Dose	No. of Doses	ISIS 396443 Concentrations (ng/mL)		
					N	Mean $\pm$ SD	Median (range) ^a
2	0.3	8	7	1	10	14.1 $\pm$ 16.9	6.55 (4.05 - 55.5)
		15	7	2	10	9.79 $\pm$ 3.20	8.62 (5.98 - 15.3)
		22	7	3	10	29.8 $\pm$ 27.4	21.8 (10.1 - 101)
		29	7	4	10	33.6 $\pm$ 36.3	20.1 (12.5 - 130)
		71	42	5	10	14.3 $\pm$ 10.4	11.3 (6.19 - 43.2)
		113	42	6	10	10.0 $\pm$ 6.72	7.94 (0 - 24.5)
		155	42	7	10	19.6 $\pm$ 29.1	10.2 (6.90 - 102)
		197	42	8	10	15.0 $\pm$ 12.6	10.5 (6.16 - 43.2)
		239	42	9	10	9.62 $\pm$ 4.97	8.07 (5.20 - 19.5)
		281	42	10	10	15.0 $\pm$ 15.8	7.90 (4.79 - 56.4)
		323	42	11	10	10.4 $\pm$ 9.48	6.38 (4.09 - 34.7)
		365	42	12	9	13.5 $\pm$ 19.4	6.85 (3.14 - 64.7)
		372	7	13	10	12.0 $\pm$ 10.6	9.85 (4.68 - 41.1)
3	1	8	7	1	10	13.4 $\pm$ 6.54	11.8 (6.58 - 29.2)
		15	7	2	10	29.8 $\pm$ 36.1	16.7 (6.83 - 128)
		22	7	3	9	28.4 $\pm$ 14.0	29.8 (8.04 - 42.6)
		29	7	4	9	32.1 $\pm$ 16.0	27.1 (16.9 - 68.4)
		71	42	5	9	18.8 $\pm$ 7.39	18.6 (9.49 - 32.9)
		113	42	6	10	19.4 $\pm$ 9.86	17.1 (8.00 - 37.4)
		155	42	7	10	26.4 $\pm$ 19.7	18.0 (9.11 - 71.1)
		197	42	8	10	20.5 $\pm$ 9.42	16.1 (10.8 - 39.0)
		239	42	9	10	28.6 $\pm$ 27.0	14.7 (9.76 - 88.8)
		281	42	10	10	28.0 $\pm$ 26.2	19.2 (10.8 - 98.4)
		323	42	11	9	21.2 $\pm$ 21.3	14.3 (5.23 - 69.4)
		365	42	12	10	16.4 $\pm$ 10.5	13.2 (8.03 - 42.1)
		372	7	13	10	23.3 $\pm$ 16.7	19.5 (8.78 - 68.2)
4	4	8	7	1	14	18.5 $\pm$ 11.3	18.3 (0 - 40.2)
		15	7	2	14	58.4 $\pm$ 63.3	30.9 (15.8 - 242)
		22	7	3	14	44.3 $\pm$ 20.7	38.7 (14.5 - 80.9)
		29	7	4	14	53.6 $\pm$ 32.5	47.7 (19.2 - 152)
		71	42	5	14	30.8 $\pm$ 10.8	31.0 (13.0 - 56.4)
		113	42	6	14	31.4 $\pm$ 15.7	26.8 (13.4 - 69.2)
		155	42	7	14	40.0 $\pm$ 21.2	33.9 (15.8 - 80.3)
		197	42	8	14	48.8 $\pm$ 39.9	39.6 (14.9 - 168)
		239	42	9	14	31.6 $\pm$ 14.0	25.4 (15.9 - 64.5)
		281	42	10	14	42.1 $\pm$ 27.5	30.3 (16.1 - 107)
		323	42	11	14	37.0 $\pm$ 23.1	34.7 (13.1 - 104)
		365	42	12	14	42.0 $\pm$ 49.9	23.8 (9.28 - 204)
		372	7	13	14	51.8 $\pm$ 37.8	40.0 (16.1 - 162)
		379	14	13	4	51.2 $\pm$ 22.4	51.0 (24.5 - 78.3)
		407	42	13	4	26.4 $\pm$ 10.2	23.7 (17.1 - 40.8)
		449	84	13	4	18.4 $\pm$ 14.4	19.3 (0 - 35.0)
		554	189	13	4	10.3 $\pm$ 8.86	10.9 (0 - 19.6)

^a Values reported as "0" are below the lower limit of quantification (1.5 ng/mL). See Section 7.4.

SD = Standard Deviation

Table IVC.10 Post-dose plasma concentrations

Group	Dose Level (mg/dose)	No. of Doses	Profile Day	Time After Dosing (hr)	ISIS 396443 Concentration (ng/mL)		
					N	Mean $\pm$ SD ^a	Median (range) ^b
2	0.3	0	1	0	10	BLQ	BLQ
		1	1	1	10	93.0 $\pm$ 41.6	79.6 (33.0 - 164)
		1	1	2	10	112 $\pm$ 44.6	110 (43.0 - 181)
		1	1	4	10	92.0 $\pm$ 25.2	93.0 (50.7 - 131)
		1	1	8	10	50.7 $\pm$ 14.9	46.0 (36.6 - 77.1)
		1	1	12	10	25.7 $\pm$ 5.41	24.6 (19.7 - 37.1)
		1	1	24	10	6.31 $\pm$ 1.32	5.99 (4.62 - 9.20)
		1	1	48	10	0.821 $\pm$ 0.871	0.763 (0 - 1.89)
		1	1	168	10	BLQ	BLQ
		12	365	0	10	0.161 $\pm$ 0.510	0 (0 - 1.61)
		13	365	0.5	10	49.3 $\pm$ 47.3	20.5 (13.6 - 142)
		13	365	1	10	72.6 $\pm$ 61.1	57.1 (20.1 - 207)
		13	365	2	10	68.5 $\pm$ 30.1	64.3 (42.5 - 145)
		13	365	4	10	49.6 $\pm$ 17.4	49.5 (25.0 - 79.0)
		13	365	6	10	39.9 $\pm$ 14.2	38.7 (18.2 - 62.2)
		13	365	8	10	30.0 $\pm$ 10.8	30.7 (15.4 - 49.9)
		13	365	12	10	18.4 $\pm$ 5.68	18.8 (8.72 - 30.3)
		13	365	24	10	4.63 $\pm$ 1.35	4.5 (2.55 - 7.27)
		13	365	48	10	0.304 $\pm$ 0.961	0 (0 - 3.04)
		13	365	168	10	0.181 $\pm$ 0.574	0 (0 - 1.81)
3	1	0	1	0	10	BLQ	BLQ
		1	1	1	10	448 $\pm$ 201	419 (204 - 874)
		1	1	2	10	561 $\pm$ 193	539 (311 - 852)
		1	1	4	10	413 $\pm$ 126	422 (204 - 605)
		1	1	8	10	156 $\pm$ 35.9	160 (94.2 - 203)
		1	1	12	10	92.4 $\pm$ 20.8	93.5 (62.1 - 132)
		1	1	24	10	20.9 $\pm$ 6.79	18.3 (12.8 - 32.5)
		1	1	48	10	3.65 $\pm$ 1.37	3.26 (2.40 - 6.51)
		1	1	168	10	0.235 $\pm$ 0.742	0 (0 - 2.35)
		12	365	0	10	2.21 $\pm$ 6.98	0 (0 - 22.1)
		13	365	0.5	10	263 $\pm$ 274	179 (31.4 - 951)
		13	365	1	10	335 $\pm$ 202	322 (59.7 - 625)
		13	365	2	10	306 $\pm$ 167	293 (123 - 665)
		13	365	4	10	250 $\pm$ 188	170 (115 - 753)
		13	365	6	10	183 $\pm$ 109	127 (70.2 - 389)
		13	365	8	10	123 $\pm$ 89.5	100 (43.1 - 296)
		13	365	12	10	73.6 $\pm$ 66.0	57.9 (19.4 - 250)
		13	365	24	10	20.7 $\pm$ 24.0	14.3 (7.27 - 88.2)
		13	365	48	10	7.53 $\pm$ 14.5	2.75 (1.54 - 48.7)
		13	365	168	10	5.96 $\pm$ 18.3	0 (0 - 58.1)

4	4	0	1	0	14	BLQ	BLQ
		1	1	1	14	2219 ± 1676	1536 (282 - 6144)
		1	1	2	14	3409 ± 1968	3128 (1010 - 7476)
		1	1	4	14	2883 ± 1501	2467 (1577 - 7173)
		1	1	8	14	869 ± 455	748 (461 - 2233)
		1	1	12	14	346 ± 92.1	321 (192 - 502)
		1	1	24	14	94.9 ± 20.7	89.6 (69.2 - 132)
		1	1	48	14	17.3 ± 5.56	16.0 (11.0 - 29.8)
		1	1	168	14	5.16 ± 2.20	4.71 (2.05 - 9.59)
		12	365	0	14	1.94 ± 3.73	0 (0 - 12.8)
		13	365	0.5	14	536 ± 375	425 (82.9 - 1287)
		13	365	1	14	1045 ± 659	1016 (346 - 2738)
		13	365	2	14	1420 ± 603	1436 (626 - 2645)
		13	365	4	14	1011 ± 220	1004 (677 - 1379)
		13	365	6	14	639 ± 182	616 (412 - 1107)
		13	365	8	14	356 ± 102	357 (189 - 549)
		13	365	12	14	208 ± 79.8	191 (113 - 393)
		13	365	24	14	62.2 ± 21.7	52.9 (35.7 - 112)
		13	365	48	14	14.7 ± 8.27	12.8 (7.04 - 39.5)
		13	365	168	14	5.66 ± 4.94	3.69 (1.90 - 18.0)
		13	365	336	4	7.94 ± 8.36	6.94 (0 - 17.9)
		13	365	672	4	5.47 ± 6.45	4.67 (0 - 12.5)
		13	365	1008	4	6.53 ± 7.55	6.46 (0 - 13.2)
		13	365	1344	4	4.98 ± 5.77	4.70 (0 - 10.5)
		13	365	1680	4	3.56 ± 4.11	3.46 (0 - 7.32)
		13	365	2016	4	2.60 ± 3.01	2.53 (0 - 5.34)
		13	365	2736	4	2.29 ± 2.67	2.03 (0 - 5.09)
		13	365	3672	2	1.38 ± 1.95	1.38 (0 - 2.76)
		13	365	3864	1	BLQ	BLQ
		13	365	3888	1	1.72	1.72 (1.72 - 1.72)
		13	365	4536	4	0.515 ± 1.03	0 (0 - 2.06)

^a Mean values reported as less than 1.5 ng/mL are below the limit of quantification and should be interpreted with caution. See Section 7.4 for handling of BLQ values.

^b Values reported as "0" are below the lower limit of quantification (1.5 ng/mL). See Section 7.4.

BLQ = Below the lower limit of quantification (1.5 ng/mL)

SD = Standard Deviation

Table IVC.11 Plasma TK parameters

Group	Dose Level (mg)	Profile Day	N	AUC _{0-168hr} (hr*µg/mL)	AUC _{0-24hr} (hr*µg/mL)	CL/F (mL/hr) ^a	C _{max} (µg/mL)	T _{max} (hr)
2	0.3	1	10	1.12 ± 0.219	0.983 ± 0.186	314 ± 54.4	0.122 ± 0.036	2 (1 - 4)
		365	10	0.714 ± 0.270	0.626 ± 0.197	521 ± 158	0.0859 ± 0.0514	2 (1 - 6)
3	1.0	1	10	4.55 ± 1.04	4.02 ± 0.915	263 ± 70.7	0.569 ± 0.2	2 (1 - 4)
		365	10	3.94 ± 3.98	2.79 ± 1.59	417 ± 125	0.454 ± 0.258	3 (0.5 - 6)
4	4.0	1	14	25.5 ± 8.55	22.8 ± 8.18	193 ± 58.3	3.74 ± 1.72	4 (2 - 8)
		365	14	11.7 ± 2.63	9.59 ± 1.90	433 ± 87.2	1.52 ± 0.535	2 (1 - 6)

^a CL/F = Dose Level (mg) / AUC_{0-24hr} (hr*µg/mL) x 1000

Note: values reported as Mean ± Standard Deviation, except T_{max} which is reported as median (range)

Table IVC.12 Tissue concentrations 7 days after last (13th) dose and at end of recovery

Tissue	Day of Collection	ISIS 396443 Concentrations (µg/g)			
		Mean ±SD			
		N	0.3 mg/dose	1.0 mg/dose	4.0 mg/dose
Brain Cerebellum	372	10	4.27 ± 1.39	20.8 ± 16.0	39.6 ± 19.7
	554	4	NSC	NSC	34.5 ± 23.2
Brain Cortex	372	10	4.23 ± 1.89	16.6 ± 5.97	71.3 ± 29.3
	554	4	NSC	NSC	43.8 ± 4.41
Brain Hippocampus	372	10	10.1 ± 2.81	28.9 ± 10.3	88.9 ± 18.0
	554	4	NSC	NSC	46.0 ± 7.75
Brain Pons	372	10	4.14 ± 3.06	8.84 ± 3.97	34.2 ± 15.5
	554	4	NSC	NSC	16.6 ± 4.27
Cervical Spinal Cord	372	10	6.68 ± 2.78	15.9 ± 5.14	35.1 ± 11.0
	554	4	NSC	NSC	18.4 ± 6.05
Thoracic Spinal Cord	372	10	14.0 ± 9.34	23.0 ± 10.6	59.8 ± 11.7
	554	4	NSC	NSC	20.3 ± 5.60
Lumbar Spinal Cord	372	10	21.4 ± 9.28	53.5 ± 33.6	99.0 ± 71.5
	554	4	NSC	NSC	39.7 ± 10.8
Kidney Cortex	372	10	23.7 ± 17.6	42.8 ± 19.0	239 ± 176
	554	4	NSC	NSC	6.17 ± 3.54
Liver	372	10	0.850 ± 0.430	2.79 ± 1.64	21.9 ± 9.16
	554	4	NSC	NSC	0.101 ± 0.0675

SD = Standard Deviation

NSC = no sample collected

Table IVC.13 Terminal Half-Life values in plasma, CSF, and tissues during recovery

Matrix	Half-Life (Days)
Plasma ^a	55.8
CSF	111
Brain Cerebellum	NR ^b
Brain Cortex	NR ^b
Brain Hippocampus ^c	191
Brain Pons ^c	174
Kidney Cortex ^c	34.5
Liver ^c	23.4
Cervical Spinal Cord ^c	195
Thoracic Spinal Cord ^c	117
Lumbar Spinal Cord ^c	138

^a Half-Life estimated using data from Day 372 to 554. Day 526 concentrations was BLQ and therefore excluded from the half-life estimation.^b No value reported due to insufficient decrease in concentration over the evaluated time interval (see Section 7.3.3)^c Half-life determined using a two-point time curve (Day 372 and Day 554 only). Data should be interpreted with caution.

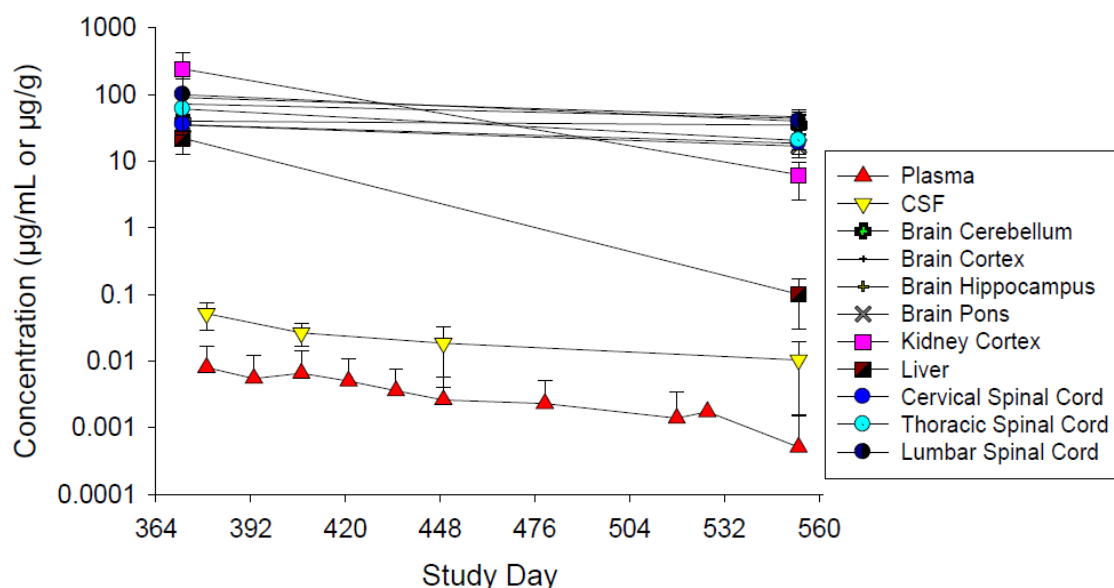
NR = Not reported

Table IVC.14 Comparison of CSF, plasma, and tissue exposure at HD

Matrix	Study Day	Concentration (ng/mL ^a or ng/g)	Ratio to Plasma Conc.	Ratio to CSF Conc.	AUC _{407-554h} (ng ^a day/mL or ng ^a day/g)	Ratio to Plasma AUC
Plasma	379	7.94 ^a	--	--	543	--
CSF	379	51.2 ^a	6.4	--	3534	6.5
Brain Cerebellum	372	39620	4990	774	--	--
Brain Cortex	372	71320	8982	1393	--	--
Brain Hippocampus	372	88940	11202	1737	--	--
Brain Pons	372	34210	4309	668	--	--
Kidney Cortex	372	238580	30048	4660	--	--
Liver	372	21920	2761	428	--	--
Cervical Spinal Cord	372	35100	4421	686	--	--
Thoracic Spinal Cord	372	59790	7530	1168	--	--
Lumbar Spinal Cord	372	98980	12466	1933	--	--
Plasma	554	0.515 ^a	--	--	--	--
CSF	554	10.3 ^a	20.2	--	--	--
Brain Cerebellum	554	34470	67588	3347	--	--
Brain Cortex	554	43800	85882	4252	--	--
Brain Hippocampus	554	46010	90216	4467	--	--
Brain Pons	554	16600	32549	1612	--	--
Kidney Cortex	554	6170	12098	599	--	--
Liver	554	100	196	10	--	--
Cervical Spinal Cord	554	18380	36039	1784	--	--
Thoracic Spinal Cord	554	20330	39863	1974	--	--
Lumbar Spinal Cord	554	39680	77804	3852	--	--

^a CSF and Plasma CSF concentrations from combined outlier and non-outlier data. See Table 1 and Table 3.

Figure IVC.2 CSF, plasma, and tissue concentrations after end of dosing in 1-year monkey study



xii. Anti-drug antibody

Following 1 year of treatment, anti-ISIS 396443 antibodies were detected in 1 MD (17345 on Day 197) and 2 HD (17348 and 17367 on Days 113, 197 and 323) animals. No anti-drug antibodies were detected following the recovery period.

Table IVC.15 Anti-drug antibody analysis

Dose (mg)	Screening		Confirmatory			Titration	
	No. of Animals with at Least 1 Positive Time Point	No. of Positive Study Samples	No. Animals with at Least 1 Confirmed Positive Time Point	No. Samples Confirmed Positive	% Animals with at Least 1 Confirmed Positive Time Point*	% Study Samples Confirmed Positive**	Range of Titer Reported
0.0	0/14	0/74	0/14	0/74	0.0%	0.0%	-
0.3	1/10	3/50	0/10	0/50	0.0%	0.0%	-
1.0	1/10	1/50	1/10	1/50	2.1%	2.0%	8
4.0	2/14	6/74	2/14	6/74	4.2%	8.1%	1-8

Overall false positive rate

(study samples confirmed negative/overall study samples screened) X 100 = 1.2%.

* % animals with at least 1 confirmed positive time point = (number of animals with at least 1 confirmed positive sample/total number of animals screened) X 100, where total number of animals screened = 48.

** % samples confirmed positive = (number of samples confirmed positive/total number of samples screened per Group) X 100.

c. Conclusions

Repeated lumbar IT administration of 0, 0.3, 1, or 4 mg to juvenile (9 to 11 months of age) cynomolgus monkeys for 53 weeks (5 weekly doses followed by 8 doses administered once every 6 weeks; total doses 0, 3.9, 13, or 52 mg) resulted in deficits in lower spinal reflexes (HD), neurobehavioral (learning and memory) deficits (HD), and neurohistopathology (neuronal vacuolation and necrotic cells with accompanying cellular debris in the hippocampal area at MD

and HD). After an extended (26-week) recovery period, neuronal vacuolation in the hippocampus was found of 3/4 HD animals, with a scoring grade similar to that at the terminal necropsy. Brain and spinal cord tissue concentrations (measured at the end of the maintenance dose phase) associated with hippocampal microscopic findings are shown in Table IVC.16. Liver concentrations are also shown for comparison. The NOAEL was 0.3 mg/dose.

Table IVC.16

Dose Group	Hippocampal Microscopic Findings ^a		Mean (and SD) Tissue Concentration (µg/g) ^b		
	Neuronal Vacuolation	Necrotic Cells ^c	Brain (cortex)	Spinal Cord (lumbar)	Liver
0.3 mg/dose (3.9 mg total) ^d	none	none	4.2 (1.9)	21.4 (9.3)	0.9 (0.4)
1 mg/dose (13 mg total) ^d	2 of 10	1 of 10	16.6 (6.0)	53.5 (33.6)	2.8 (1.6)
4 mg/dose (52 mg total) ^d	6 of 10	3 of 10	71.3 (29.2)	106 (78.9)	21.9 (9.2)

^a Incidence using combined male and female results

^b Tissue concentrations measured on Day 372 (main necropsy)

^c Necrotic cells were not neurons

^d Total dose in 53 week period

D. GENOTOXICITY

1. Bacterial Reverse Mutation Assay (Study No. 396443-IS02, (b) (4) AD25FG.503.BTL, conducted by (b) (4), report dated 9/9/11, GLP)

ISIS 396443 (Lot #RP396443-001, PBS solvent) was tested in the Ames assay using tester Salmonella strains TA98, TA100, TA1535 and TA1537, and E. coli strain WP2 uvrA in the presence and absence of rat liver S9 using the plate incorporation method. In the initial cytotoxicity-mutation assay, the concentrations tested were 1.5, 5, 15, 50, 150, 500, 1500, and 5000 µg per plate. No positive mutagenic responses were observed with any of the tester strains in either the presence or absence of S9 activation. Neither precipitate nor appreciable toxicity was observed. Based on the findings of the initial assay, concentrations of 50, 150, 500, 1500, and 5000 µg per plate were tested in the confirmatory mutagenicity assay. No positive mutagenic responses were observed with any of the tester strains in either the presence or absence of S9 activation. Neither precipitate nor appreciable cytotoxicity was observed. Under the conditions of this study, ISIS 396443 was concluded to be negative in the Ames assay.

2. In Vitro Mammalian Chromosome Aberration Test (Study No. 396443-IS03, (b) (4) AD25FG.331.BTL, conducted by (b) (4), report dated 9/6/11, GLP)

ISIS 396443 (Lot #RP396443-001) was tested in the CHO chromosome aberration assay in the absence and presence of rat liver S9. In the preliminary cytotoxicity assay, the maximum concentration tested was 5000 µg/mL. CHO cells were treated for 4 and 20 hours in the non-activated test system and for 4 hours in the S9-activated test system. All cells were harvested 20 hours after treatment initiation. Substantial cytotoxicity (i.e., at least 50% cell growth inhibition, relative to the solvent control) was not observed at any concentration in all three exposure groups. Based on these findings, the concentrations chosen for the chromosome aberration assay ranged from 750 to 5000 µg/mL for all three treatment groups. In the chromosome aberration assay, the cells were incubated with ISIS 396443 for 4 and 20 hours in the non-activated test system and for 4 hours in the S9-activated test system. All cells were harvested 20 hours after treatment initiation. In the absence of precipitation in the treatment medium and at least 50% toxicity, the highest concentration evaluated was 5000 µg/mL. Two additional lower concentrations (2000 and 3500 µg/mL) were included. The percentage of cells with structural or numerical aberrations in the drug-exposed cultures was not significantly increased relative to solvent control at any dose. The positive and solvent controls fulfilled the requirements for a valid test. Based on the findings of this study, ISIS 396443 was concluded to be negative for the induction of structural and numerical chromosome aberrations in CHO cells in non-activated and S9-activated test systems.

3. ISIS 396443: Mammalian Micronucleus Assay in Bone Marrow Erythrocytes Following Subcutaneous Injection in ICR Mice (Study No. 396443-AS10, (b) (4) G115069, conducted by (b) (4), report dated 10/22/15, GLP)

ISIS 396443 (Lot #CA396443-004) was evaluated for its potential to increase the incidence of micronucleated polychromatic erythrocytes (MNPCEs) in bone marrow cells of male and female ICR mice. Sterile 0.9% sodium chloride vehicle was the negative control and cyclophosphamide monohydrate (CPA) was the positive control. The number of MNPCEs was counted in 4000 PCEs (Polychromatic Erythrocytes or Immature Erythrocytes) per animal. There was no statistically significant increase in the frequencies of MNPCEs at any dose tested compared to negative C and no significant difference in the ratio of PCEs to total erythrocytes. Concentrations in liver (from the first 3/sex/time point) at approximately 24 and 48 hours post-dose ranged from 65 to 183 µg/g and 72 to 188 µg/g, respectively. Under the conditions of this study, ISIS 396443 did not induce an increased frequency of micronuclei in bone marrow cells of ICR mice.

E. CARCINOGENICITY

Not conducted.

F. REPRODUCTIVE AND DEVELOPMENTAL TOXICOLOGY

1. A Combined Fertility and Developmental Toxicity Study of ISIS 396443 Administered by Subcutaneous Injection in CD-1 Mice (Report No. 396443-AS08; dated 4/01/15; conducted by (b) (4) GLP)

a. Methods

Male and female CD-1 mice (25/sex/grp + 6/sex/grp TK) were administered ISIS 396443 (lot number: CA396443-001) subcutaneously (10 mL/kg) at doses of 0 (0.9% sterile saline), 3, 10, or 25 mg/kg every other day to deliver weekly doses of 10.5, 35, and 87.5 mg/kg/week. Males were dosed for 4 weeks prior to the mating and during and after the 2-week mating period for a total of 10 weeks. Females were dosed for 2 weeks prior to mating, during the mating period, and until GD15 (i.e., throughout organogenesis).

Group	Test Substance	Dose Level (mg/kg)	Weekly Dose (mg/kg/week) ^c	Number of Animals	
				Males	Females
1	0.9% Sterile Saline	0	0	25 ^a + 6 ^b	25 ^a + 6 ^b
2	ISIS 396443	3	10.5	25 ^a + 6 ^b	25 ^a + 6 ^b
3	ISIS 396443	10	35	25 ^a + 6 ^b	25 ^a + 6 ^b
4	ISIS 396443	25	87.5	25 ^a + 6 ^b	25 ^a + 6 ^b

^a Toxicology animals

^b TK animals

^c Animals were dosed every other day and received 3 or 4 doses per week (except as noted in the Deviations Summary in [Appendix 2](#)). Thus, the weekly dose (mg/kg/week) was equivalent to 3.5 times the single mg/kg dose.

Clinical observations (daily), body weight and food consumption (weekly), and estrous cyclicity were assessed and animals were mated (1:1) starting on Day 15 (females) or Day 29 (males) until confirmation of mating or for 14 days. Terminal necropsy for males was on Day 71 and for pregnant dams on GD18. For males, gross observations were recorded and selected organs were weighed and tissues collected. For pregnant females, typical litter parameters were assessed, external fetal examinations were conducted, and half the fetuses in each litter were assigned to skeletal evaluations (alizarin red staining) and the other half to visceral evaluations (Wilson's [head] and Staples [bodies] methods). Blood samples for hematology or serum chemistry were taken from animals at necropsy, with half the animals in each group assigned for each. Liver and kidney samples were harvested from adult TK males and females and the placenta and fetal kidneys and livers were harvested from up to 10 fetuses per litter. Histopathology was performed on the kidney, liver, and spleen of 4/sex/grp. Histopathology of additional tissues was performed only on C and HD animals. Sperm was collected from the left cauda epididymis of TK males for total sperm count, motility, concentration, and morphology analysis.

b. Results

i. Mortality and Clinical Observations

There were 3 early deaths (1 LD and 1 MD males, 1 MD female), but these were not considered drug-related. There were no clinical observations.

ii. Body Weight and Clinical Pathology

Slight increases in body weight (up to 8.9% compared to C, SS) were seen in males at the MD and HD. There were no BW effects in females. In males, minimal (but SS) decreases in RBC indices (hemoglobin, mean corpuscular hemoglobin, mean corpuscular volume, and red blood cell distribution width) were observed at the HD and more marked increases in white blood cell, neutrophil, and/or monocytes counts were seen at the MD and HD. Albumin and total protein were dose-dependently increased (up to 13% at HD), triglycerides were decreased (34% at the HD), and ALT and AST were increased (up to 2X at HD). There were no effects on clinical pathology parameters in females.

iii. Necropsy

Liver and spleen weights were dose-dependently increased (up to 20 and 66% at HD, respect). Drug-related gross findings included kidney discoloration seen in a HD male (#66) and MD (#231) and HD (#166) females. Increased spleen size was observed in two HD males (#61 and #66). Discolorations of the kidney and increases in spleen size were considered drug-related based on the similarity of these findings to studies of other ASOs.

Histopathological changes observed in kidney, liver, spleen, testes, and epididymides were considered the result of oligonucleotide accumulation in these tissues (Table IVF.1). Findings included minimal to mild basophilic intracytoplasmic granules in cortical tubular cells of the kidneys, hypertrophied Kupffer cells in the liver, minimal to moderate macrophage infiltrates in the spleen, mild to moderate vacuolation of Leydig cells in testes, and minimal to mild macrophage infiltrates in the epididymides, prostate glands and seminal vesicles.

Table IVF.1 Histopathology findings in mouse fertility and developmental toxicity study

Removal Reason: Terminal necropsy	Male				Female			
	Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg	Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Number of Animals:	12	12	12	12	12	12	12	12
ADRENAL GLANDS								
Examined	1	0	0	0	0	0	0	0
Decreased Size; bilateral	1	-	-	-	-	-	-	-
.... moderate	1	-	-	-	-	-	-	-
CERVIX								
Examined	-	-	-	-	12	0	0	12
No Visible Lesions	-	-	-	-	12	-	-	12
EPIDIDYIMIDES								
Examined	12	0	0	12	-	-	-	-
No Visible Lesions	12	-	-	0	-	-	-	-
Infiltration, Macrophages; interstitial, multifocal	0	-	-	12	-	-	-	-
.... minimal	0	-	-	7	-	-	-	-
.... mild	0	-	-	5	-	-	-	-
GALLBLADDER								
Examined	12	0	0	12	12	0	0	12
No Visible Lesions	12	-	-	12	12	-	-	12
KIDNEYS								
Examined	12	12	12	12	12	12	12	12
No Visible Lesions	0	1	1	1	8	3	1	0
tubules; Cast; hyaline, few	0	2	0	0	0	0	0	0
.... minimal	0	2	0	0	0	0	0	0
tubules; Cast; hyaline, focal	1	0	0	0	0	0	0	0
.... minimal	1	0	0	0	0	0	0	0

cortex; Cyst; tubular, single	0	0	0	0	0	0	0	1
.... moderate	0	0	0	0	0	0	0	1
cortex; Cyst; focal	0	0	1	0	0	0	0	0
.... mild	0	0	1	0	0	0	0	0
cortex; Cyst; single	1	0	0	0	0	0	0	0
.... moderate	1	0	0	0	0	0	0	0
tubules; Dilation; multifocal	0	0	0	0	0	0	0	1
.... minimal	0	0	0	0	0	0	0	1
Glomerulosclerosis; focal	0	0	0	1	0	0	0	0
.... minimal	0	0	0	1	0	0	0	0
tubules; cortex; Granularity; basophilic-intracytoplasmic, few	0	0	1	1	0	0	0	0
.... minimal	0	0	1	1	0	0	0	0
tubules; cortex; Granularity; basophilic-intracytoplasmic, multifocal	0	0	3	8	0	6	11	12
.... minimal	0	0	3	5	0	6	8	1
.... mild	0	0	0	3	0	0	3	11
Infiltration, Mononuclear Cells; few	0	0	0	1	0	0	0	0
.... minimal	0	0	0	1	0	0	0	0
Infiltration, Mononuclear Cells; focal	7	4	3	1	3	2	1	2
.... minimal	7	4	3	1	3	2	1	2
Infiltration, Mononuclear Cells; multifocal	4	4	2	4	1	0	0	0
.... minimal	4	4	2	4	1	0	0	0
medulla; Mineralization; few	0	2	0	1	1	2	3	2
.... minimal	0	2	0	1	1	2	3	2
medulla; Mineralization; focal	0	0	1	0	0	0	0	0
.... minimal	0	0	1	0	0	0	0	0
medulla; Mineralization; multifocal	0	1	2	0	0	0	0	0
.... minimal	0	1	2	0	0	0	0	0
tubules; Degeneration/Regeneration; few	0	0	0	4	0	0	0	0
.... minimal	0	0	0	4	0	0	0	0
tubules; Degeneration/Regeneration; focal	3	1	3	3	0	1	0	1
.... minimal	2	1	3	3	0	1	0	1
.... mild	1	0	0	0	0	0	0	0
tubules; Degeneration/Regeneration; multifocal	0	0	0	1	0	0	0	0
.... minimal	0	0	0	1	0	0	0	0
tubules; cortex; Hypertrophy; focal	0	0	0	0	0	1	0	1
.... minimal	0	0	0	0	0	1	0	1
LIVER								
Examined	12	12	12	12	12	12	12	12
No Visible Lesions	3	2	0	0	0	0	0	0
Cyst; biliary, multiple	0	0	0	0	0	0	0	1
.... mild	0	0	0	0	0	0	0	1
Granuloma; few	0	0	4	4	0	0	0	0
.... minimal	0	0	4	4	0	0	0	0
Granuloma; focal	0	0	0	0	0	1	0	1
.... minimal	0	0	0	0	0	0	0	1
.... mild	0	0	0	0	0	1	0	0
Granuloma; multifocal	0	2	0	7	0	0	0	0
.... minimal	0	2	0	1	0	0	0	0
.... mild	0	0	0	1	0	0	0	0
.... moderate	0	0	0	3	0	0	0	0
.... marked	0	0	0	2	0	0	0	0
Hematopoiesis; extramedullary	0	0	0	2	1	0	1	1
.... minimal	0	0	0	2	1	0	1	1
kupffer cell; Hypertrophy; few	0	4	5	0	0	0	0	1
.... minimal	0	4	5	0	0	0	0	1
kupffer cell; Hypertrophy; focal	0	0	0	0	0	0	0	1
.... minimal	0	0	0	0	0	0	0	1
kupffer cell; Hypertrophy; multifocal	0	0	7	12	0	0	0	0
.... minimal	0	0	7	4	0	0	0	0
.... mild	0	0	0	8	0	0	0	0
Infiltration, Mixed Inflammatory Cells; focal	0	0	0	0	0	1	0	0
.... mild	0	0	0	0	0	1	0	0
Infiltration, Mixed Inflammatory Cells; multifocal	0	1	0	2	0	0	0	0
.... minimal	0	1	0	1	0	0	0	0
.... mild	0	0	0	1	0	0	0	0
Infiltration, Mononuclear Cells; interstitial, multifocal	1	0	0	1	0	0	0	0

.... minimal	1	0	0	1	0	0	0	0
Infiltration, Mononuclear Cells; few	0	0	5	0	0	0	0	0
.... minimal	0	0	5	0	0	0	0	0
Infiltration, Mononuclear Cells; multifocal	2	1	2	3	0	0	1	1
.... minimal	2	1	2	3	0	0	1	1
hepatocyte; Necrosis; few	0	1	1	2	0	1	2	2
.... minimal	0	0	1	2	0	1	2	2
.... mild	0	1	0	0	0	0	0	0
hepatocyte; Necrosis; focal	0	0	3	0	3	0	0	1
.... minimal	0	0	2	0	2	0	0	0
.... mild	0	0	1	0	1	0	0	1
hepatocyte; Necrosis; multifocal	0	0	0	0	1	0	0	0
.... mild	0	0	0	0	1	0	0	0
hepatocyte; Vacuolation; multifocal	9	7	8	4	12	12	12	12
.... minimal	8	6	8	4	3	5	4	4
.... mild	1	1	0	0	5	5	8	6
.... moderate	0	0	0	0	4	2	0	2
hepatocyte; Increased Mitosis	0	0	0	0	6	10	9	6
.... minimal	0	0	0	0	4	5	5	2
.... mild	0	0	0	0	1	4	4	4
.... moderate	0	0	0	0	1	1	0	0

Removal Reason: Terminal necropsy	Male				Female			
	Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg	Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Number of Animals:	12	12	12	12	12	12	12	12
SEMINAL VESICLES (Continued...)								
Infiltration, Macrophages; interstitial, focal	0	-	-	1	-	-	-	-
.... minimal	0	-	-	1	-	-	-	-
SPLEEN								
Examined	12	12	12	12	12	12	12	12
No Visible Lesions	12	11	0	0	12	12	7	0
Hematopoiesis; extramedullary	0	0	0	1	0	0	0	1
.... minimal	0	0	0	0	0	0	0	1
.... mild	0	0	0	1	0	0	0	0
Infiltration, Macrophages; few	0	1	2	1	0	0	4	3
.... minimal	0	1	2	1	0	0	4	3
Infiltration, Macrophages; multifocal	0	0	10	11	0	0	1	9
.... minimal	0	0	10	4	0	0	1	8
.... mild	0	0	0	5	0	0	0	1
.... moderate	0	0	0	2	0	0	0	0
TESTES								
Examined	12	0	0	12	-	-	-	-
No Visible Lesions	12	-	-	0	-	-	-	-
Interstitial cell; Vacuolation; multifocal	0	-	-	12	-	-	-	-
.... mild	0	-	-	6	-	-	-	-
.... moderate	0	-	-	6	-	-	-	-
UTERUS								
Examined	-	-	-	-	12	0	0	12

iv. Male and female reproductive indices

Estrous cyclicity did not differ among groups, but fertility parameters appeared to be dose-dependently decreased. However, fertility and fecundity were lower than expected in the C group, so differences were not observed. There were no effects on sperm parameters that were considered drug-related by the sponsor. There was an apparent increase (SS) in abnormal sperm in HD males (13%) compared to C (6%); however, this was primarily due to a single HD animal (104) with an extremely high (99%) percentage of abnormal sperm which skewed the mean for the high dose group. Additionally, there was no clear pattern supporting an increase in sperm abnormalities with increased dose. Thus, this finding was considered unlikely to be related to drug.

Table IVF.2 Fertility parameters

Sex: Female		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Paired	N+ve	31	31	31	31
F Mated	N+ve	30	28	29	31
Pregnant?	ProA	22/31	29/31	27/31	26/31
Female Ferti	%	71.0	93.5	87.1	83.9
Fec Index	%	70.0	92.9	86.2	83.9
Index	ProA	21/30	26/28	25/29	26/31

Table IVF.3 Sperm Morphology

Sex: Male		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Total Sperm	Mean	208	200	200	200
Evaluated	SD	15	0	0	0
	n	25	24	24	25
Normal Sperm	Mean	195 ¹	183 ^{dd2}	184 ^{dd2}	174 ^{ddd3}
	SD	14	8	9	37
	n	25	24	24	25
Intact Sperm	Mean	0 ^{R4}	0	1	8
Ab. Head	SD	1	0	2	39
	n	25	24	24	25
Intact Sperm	Mean	7 ^{I5}	8	7	9
Ab. Tail	SD	5	5	5	5
	n	25	24	24	25
Intact Sperm	Mean	0 ^{R4}	0	0	0
Head&Tail	SD	0	0	0	0
	n	25	24	24	25
Tailless	Mean	6 ^{I5}	8	8	8
Normal Head	SD	6	5	6	5
	n	25	24	24	25
Sperm	Mean	0 ^{R1}	0	0	0
AbnormalHead	SD	0	0	0	0
	n	25	24	24	25
Total Ab'mal	Mean	13 ^{L,a2}	17	16	26 ^{dd3}
Sperm	SD	9	8	9	37
	n	25	24	24	25
% Abnormal	Mean	6.2 ⁴	8.3	8.2	13.0 ^{dd3}
Sperm	SD	4.2	4.2	4.4	18.3
	n	25	24	24	25

1 [R - Automatic Transformation: Rank] 2 [L,a - Automatic Transformation: Log, Group Factor Test:

Analysis of Variance p < 0.05]

3 [dd - Test: Dunnett 2 Sided p < 0.01] 4 [L,aa - Automatic Transformation: Log, Group Factor Test:

Analysis of Variance p < 0.01]

v. Litter parameters

No drug effects on C-sectioning parameters or fetal exams were apparent.

Table IVF.4 Caesarean-Sectioning Observations

Sex: Female		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Gravid Uterus (g)	Mean	22.364 I ¹	23.729	23.848	23.311
	SD	5.925	4.527	4.570	3.943
	n	15	21	18	20
Number of CorporaLutea	Mean	15 I ¹	16	15	17
	SD	4	2	2	3
	n	15	21	18	20
Number of Implantat'ns	Mean	14 I ¹	14	14	15
	SD	3	3	2	2
	n	15	21	18	20
Implantation Index (Percentage)	Mean	109 R ²	90	94	90
	SD	68	16	9	10
	n	15	21	18	20
Live Fetuses	Mean	13 I ¹	13	13	13
	SD	4	3	3	3
	n	15	21	18	20
Live Male Fetuses	Mean	7 I ¹	7	7	7
	SD	2	2	3	2
	n	15	21	18	20
Sex Ratio	Mean	55.1 I ¹	55.3	51.1	50.6
	SD	18.0	13.2	13.8	11.9
	n	15	21	18	20
Fetal Viability (Percentage)	Mean	86 R ²	93	91	91
	SD	18	9	12	11
	n	15	21	18	20
Mean Fetal Weight (M) (g)	Mean	1.51 I ¹	1.52	1.54	1.46
	SD	0.08	0.11	0.15	0.12
	n	15	21	18	20
Mean Fetal Weight (F) (g)	Mean	1.45 I ¹	1.47	1.48	1.41
	SD	0.09	0.11	0.14	0.11
	n	14	21	18	20
Mean Fetal Weight(both) (g)	Mean	1.49 I ¹	1.50	1.51	1.43
	SD	0.08	0.11	0.13	0.11
	n	15	21	18	20

Table IVF.5 Fetal examinations

Exam Type: External		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Number of Fetuses Examined:		189	283	238	272
Number of Litters Examined:		15	21	18	20
Malformation	Number of Fetuses	0	2	1	2
	% of Fetuses	0.0 I ¹	0.7	0.3	0.8
	Number of Litters	0	2	1	2
Totals	Number of Fetuses	0	2	1	2
	% of Fetuses	0.0	0.7	0.4	0.7
	Number of Litters	0	2	1	2
Exam Type: Bouins		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Number of Fetuses Examined:		93	136	114	132
Number of Litters Examined:		15	21	18	20
Variation	Number of Fetuses	15	11	12	13
	% of Fetuses	15.2 L ¹	7.9	11.3	10.0
	Number of Litters	8	9	9	9
Malformation	Number of Fetuses	4	1	1	0
	% of Fetuses	4.5 R ²	0.8	0.7	0.0
	Number of Litters	3	1	1	0
Totals	Number of Fetuses	17	12	13	13
	% of Fetuses	18.3	8.8	11.4	9.8
	Number of Litters	9	10	10	9
Exam Type: Skeletal-Small Animal		Group 1 0 mg/kg	Group 2 3 mg/kg	Group 3 10 mg/kg	Group 4 25 mg/kg
Number of Fetuses Examined:		97	143	123	137
Number of Litters Examined:		15	21	18	20
Variation	Number of Fetuses	48	76	70	77
	% of Fetuses	46.9 I ¹	54.4	56.1	57.4
	Number of Litters	14	21	17	18
Malformation	Number of Fetuses	0	4	1	1
	% of Fetuses	0.0 R ²	2.6	0.6	0.5
	Number of Litters	0	3	1	1
Totals	Number of Fetuses	48	76	70	77
	% of Fetuses	49.5	53.1	56.9	56.2
	Number of Litters	14	21	17	18

The % fetuses for the Variation or Malformation represents the group mean litter percentage.

The % fetuses for the Total percentage of findings represents the percentage of fetuses affected of all fetuses examined

vi. Tissue exposure

Concentrations in the maternal and paternal mouse liver increased in a dose-dependent manner. ISIS 396443 was only detected in the pooled placenta tissue from HD litters, with an organ tissue concentration of ~10 ug/g, which was ~2% of the maternal liver concentration.

Table IVF.6 Parental and fetal ISIS 396443 tissue exposure

Group	Dose Level	Study Day	Gender	N	Number of Doses	Tissue Concentration (µg/g)		
						Mean ± SD		
						Fetal Liver	Liver	Placenta
1	0 mg/kg	71	Male	6	35	NA	BLQ	NA
		GD18	Female	5 ^a	16-20	NA ^b	BLQ	NA ^b
2	3 mg/kg	71	Male	6	35	NA	185 ± 23.6	NA
		GD18	Female	5 ^a	16-17	BLQ	69.2 ± 10.9	BLQ
3	10 mg/kg	71	Male	6	35	NA	478 ± 131	NA
		GD18	Female	6	16-17	BLQ	164 ± 32.7	BLQ
4	25 mg/kg	71	Male	6	35	NA	989 ± 255	NA
		GD18	Female	6	16-17	BLQ	417 ± 57.0	9.60 ± 4.82

Note: The lower limit of quantitation (LLOQ) for the assay was 10.0 µg/g. All values below the lower limit of quantitation (BLQ) were treated as zero (0) for the statistical summary analysis.

^a One female in Group 1 and one female in Group 2 were not pregnant at necropsy and samples were not collected (see toxicology report for additional details).

^b Not analyzed as per protocol

NA = Not applicable; SD = Standard Deviation

c. Conclusions

Administration of ISIS 396443 (0, 3, 10, or 25 mg/kg sc) to CD-1 male and female mice every other day (10.5, 35, 87.5 mg/kg/week) prior and during mating and pregnancy (through organogenesis) resulted in histopathology findings in treated parental animals that were consistent with those previously observed in mice treated with antisense oligonucleotides and which were attributed to the proinflammatory effects and intracellular accumulation of the drug. These included basophilic intracytoplasmic granules in renal cortical tubular cells, hypertrophied Kupffer cells and granulomas in the liver, and infiltrates of tissue macrophages in the spleen, epididymides, prostate glands and testes (with associated vacuolation of Leydig cells). There was no NOEL for histopathology. There was some suggestion of effects on sperm morphology and fertility parameters; however, the effect on sperm was primarily driven by a single male and the possible fertility effect could not be adequately evaluated because of an aberrant control response. There were no clear effects on embryofetal development. Based on placental tissue levels, it appeared the placental transfer was low.

2. Embryo-Fetal Development Toxicity Study of ISIS 396443 Following Subcutaneous Injection in New Zealand White Rabbits (Report No. 396443-AS09; dated 1/12/15; conducted by (b) (4); GLP)

a. Methods

Mated female NZW rabbits were dosed sc (0.30 (C & LD) and 0.21 (MD & HD) mL/kg) with 0 (0.9% Sodium Chloride Injection vehicle), 6, 12.6, or 25 mg/kg ISIS 396443 (lot CA396443-004) every other day from GD 6 through 19 (GDs 6, 8, 10, 12, 14, 16 and 18; 21, 44.1, and 87.5 mg/kg/wk). Does were observed for viability, clinical signs, premature deliveries, and deaths. Body weights (BW) and food consumption were recorded during the dosing and post-dosing period. Blood samples were collected for hematology and serum chemistry prior to necropsy on GD 28. Animals were sacrificed on GD 28 and C-sectioned. Liver samples were collected from 5 tox group does that were pregnant with at least five fetuses. Pregnancy status was assessed, the gravid uterus was weighed, and the numbers of corpora lutea, implantations, and live fetuses recorded. Live fetuses were weighed, sexed, and examined for external abnormalities. The placental weights were also recorded. Visceral examinations were performed on fresh specimens on the day of C-section. Following visceral examination, fetuses were fixed in ethyl alcohol, stained with alizarin red and examined for skeletal variations and malformations. Portions of liver, kidney, and spleen for TK evaluation were collected from pregnant does with at least 5 fetuses at the time of necropsy on GD20. Placentae, fetal liver, and fetal kidney samples were also collected from the litters of these same does.

Group	Dose Level (mg/kg) ^a	Tox Necropsy (GD28)		TK Necropsy (GD20)	
		Pregnant	Non-pregnant	Pregnant	Non-pregnant
1	0	18	4 ^{b,c}	-	-
2	6	17	5 ^b	3	-
3	12.6	18	4 ^b	3	-
4	25	18	4	3	-

^a Animals were dosed every other day between GD6 and GD18 and received 3 or 4 doses per week. Thus, the weekly dose (mg/kg/wk) was equivalent to 3.5 times the single mg/kg dose: 21, 44.1, and 87.5 mg/kg/wk.

^b Of the animals not pregnant at necropsy, the following ones gave birth/aborted prior to the scheduled necropsy: 0 mg/kg/wk group – Tox Animal 5312 on GD26; 21 mg/kg/wk – Tox Animal 5335 on GD17; 44.1 mg/kg/wk – Tox Animal 5394 in the morning of GD28 (prior to the time of the scheduled necropsy).

^c Animal 5328 (0 mg/kg/wk) appeared to have a non-gravid uterus at necropsy on GD28, however, one implantation site was noted upon examination.

Dose selection: Doses were based on a dose range-finding study in pregnant NZW rabbits in which sc doses of 5, 15, and 30 mg/kg were given every other day (17.5, 52.5, and 105 mg/kg/week) from GD 6-18. Drug-related changes at the HD included premature deliveries (2 HD does delivered 1 and 3 dead fetuses on GD27) and decreases (up to ~80% of C) in red cell parameters (RBCs, hemoglobin, and/or hematocrit). There were no effects on the gross pathology examination of the does, gravid uterine weights, number of live fetuses on GD28, or external fetal and placental evaluations. The HD was considered by the sponsor to be associated with minimal to mild maternal toxicity and no effects on embryofetal development.

b. Results

i. Maternal effects

There were no deaths and no drug-related clinical signs or changes in BW or BW gain. Four previously pregnant animals (2C, 1 LD, and 1 MD) were not pregnant at the time scheduled necropsy on GD28 but these were not considered drug-related. Slight (values generally within 85% of C) but SS decreases in red cell parameters (RBCs, hemoglobin,

and hematocrit) were noted in all treated groups. Increases (SS) in platelets (up to 49% compared to C), white blood cells (up to 29%), neutrophils (up to 81%), and lymphocytes (up to 26%) were noted in all treatment groups, except for lymphocytes at the HD. Decreases in albumin (<10%) and A/G ratio ($\leq 21\%$), and increases in globulin (<20%) were observed in drug-treated groups. The changes in serum chemistry parameters in association with the WBC and differential parameters were thought to be indicative of pro-inflammatory class effects.

Increases (1.1-1.4-fold) in liver, spleen, and kidney weights were observed, primarily at the HD, but there were no macroscopic changes noted. Tissue bioanalysis data demonstrated dose-dependent concentrations in maternal liver at both GD20 and GD28 (Table IVF.7). On GD28 (10 days post last dose), maternal liver concentrations had decreased ~40% compared to the levels on GD20 (2 days post last dose), consistent with the long tissue half-life of this compound. Minimal exposure ($\leq 9\%$ of maternal liver exposure) was observed in the placenta and no measurable exposure (<10.00 $\mu\text{g/g}$) was detected in fetal liver at any dose.

Table IVF.7 Parental and fetal tissue exposure

Group	Dose Level (mg/kg) ^a	Study Day	Number of Doses	N	Tissue Concentration ($\mu\text{g/g}$)		
					Fetal Liver	Liver	Placenta
1	0	GD28	7	5	NA	BLQ	NA
2	6	GD20	7	3	BLQ	90.5 $\pm$ 38.3	BLQ
		GD28	7	5	BLQ	40.1 $\pm$ 11.4	BLQ
3	12.6	GD20	7	3	BLQ	224 $\pm$ 40.9	18.4 $\pm$ 3.23
		GD28	7	5	BLQ	120 $\pm$ 57.1	BLQ
4	25	GD20	7	3	BLQ	374 $\pm$ 128	28.3 $\pm$ 9.61
		GD28	7	5	BLQ	281 $\pm$ 60.1	17.7 $\pm$ 5.17

Note: The lower limit of quantitation (LLOQ) for the assay was 10.0 $\mu\text{g/g}$, for all evaluated tissues. All values below the lower limit of quantitation (BLQ) were treated as zero (0) for the statistical summary analysis. Values are presented as mean $\pm$ standard deviation

For all dose groups, doses were administered on GD 6, 8, 10, 12, 14, 16, and 18.

GD = Gestation Day; NA = Not Applicable; BLQ = below the lower limit of quantification (10.0 $\mu\text{g/g}$)

ii. Litter parameters and fetal evaluations

A decrease (16%) in gravid uterine weights at the HD was considered drug-related in the report. There was a slight decrease in fetal weights, but this was not dose-related. The effect on placental weight may have been due to a decrease in implants at the HD which would not be expected to be due to dosing starting after implantation. There were no other drug-related changes in litter parameters (Table IVF.8).

There were no clear effects on embryofetal development based on the pattern and incidences of fetal structural abnormalities (Table IVF.9) and lack of effect on fetal BWs. An external malformation (exencephaly) was seen only at the LD. Visceral malformations were seen only in MD (1 with misshapen kidney) and HD (2 with misshapen kidney, 1 with multiple heart and great vessel malformations) fetuses. Skeletal malformations were seen in C (1 with fused ribs, 1 with fused sternbrae, 1 with thoracic hemivertebra, 1 with both fused sternbrae and thoracic hemivertebra), LD (1 with open skull [exencephalic fetus], 3 with fused sternbrae), MD (1 with skull openings, 1 with fused sternbrae), and HD (2 with fused sternbrae) fetuses. Incidences of skeletal variations such as supernumerary ribs and incomplete ossification were similar across groups.

Table IVF.8 Caesarean-sectioning observations

Sex: Female		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Gravid Uterus (g)	Mean	424.7 I ¹	405.2	405.9	358.5
	SD	86.4	93.2	103.9	80.1
	n	18	17	18	18
Number of CorporaLutea	Mean	10 I ¹	9	10	9
	SD	2	2	2	1
	n	18	17	18	18
Number of Implantat'ns	Mean	9 R, k ²	8	9	7
	SD	2	2	3	1
	n	18	17	18	18
Implantation Index (Percentage)	Mean	93 R ³	90	88	85
	SD	12	9	20	13
	n	18	17	18	18
Live Fetuses	Mean	8 I ¹	7	9	7
	SD	2	2	3	2
	n	18	17	18	18
Live Male Fetuses	Mean	4 I ¹	4	5	3
	SD	2	2	2	2
	n	18	17	18	18
Sex Ratio	Mean	49.4 I ¹	55.8	53.7	44.2
	SD	17.9	21.4	16.4	19.3
	n	18	17	18	18
Fetal Viability (Percentage)	Mean	95 R ²	91	96	97
	SD	6	14	7	7
	n	18	17	18	18
Mean Fetal Weight (M) (g)	Mean	34.92 I ¹	36.14	32.28	32.89
	SD	4.50	4.78	5.18	3.88
	n	18	17	18	18
Mean Fetal Weight (F) (g)	Mean	33.47 I, a ³	35.33	30.58	32.11
	SD	4.69	4.94	5.06	5.13
	n	18	17	17	18
Mean Fetal Weight(both) (g)	Mean	34.22 I ¹	35.92	31.94	32.43
	SD	4.30	4.48	5.48	4.21
	n	18	17	18	18

Table IVF.9 Fetal abnormalities in rabbit embryofetal development study

Exam Type: External		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined:		153	128	160	131
Number of Litters Examined:		18	17	18	18
Malformation	Number of Foetuses	0	1	0	0
	% of Foetuses	0.0 I ¹	0.8	0.0	0.0
	Number of Litters	0	1	0	0
Totals	Number of Foetuses	0	1	0	0
	% of Foetuses	0.0	0.8	0.0	0.0
	Number of Litters	0	1	0	0

Exam Type: Fresh Visceral		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined:		152	127	158	130
Number of Litters Examined:		18	17	18	18
Variation	Number of Fetuses	0	2	0	3
	% of Fetuses	0.0	1.5	0.0	2.1
	Number of Litters	0	2	0	3
Malformation	Number of Fetuses	0	0	1	3
	% of Fetuses	0.0	0.0	0.7	2.3
	Number of Litters	0	0	1	2
Totals	Number of Fetuses	0	2	1	5
	% of Fetuses	0.0	1.6	0.6	3.8
	Number of Litters	0	2	1	4

Exam Type: Skeletal-Small Animal		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined:		152	118	130	130
Number of Litters Examined:		18	16	16	18
Variation	Number of Fetuses	133	107	121	116
	% of Fetuses	88.9	91.4	93.5	88.8
	Number of Litters	18	16	16	18
Malformation	Number of Fetuses	4	4	2	2
	% of Fetuses	2.4	3.2	1.4	1.5
	Number of Litters	4	3	2	2
Totals	Number of Fetuses	135	107	121	117
	% of Fetuses	88.8	90.7	93.1	90.0
	Number of Litters	18	16	16	18

Exam Type: External		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined:		153	128	160	131
Number of Litters Examined:		18	17	18	18
head					
Cranium, Exencephaly-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	1(0.8+3.5) 1(5.9)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)

Exam Type: Fresh Visceral		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined:		152	127	158	130
Number of Litters Examined:		18	17	18	18
abdomen					
Kidney, Left, Misshapen-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) L ¹ 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.6)	2(1.6+5.7) 1(5.5)
Liver, Abnormal lobation-Variation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) L ¹ 0(0.0)	1(0.7+2.7) 1(5.9)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)
gonadal					
Artery, Malpositioned-Variation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	1(0.8+3.5) 1(5.9)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)
heart					
Ventricular chamber, Enlarged-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.5)
Ventricular chamber, Small-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.5)
thorax (excluding heart)					
Aortic arch, Dilated-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.5)
Ductus arteriosus, Narrowed-Malformation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.5)
Pulmonary artery, Narrowed-Variation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	1(0.7+2.9) 1(5.5)
Pulmonary artery, Narrowed-Variation	Litters n(%)	0(0.0)	0(0.0)	0(0.0)	1(5.6)
Subclavian artery, Suprumerary-Variation	Fetuses n(%+SD) Litters n(%)	0(0.0+0.0) 1 ⁺ 0(0.0)	0(0.0+0.0) 0(0.0)	0(0.0+0.0) 0(0.0)	2(1.4+4.1) 2(11.1)

Exam Type: Skeletal-Small Animal		Group 1 Control 0 mg/kg	Group 2 ISIS396443 6 mg/kg	Group 3 ISIS396443 12.6mg/kg	Group 4 ISIS396443 25 mg/kg
Number of Fetuses Examined: Number of Litters Examined:		152 18	118 16	130 16	130 18
ribs					
5th rib, Right, Fused-Malformation	Foetuses n(%±SD) Litters n(%)	1(0.6±2.6) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
13th rib, Bilateral, Full supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	83(54.3±34.5) 17(94.4)	1 ^I 54(46.5±32.1) 13(81.3)	66(52.0±33.5) 13(81.3)	80(59.5±32.9) 17(94.4)
13th rib, Bilateral, Short supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	3(1.4±4.4) 2(11.1)	1 ^L 3(2.3±5.1) 3(18.8)	4(4.4±15.1) 2(12.5)	6(5.6±10.2) 5(27.8)
13th rib, Left, Full supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	8(4.9±7.8) 6(33.3)	1 ^I 13(10.6±10.7) 9(56.3)	13(9.1±11.9) 8(50.0)	8(6.0±9.0) 7(38.9)
13th rib, Left, Short supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	5(2.8±5.9) 4(22.2)	1 ^I 7(6.4±10.9) 6(37.5)	8(4.5±8.2) 4(25.0)	6(4.7±7.1) 6(33.3)
13th rib, Right, Full supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	10(6.5±8.6) 8(44.4)	1 ^I 9(8.3±10.9) 8(50.0)	11(7.6±10.6) 7(43.8)	7(6.5±11.9) 5(27.8)
13th rib, Right, Short supernumerary-Variation	Foetuses n(%±SD) Litters n(%)	7(4.5±7.1) 6(33.3)	1 ^I 10(8.7±9.4) 8(50.0)	8(5.3±10.6) 5(31.3)	8(5.9±8.8) 6(33.3)
skull					
Frontal, Bilateral, Hole(s)-Malformation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^I 0(0.0±0.0) 0(0.0)	1(0.5±1.9) 1(6.3)	0(0.0±0.0) 0(0.0)
Frontal, Bilateral, Incomplete ossification-Variation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	1(0.6±2.6) 1(5.6)
Frontal, Bilateral, Misshapen-Variation	Foetuses n(%±SD) Litters n(%)	1(0.6±2.6) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
General, Craniofenestria-Malformation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^I 1(0.9±3.6) 1(6.3)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
Hyoid, Bent-Variation	Foetuses n(%±SD) Litters n(%)	2(1.6±6.7) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	2(1.5±2.3) 2(11.1)
Hyoid, Incomplete ossification-Variation	Foetuses n(%±SD) Litters n(%)	46(32.4±28.4) 15(83.3)	1 ^I 38(30.0±33.1) 11(68.8)	52(40.2±30.0) 13(81.3)	41(31.0±28.9) 12(66.7)
Hyoid, Unossified-Variation	Foetuses n(%±SD) Litters n(%)	2(1.2±3.4) 2(11.1)	1 ^k 0(0.0±0.0) 0(0.0)	7(5.3±9.7) 4(25.0)	10(6.9±13.6) 6(33.3)
Parietal, Hole(s)-Malformation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^I 0(0.0±0.0) 0(0.0)	1(0.5±1.9) 1(6.3)	0(0.0±0.0) 0(0.0)
Parietal, Incomplete ossification-Variation	Foetuses n(%±SD) Litters n(%)	1(0.6±2.6) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
sternebrae					
2nd sternebra, Misaligned-Variation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^L 1(0.7±2.8) 1(6.3)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
2nd sternebra, Asymmetry-Variation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^L 1(0.7±2.8) 1(6.3)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
5th sternebra, Misaligned-Variation	Foetuses n(%±SD) Litters n(%)	1(0.9±3.9) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	1(1.3±5.0) 1(6.3)	0(0.0±0.0) 0(0.0)
5th sternebra, Unossified-Variation	Foetuses n(%±SD) Litters n(%)	14(10.6±19.6) 5(27.8)	1 ^L 7(7.4±15.6) 4(25.0)	7(5.4±9.4) 5(31.3)	9(6.4±11.1) 6(33.3)
6th sternebra, Unossified-Variation	Foetuses n(%±SD) Litters n(%)	3(2.1±8.8) 1(5.6)	1 ^L 9(7.7±12.7) 5(31.3)	3(1.7±4.6) 2(12.5)	4(3.0±7.2) 3(16.7)
6th sternebra, Asymmetry-Variation	Foetuses n(%±SD) Litters n(%)	1(0.6±2.6) 1(5.6)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	1(0.9±3.9) 1(5.6)
4th & 5th sternebrae, Fused-Malformation	Foetuses n(%±SD) Litters n(%)	2(1.3±3.8) 2(11.1)	1 ^L 1(0.9±3.6) 1(6.3)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
5th & 6th sternebrae, Fused-Malformation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^I 1(0.7±2.8) 1(6.3)	1(0.9±3.6) 1(6.3)	2(1.5±4.6) 2(11.1)
3rd, 4th & 5th sternebrae, Fused-Malformation	Foetuses n(%±SD) Litters n(%)	0(0.0±0.0) 0(0.0)	1 ^L 1(0.7±2.8) 1(6.3)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)
thoracic vertebrae					
Thoracic vertebra, Thoracic hemivertebra-Malformation	Foetuses n(%±SD) Litters n(%)	2(1.1±3.3) 2(11.1)	1 ^I 0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)	0(0.0±0.0) 0(0.0)

c. Conclusions

SC administration of ISIS 396443 to pregnant NZW rabbits every other day throughout organogenesis (GD6-18) at doses up to 25 mg/kg (87.5 mg/kg/wk) did not produce any clear evidence of maternal or embryofetal toxicity.

G. OTHER TOXICITY STUDIES

1. Evaluation of oligonucleotide-associated hippocampal vacuolation following intrathecal lumbar bolus administration using several tissue fixation methods in cynomolgus monkeys (Report #: 396443-AS11; conducted by (b) (4), report dated 4/14/15; Non-GLP)

a. Methods

Cynomolgus monkeys (3/sex/grp) were administered weekly IT bolus doses of 5 mg/dose (1.25 mL) for 36 days (Days 1, 8, 15, 22, 29 and 36; total of 6 doses). Dosing was via indwelling lumbar catheters lumbar space in a volume of 1.25 mL. On Day 50 (approximately 2 weeks after the last dose), all animals were sacrificed and samples were taken for postmortem examination. Observations included body weight and food consumption, clinical signs, and neurological (spinal reflexes: knee jerk, cutaneous reflex, proprioception, sensory foot, tail reflex) and physical (heart rate, respiration, temperature, gait, disposition, abdominal palpation, lymph nodes, and general appearance of the eyes, ears, oral cavity, and skin and nails) examinations.

Study Design								
Treatment Regimen ^a				Total Dose (mg)	Total Number of Animals	Sacrifice (14 Days after final dose; Day 50)		Brain Fixation
Group	Dose (mg)	Dosing Solution Conc. (mg/mL)	Volume (mL/dose)			Number		
						M	F	
1	5	4	1.25	30	6	3	3	Immersion ^b /Frozen ^c
2	5	4	1.25	30	6	3	3	Perfusion ^d

^a IT Bolus doses administered over 3 minutes once-weekly on Days 1, 8, 15, 22, 29, and 36

^b Alternate brain slices taken from the left hemisphere of the brain and then immersed in 10% neutral buffered formalin or Carnoy's solution

^c Frozen sections taken from the right hemisphere of the brain

^d Perfusion fixation using modified Karnovsky's Fixative (2% paraformaldehyde/2.5% glutaraldehyde)

b. Results

i. Clinical observations and body weight

There were no deaths and no clinical signs or changes in body weight were reported.

ii. Neurological and physical examinations

No changes in physical examinations were noted. Transient decreases in lower spinal reflexes (negative responses to the knee jerk, cutaneous, sensory foot and tail reflexes) were observed in 8/12 animals at one or more evaluations following dosing. While their onset varied, decreases in lower spinal reflexes were generally observed within the first 4 hours of dosing and resolved within 24 to 48 hours of dosing.

Table IVG.1 Neurological examination summary (# positive/# per group) Groups 1 and 2 - 5 mg/dose

Dose Number			1													
Study Day			1													
Time Interval			Pre		1 hr		2 hr		4 hr		8 hr		24 hr		48 hr	
Sex			Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
Spinal Reflexes	Knee Jerk	Left Hind Limb	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	5/6	6/6	6/6	6/6
		Right Hind Limb	6/6	6/6	5/6	6/6	6/6	6/6	6/6	5/6	6/6	6/6	6/6	5/6	6/6	6/6
	Proprioceptive Reflex	Left Hind Limb	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
		Right Hind Limb	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
	Cutaneous Reflex	Left Side	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
		Right Side	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
	Sensory Foot Reflex	Left Hind Limb	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
		Right Hind Limb	6/6	6/6	6/6	6/6	6/6	6/6	5/6	6/6	6/6	6/6	6/6	6/6	6/6	6/6
	Tail Reflex		6/6	6/6	6/6	6/6	5/6	6/6	3/6	5/6	5/6	6/6	6/6	6/6	6/6	6/6

iii. Neurohistopathology

In the 10% neutral buffered formalin immersion-fixed brain, minimal vacuolation in the hippocampus was noted in 3/6 monkeys; no additional histological alterations were reported. Immunohistochemistry demonstrated positive staining of neuronal cytoplasm in the region of the hippocampus and corresponded to the location of the vacuoles. The vacuolated cells present also stained positive for the presence of ISIS 396443.

In the brain fixed using the alternative methods (immersion fixation in Carnoy's solution, perfusion fixation with modified Karnovsky's solution and fixation by snap freezing), no vacuolation of the hippocampus or any other histological alterations were observed. Additionally, there were no histological alterations present in the spinal cord regardless of the fixation method utilized. The pathology report contained only 4 photomicrographs and the following descriptions:

Immersion Fixation in Carney's Solution: No histological alterations were present in the brain of 6/6 monkeys.

Perfusion Fixation with Modified Karnovsky's Solution: No histological alterations were present in the brain of 6/6 monkeys.

Fixation by snap freezing: No histological alterations were present in the brain of 6/6 monkeys.

There were no histological alterations present in the spinal cord regardless of the fixation method utilized.

Bioanalysis indicated that the highest concentrations were observed in the lumbar spinal cord with lower concentrations in the thoracic and cervical regions (Table IVG.2). Concentrations were slightly lower in the cortical regions of the brain, with the lowest concentration observed in the hippocampus. Approximately 14 days after the 6th dose (Day 50), the highest mean CNS tissue concentrations of ISIS 396443 were generally observed in the lumbar spinal cord (99.95 µg/g), consistent with the IT route of administration and catheter placement; with similar concentrations in the thoracic spinal cord (84.86 µg/g), and somewhat lower concentrations in the cervical section (44.84 µg/g). In brain tissues, ISIS 396443 concentrations were highest in the occipital (60.56 µg/g) and frontal (58.71 µg/g) cortex. Lower concentrations were measured in the hippocampus (9.87 µg/g).

Table IVG.2

Group	Animal #	Study Day	Time Point	Gender	# of Doses	Matrix	Dose Level	Route	ISIS 396443 (µg/g)	Group Mean (µg/g)	SD
1	001	50	14	M	6	Lumbar SC	5 mg	IT	248.71	99.95	78.4
1	002A	50	14	M	6	Lumbar SC	5 mg	IT	41.61		
1	003	50	14	M	6	Lumbar SC	5 mg	IT	119.06		
1	004	50	14	F	6	Lumbar SC	5 mg	IT	77.44		
1	005	50	14	F	6	Lumbar SC	5 mg	IT	40.53		
1	006	50	14	F	6	Lumbar SC	5 mg	IT	72.34		
1	001	50	14	M	6	Thoracic SC	5 mg	IT	105.32	84.86	27.4
1	002A	50	14	M	6	Thoracic SC	5 mg	IT	83.00		
1	003	50	14	M	6	Thoracic SC	5 mg	IT	82.02		
1	004	50	14	F	6	Thoracic SC	5 mg	IT	100.53		
1	005	50	14	F	6	Thoracic SC	5 mg	IT	33.25		
1	006	50	14	F	6	Thoracic SC	5 mg	IT	105.03		
1	001	50	14	M	6	Cervical SC	5 mg	IT	80.60	44.84	26.5
1	002A	50	14	M	6	Cervical SC	5 mg	IT	44.08		
1	003	50	14	M	6	Cervical SC	5 mg	IT	66.90		
1	004	50	14	F	6	Cervical SC	5 mg	IT	22.63		
1	005	50	14	F	6	Cervical SC	5 mg	IT	9.36		
1	006	50	14	F	6	Cervical SC	5 mg	IT	45.44		
1	001	50	14	M	6	Hippocampus	5 mg	IT	16.97	9.87	7.7
1	002A	50	14	M	6	Hippocampus	5 mg	IT	14.59		
1	003	50	14	M	6	Hippocampus	5 mg	IT	12.09		
1	004	50	14	F	6	Hippocampus	5 mg	IT	0.12		
1	005	50	14	F	6	Hippocampus	5 mg	IT	0.11		
1	006	50	14	F	6	Hippocampus	5 mg	IT	15.32		
1	001	50	14	M	6	Frontal Cortex	5 mg	IT	91.31	58.71	46.3
1	002A	50	14	M	6	Frontal Cortex	5 mg	IT	65.74		
1	003	50	14	M	6	Frontal Cortex	5 mg	IT	111.21		
1	004	50	14	F	6	Frontal Cortex	5 mg	IT	2.42		
1	005	50	14	F	6	Frontal Cortex	5 mg	IT	1.95		
1	006	50	14	F	6	Frontal Cortex	5 mg	IT	79.62		
1	001	50	14	M	6	Occipital Cortex	5 mg	IT	89.281	60.56	50.7
1	002A	50	14	M	6	Occipital Cortex	5 mg	IT	73.751		
1	003	50	14	M	6	Occipital Cortex	5 mg	IT	131.690		
1	004	50	14	F	6	Occipital Cortex	5 mg	IT	2.867		
1	005	50	14	F	6	Occipital Cortex	5 mg	IT	1.682		
1	006	50	14	F	6	Occipital Cortex	5 mg	IT	64.069		

c. Conclusions

These results were interpreted by the sponsor to indicate that the mechanism for hippocampal vacuolation in monkeys treated with ISIS 396443 is the same as that for vacuolation seen in the renal proximal tubule, i.e., "oligonucleotide taken up into the endosomal/lysosomal system is eluted during the fixation/processing of these tissues leaving a dilated vacuolar system remnant to be seen by light microscopy." They further state that "the presence of vacuoles in the hippocampus following treatment with oligonucleotides does not represent an adverse toxicological finding, but rather represents an artifact of tissue preservation that occurs with the presence of drug."

2. A 13-week repeat dose toxicity and impurity qualification study of ISIS 396443 in CD-1 mice (Study # 396443-AS13, (b) (4) 727-083, conducted by (b) (4) report dated, 6/10/16, GLP)

a. Methods

CD-1 mice (6/sex/grp, ~8 weeks old) were administered (SC, 5 mL/kg) saline vehicle or 3 different formulations of drug/impurities (ISIS 396443 Impurity Mixtures 1, 2, and 3; TAM: Test Article Mixture) at two doses (5 and 25 mg/kg/week) weekly for 13 weeks (Days 1, 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, and 91). Observations consisted of mortality, clinical observations, body weight, food consumption, ophthalmoscopic examinations, clinical and anatomic pathology, and tissue TK.

Table A: Group Assignments			
Group Number	Dose Level (mg/kg/week) ^a	Treatment	Number of Animals
			Male/Female
Main Study			
1	0	Saline Control	6/6
2	5	ISIS 396443	6/6
3	25	ISIS 396443	6/6
4	5	ISIS 396443 TAM 1 ^b	6/6
5	25	ISIS 396443 TAM 1 ^b	6/6
6	5	ISIS 396443 TAM 2 ^b	6/6
7	25	ISIS 396443 TAM 2 ^b	6/6
8	5	ISIS 396443 TAM 3 ^b	6/6
9	25	ISIS 396443 TAM 3 ^b	6/6

^aThe control and test articles were administered once weekly via subcutaneous injection (Days 1, 7, 14, 21, 28, etc. through Day 91), with the necropsy occurring on Day 93 (48 hours post the Day 91 dose).

^bImpurity mixture

Doses were based on the previous 13-week SC toxicology study (396443-AS07) in which doses up to 50 mg/kg produced changes consistent with the uptake of drug into tissues. The doses selected for the current study were said to provide exposure margins, based on body surface area, of 1- and 20-fold over doses being used in clinical trials in children and <1- and 4-fold in infants. Levels of impurities exceeded those normally found in GLP/GMP lots of oligonucleotides to set manufacturing specifications.

b. Results

- i. Mortality, clinical signs, body weight, ophthalmoscopic examinations

No drug-related effects on survival, clinical findings, body weights, or ophthalmoscopic exams were observed.

- ii. Clinical pathology

There were no effects on hematology parameters. Mild increases in AST and ALT seen in a few individual animals were thought to possibly have been related to the microscopic findings in the liver (e.g. vacuolated/granular macrophages) which were considered drug-related.

iii. Histopathology

Dose-dependent accumulation of vacuolated/granular macrophages was seen in tissues throughout the body of drug (alone or with impurities)-treated animals. There was no associated cellular degeneration or necrosis. The infiltrates were most prominent in the liver (Table IVG.3), lymph nodes, and injection sites, and were often barely detectable in other tissues. Vacuolated/granular macrophages were present in the sinusoids of the liver, the dermis and subcutis of the injection sites and skin, the sinuses of the lymph nodes and spleen, and interstitial connective tissue of the gastrointestinal and urogenital tracts, glandular organs, heart, skeletal muscle, and choroid plexus of the brain.

The kidneys of drug (alone or with impurities)-treated animals had minimal dose-related accumulation of basophilic granules within tubular epithelial cells (Table IVG.4), consistent with oligonucleotide uptake. Incidences were similar among groups at the same dose. There was no associated cellular degeneration.

Minimal to mild monocyte/macrophage hyperplasia was observed in the liver, bone marrow, and mandibular lymph node of 1 female (#7504) from the HD TAM2 group. According to the report, this was “characterized by multifocal aggregates of macrophages with abundant pale cytoplasm and indistinct cell margins. In the liver, the monocyte/macrophage hyperplasia was randomly distributed with occasional small numbers of admixed leukocytes and/or rare attendant single cell necrosis of adjacent hepatocytes.” The widespread finding was not observed in other treated animals and the relationship to the test article and/or TAM 2 impurity was considered unclear in the report. However, monocyte/macrophage hyperplasia was also identified in the lymph nodes of other animals treated with ISIS 396443 with or without impurities (animal numbers 3002 (HD ISIS 396443 M), 7003 (HD TAM2 M), and 9503(HD TAM3 F)), but in these cases was isolated to a single lymph node.

Table IVG.3 ISIS 396443–related microscopic observations in the liver

Dose Level: mg/kg/week	5		25		TAM 1				TAM 2				TAM 3			
Sex	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F
Number Examined	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
macrophages, vacuolated/granular	2	3	6	6	3	2	6	6	1	1	6	6	0	1	6	6
-minimal	2	3	5	4	3	2	6	2	1	1	5	5	0	1	6	4
-mild	0	0	1	2	0	0	0	4	0	0	1	1	0	0	0	2
M-Male; F-Female																

Table IVG.4 ISIS 396443–related microscopic observations in the kidney

	ISIS 396443				TAM 1				TAM 2				TAM3			
Dose (mg/kg/week)	5		25		5		25		5		25		5		25	
Sex	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F
Number	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
basophilic granules, tubular cell - minimal	0	2	6	6	0	0	4	6	0	1	4	1	0	1	6	5
basophilic tubules – minimal	0	1	2	0	1	1	1	0	0	1	1	1	0	1	0	1
degeneration, tubular	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Dose-dependent exposure in liver was observed (Table IVG.5). Lower concentrations were observed in liver of mice receiving the test article mixtures (TAM) containing impurities (TAM 1, 2, and 3) compared to the drug without impurities due to the same mg/kg dose of total oligonucleotide being given in all groups and the resulting reduced percentage of full-length ISIS 396443 in the formulation of TAM 1, 2, and 3.

Table IVG.5 ISIS 396443 liver tissue concentrations

Impurity	Study Day	Gender	N	Number of Doses	Total Oligonucleotide Dose Level (mg/kg)				
					0 mg/kg	5 mg/kg	% of ISIS 396443 Only Formulation	25 mg/kg	% of ISIS 396443 Only Formulation
NA	93	Female	6	14	BLQ	43.4 ± 6.64	100	202 ± 37.3	100
	93	Male	6	14	BLQ	53.4 ± 5.59	100	269 ± 49.7	100
TAM 1	93	Female	6	14	NA	33.1 ± 7.26	76.2	151 ± 28.3	74.7
	93	Male	6	14	NA	42.3 ± 4.12	79.1	179 ± 31.7	66.4
TAM 2	93	Female	6	14	NA	31.3 ± 7.24	72.1	172 ± 39.6	85.3
	93	Male	6	14	NA	45.4 ± 16.6	85.0	216 ± 27.7	80.2
TAM 3	93	Female	6	14	NA	32.3 ± 17.1	74.5	158 ± 45.8	78.3
	93	Male	6	14	NA	29.4 ± 4.18	54.9	141 ± 23.5	52.5

Values reported as Mean ± SD

Note: The lower limit of quantitation (LLOQ) for the assay was 10.0 µg/g. All values below the lower limit of quantitation (BLQ) were treated as zero (0) for the statistical summary analysis.

NA = Not Applicable; SD = Standard Deviation

c. Conclusions

Weekly SC administration of ISIS 396443 with and without added impurities to adult CD-1 mice for 13 weeks resulted in accumulation of vacuolated/granular macrophages in tissues throughout the body and accumulation of basophilic granules within the tubular epithelium of the kidneys. These findings were consistent with oligonucleotide uptake and/or cellular activation and cytokine production due to proinflammatory effects of the oligonucleotide and were not associated with cell degeneration. Since there were no apparent differences among the groups receiving drug alone or with impurities, no additional toxicity could be attributed to the impurities evaluated.

V. SUMMARY AND EVALUATION

Pharmacology

ISIS 396443 is a uniformly modified 2'-MOE antisense oligonucleotide (ASO) intended to modulate alternative splicing of the SMN2 mRNA transcript to include exon 7 and produce full-length SMN protein in patients with spinal muscular atrophy. The SMN2 transcript undergoes alternative splicing in which exon 7 is either skipped or included. If exon 7 is skipped, a defective truncated protein product is produced. Typically, the majority of transcripts derived from the SMN2 gene are missing exon 7. ISIS 396443 is designed to bind to the pre-mRNA on intron 7 and promote exon 7 inclusion resulting in production of full length protein. ISIS 396443 was shown to target a site in intron 7, termed ISS-N1, that was previously known to repress SMN2 exon 7 inclusion. By hybridizing to ISS-N1 it is thought that the ASO promotes the inclusion of exon 7 by displacing inhibitory heterogeneous nuclear ribonucleoproteins (hnRNPs) on the pre-mRNA allowing assembly of splicing factors on exon 7. In cell culture and animals, ISIS 396443 increased SMN2 exon 7 inclusion and had beneficial effects in mouse models of SMA. Safety pharmacology consisted of a respiratory and cardiovascular safety pharmacology study in rats treated by continuous intrathecal (IT) infusion. CNS safety was assessed as part of the acute and subchronic toxicity studies in monkeys using IT lumbar bolus injection, the intended route of administration in man. These did not indicate any adverse CV or respiratory effects, but acute and possibly long-term neurotoxicity was observed in monkeys (described below).

ADME

PK/ADME data were collected in a dedicated study in adult monkeys and in the toxicity studies. Following IT administration to monkeys, the estimated CSF terminal elimination half-life ($t_{1/2}$) was consistent between adult and juvenile monkeys, with observed values of 102-111 days. Concentrations in plasma peaked 2-5 hours after the IT bolus injection and then rapidly declined during the first 24-48 hours post-dose, followed by a slower elimination phase. The estimated plasma terminal elimination half-life (35-56 days) was shorter than that in CSF (102 to 111 days) but similar to that in systemic tissues (12-23 days in liver and 35 days in kidney).

Concentrations in systemic tissues like liver and kidney following IT administration were similar to those seen in the spinal cord indicating that a substantial proportion of the administered dose was transferred from CSF to the systemic circulation. In both adult and juvenile monkeys, ISIS 396443 appeared to be cleared slowly from CNS and systemic tissues. The estimated median CNS tissue terminal elimination half-life ($t_{1/2}$) was consistent between adult and juvenile monkeys, with observed values of 116 days (range 74-275 days) and 174 days (range 117-195 days), respectively, which are similar to the observed terminal elimination half-life in CSF.

Comparison of plasma, kidney cortex, and liver exposure after IT and IV administration suggested minimal metabolic clearance of ISIS 396443 in the CNS prior to eventual transfer to plasma. Drug exposure in liver and kidney appeared to be independent of the route of administration. ISIS 396443 is highly bound to plasma proteins in mice, monkeys, and humans (>94%), which reduces glomerular filtration (limiting urinary excretion) and therefore facilitates distribution to systemic tissues. Binding to CSF proteins (monkeys and humans) was lower (<25% at 5 µg/mL and none bound at 150 µg/mL), presumably due to lower CSF protein concentrations. Following SC administration in mice and rabbits, a lack of measurable fetal liver exposure indicated limited placental transfer.

Metabolite profiling in monkeys following IT administration indicated that intact ISIS 396443 was the primary oligonucleotide present, accounting for 100% of the total oligonucleotides in CSF, >95% in plasma and 79% to 81% in CNS and systemic tissues. Metabolites observed in CSF (none in monkey CSF), plasma (N-1 from 3'-end only) and tissues (N-1, N-2, N-3, and N-4 from

3'-end, as well as the N-1, N-2, and N-3 from 5'-end) were chain-shortened oligonucleotides thought to indicate predominant exonuclease (3'- and 5')-mediated hydrolysis.

Nonclinical urinary excretion evaluation of ISIS 396443 was not conducted, but metabolism of similar classes of ASOs in tissues has been reported to be a slow process, followed by urinary excretion of mainly metabolites (chain-shortened oligonucleotides) and a minor fraction being parent drug. The similar rate of elimination of ISIS 396443 from CSF (t1/2: 102 to 111 days) and spinal cord tissues (t1/2: 115 to 138 days) suggesting equilibrium was thought to indicate that CSF concentrations could be used as a surrogate for spinal cord tissue concentrations. The long CSF and CNS tissue elimination half-life observed in monkeys was used as support for the proposed clinical dosing regimen following IT administration.

Safety Margin

Total dose over a 1 year period and application of a scaling factor to account for the CSF volume difference was used for animal to clinical dose comparisons. Safety margins using the cumulative doses from the 53-week monkey study and a 10-fold difference in CSF volume between the species are shown in Table V.1. As proposed, patients would receive loading doses in the first year of treatment to reach CNS steady-state tissue concentrations as rapidly as possible (12 mg adult equivalent dose on Days (b) (4)), followed by maintenance dosing every 4 months. This would result in a cumulative dose of 72 mg during the first year of treatment and 36 mg in subsequent years. (b) (4)

(b) (4) Using the NOEL for hippocampal vacuolation of 0.3 mg/dose in the 1-year study instead of the HD used by the sponsor, the monkey NOAEL is about half the human first year dose and similar to the human maintenance dose, when calculated on a yearly basis. CNS tissue concentrations achieved with the IT dosing regimen used in the juvenile monkey toxicity studies were generally in the range (2-10 µg/g) associated with pharmacological activity in mouse models at the 0.3 mg NOAEL and were similar to those measured in autopsy samples from treated patients (Tables V.2-3). CSF concentrations at the NOAEL (0.3 mg) in the 53-week monkey study (Table IVC.9, p. 41) were also similar to those measured in patients receiving the proposed clinical dose of 12 mg every 4 months (Table V.4).

Table V.1: Safety margin calculations for first year and chronic (maintenance) dosing

Study Duration	Dose Level (mg/dose)	Toxicology Assessment	Cumulative Dose	HED ^a	Safety Margin: Ratio of NOAEL to Adult Dose	
					Total First Year (72 mg) ^b	Chronic Yearly Maintenance Dose (36 mg) ^c
53 Weeks	0.3	NOEL	3.9 mg	39 mg		
	1.0	NOAEL	13 mg	130 mg		
	4.0	NOAEL	52 mg	520 mg	7.2-fold	14.4-fold

^a HED, human equivalent dose is 10 x cumulative dose in monkeys (approximate 10-fold difference in CSF volume)

^b Patients with SMA will receive loading doses on Days 1, 15, 29 and 64 followed by maintenance doses every 4 months (72 mg in first year, adult equivalent)

^c Maintenance dosing will be 36 mg per year (12 mg every 4 months, adult equivalent)

Table V.2 Histopathology and tissue concentrations in 1-year IT monkey toxicity study

Dose Group	Hippocampal Microscopic Findings ^a		Mean (and SD) Tissue Concentration (µg/g) ^b		
	Neuronal Vacuolation	Necrotic Cells ^c	Brain (cortex)	Spinal Cord (lumbar)	Liver
0.3 mg/dose (3.9 mg total) ^d	none	none	4.2 (1.9)	21.4 (9.3)	0.9 (0.4)
1 mg/dose (13 mg total) ^d	2 of 10	1 of 10	16.6 (6.0)	53.5 (33.6)	2.8 (1.6)
4 mg/dose (52 mg total) ^d	6 of 10	3 of 10	71.3 (29.2)	106 (78.9)	21.9 (9.2)

^a Incidence using combined male and female results^b Tissue concentrations measured on Day 372 (main necropsy)^c Necrotic cells were not neurons^d Total dose in 53 week period

Table V.3 ISIS 396443 tissue concentrations in human tissues after treatment

Tissue	Observed ISIS 396443 Concentration (µg/g)			
	Patient 2302	Patient 1303	Patient 2311	Mean ± SD
Lumbar Spinal Cord	19.46	26.6	31.8	26.0 ± 6.21
Thoracic Spinal Cord	15.92	23.7	14.2	18.0 ± 5.07
Cervical Spinal Cord	11.93	15.5	23.3	16.9 ± 5.81
Frontal Cortex	2.56	1.48	14.73	6.26 ± 7.36
Temporal Cortex	3.63	2.09	3.45	3.06 ± 0.842
Cervical Cortex	5.80	8.74	10.54	8.36 ± 2.39
Combined cortex (all regions)				5.89 ± 4.53
Cerebellum	4.36	4.16	2.77	3.76 ± 0.87
Hippocampus	2.10	2.80	7.30	4.07 ± 2.82
Thalamus	0.23 ^a	1.31	3.95	1.83 ± 1.91
Brain Stem	3.68	13.82	8.13	8.54 ± 5.08
Liver	1.33	12.88	2.39	5.53 ± 6.38
Kidney	18.44	36.51	25.07	26.67 ± 9.14

Patient Number	Dose	Dosing Days	Days from Last Dose to Autopsy
2302	12 mg eq ^a	1, 15, 85	78
1303	6 mg eq ^a	1, 15, 85	8
	12 mg eq ^a	253	
2311	12 mg eq ^{a,b}	1, 15, 85	30

^a Dose volume (and therefore total dose) was adjusted based on patient age and CSF volume to be equivalent to a 2-year old or older child receiving a 6 mg or 12 mg dose.^b Patient 2311 received 9.6 mg on Days 1 and 15, and 10.32 mg on Day 85. All doses were equivalent to 12 mg in a 2-year old child.

Table V.4 Pre-dose CSF concentrations (ng/mL) in subjects receiving single and multiple IT administration(s) of ISIS 396443, Study ISIS 396443-CS3A

Study Day	Number of Doses	Cohort 1 / 6 mg or 12 mg Equivalent ^a		Cohort 2 / 12 mg Equivalent ^a	
		N	Mean $\pm$ SD	N	Mean $\pm$ SD
1	0	4	BLQ	14 ^b	BLQ ^c
15	1	4	1.68 $\pm$ 0.872	16	3.57 $\pm$ 3.51
85	2	4	1.11 $\pm$ 1.28	13 ^d	3.84 $\pm$ 2.99
253	3	4	1.24 $\pm$ 0.526	13	4.93 $\pm$ 3.99
379	4	2	2.47 $\pm$ 0.849	13	9.29 $\pm$ 9.19
505	5	2	4.74 $\pm$ 2.58	11	5.91 $\pm$ 4.26
631	6	2	7.46 $\pm$ 4.49	5	10.3 $\pm$ 4.04
757	7	2	6.30 $\pm$ 3.38	4	13.6 $\pm$ 10.2
883	8	2	11.3 $\pm$ 8.70	NSC	NSC

Abbreviations: BLQ = below the limit of quantification (50 pg/mL); CSF = cerebrospinal fluid; N = number of patients; NSC = no samples collected; SD = standard deviation.

^aPatients 2315 and 2316 were removed from summary statistics (see Section 11.1.1).

^bPatients 2312 was removed from summary statistics (see Section 11.1.1).

^cMedian value was BLQ with a range of values from BLQ to 0.21 ng/mL (see Section 11.1.1)

^dActual dose (mg) was adjusted for each patient by age, see protocol and Section 9.4.1 for additional details. Patients in Cohort 1 received 6 mg equivalent doses on Days 1, 15, and 85 and 12 mg equivalent on all subsequent dosing days.

Toxicology

Based on the intended patient population (neonates to adolescents), 14- and 53-week repeat dose toxicology studies were conducted in juvenile cynomolgus monkeys using IT dose administration. A 13-week study was also conducted in juvenile CD-1 mice using SC dosing to evaluate systemic effects. Reproductive and developmental toxicity studies were conducted in mice (fertility and developmental toxicity) and rabbits (embryo-fetal development) using SC dosing to provide maximum systemic exposure. A study to evaluate pre-natal and post-natal development effects is ongoing. Standard in vitro and in vivo genotoxicity studies were conducted. Carcinogenicity studies were not conducted and are not planned.

In a single IT dose study in adult monkeys at up to 7 mg/dose, transient effects on lower spinal reflexes were seen in 1 of 6 HD animals, but there was no mortality and no effects on body weight, no changes to ECGs, no effects on CSF chemistry or clinical pathology, no effects on complement activation, no evidence of systemic inflammation, and no other drug-related microscopic changes.

In the juvenile monkey studies in which IT bolus doses were administered weekly or every other week for 14 (0.3, 1, or 3 mg/dose) or 53 (0.3, 1, or 4 mg/dose) weeks, acute neurotoxicity (decreased lower spinal reflexes) was observed at the HD, which appeared to be reversible in most animals (decreased reflex responses persisted in 1 HD male), but no clearly drug-related mortality or effects on body weight, ECG, CSF chemistry/cell counts, clinical pathology, or complement activation (as measured by concentrations of Bb split product) were observed, and there was no evidence of systemic inflammation or non-CNS histopathology. In the 1-year study, an apparent long-term effect on performance in a learning and memory test (WGTA) was observed at the HD at the end of the dosing period and after a 22-25-week recovery period. However, due to data variability and the small group sizes, these effects did not reach statistical significance and were not recognized as drug-related by the sponsor. But in view of the hippocampal neurohistopathology observed in these same animals, such effects would not seem to be out of the realm of possibility despite the study pathologist's opinion that the brain findings "were unlikely to result in any clinical signs or any alteration that would in any way interfere with the animals ability to function normally in its environment." Neuronal vacuolation (graded slight to minimal) in the inferior region of the hippocampus was observed at the MD and HD in both the 14- and 53- week studies, but was not seen at the 1-month interim sacrifice in the 14-week study. In some animals at both doses, this vacuolation was associated with low incidences of neuronal and glial cell necrosis and cellular debris. Hippocampal vacuolation and necrotic cells/cellular debris were still present following 12 or 24 weeks of recovery in the 14- and 53-week studies, respectively.

The sponsor actually dismissed these neurohistopathology findings as fixation artifacts, based on an exploratory study comparing fixation methods, or considered them non-adverse, based on the pathologist's statement about the functional consequences. In the exploratory study, in which monkeys (3/sex/grp) were administered weekly IT bolus doses of 5 mg/dose for 36 days and sacrificed 2 weeks later, neuronal vacuolation in the hippocampus was seen in formalin immersion-fixed brains (as used in the tox studies) but not in brains fixed using the alternative methods (immersion fixation in Carnoy's solution, perfusion fixation with modified Karnovsky's solution and fixation by snap freezing). The vacuolated cells present in the formalin-fixed brains also stained positive for the presence of ISIS 396443. These results were interpreted by the sponsor to indicate that the mechanism for hippocampal vacuolation in monkeys treated with ISIS 396443 is the same as that for vacuolation seen in the renal proximal tubule, i.e., "oligonucleotide taken up into the endosomal/lysosomal system is eluted during the fixation/processing of these tissues leaving a dilated vacuolar system remnant to be seen by light microscopy." They further stated that "the presence of vacuoles in the hippocampus following treatment with oligonucleotides does not represent an adverse toxicological finding, but rather represents an artifact of tissue preservation that occurs with the presence of drug." However, in reports of renal tubular vacuolation associated with other oligonucleotides, in which cytoplasmic vacuoles in tubular epithelial cells appeared to be related to basophilic granulation and vacuoles were found to contain basophilic material (presumably oligonucleotide), degenerative changes were also seen at higher tissue oligonucleotide concentrations. And in the study referred to (Henry et al., Toxicology 301:13-20, 2012), renal tubular cell vacuolation in monkeys dosed with another 2'-MOE modified oligonucleotide was not considered a fixation artifact per se, but the result of oligonucleotide taken up by the endosomal/lysosomal system of the epithelial cells being eluted in the fixation process to leave clear vacuoles. As stated in the publication, the "fixation process may influence the appearance of the oligonucleotide within the phagolysosomal compartment." The accumulation of the oligonucleotide within the tubule epithelial cells was only considered non-adverse in the absence of other morphologic changes, but was clearly deleterious at doses that resulted in degenerative changes. The dose- and duration-dependent vacuolation observed only in a specific region of the brain in the monkey studies of ISIS 396443 was not considered an artifact by the study pathologist. While the evidence of associated neuronal necrosis was minimal, the difficulty of detecting neurodegeneration at a single time point in a subchronic or chronic study is well-known. The very specific location of the vacuolation could not be explained by tissue concentrations, since higher concentrations were measured in other parts of the CNS. Taken together, these considerations as well as the data suggesting a neurobehavioral deficit indicate that the hippocampal vacuolation observed in monkeys following IT administration of ISIS 396443 should be considered an adverse neurotoxic effect.

In the 13-week SC mouse study, histopathologic findings attributed to the expected effects of local and systemic accumulations of oligonucleotide and/or their pro-inflammatory effects were observed and included basophilic granules in the kidney, liver Kupffer cell hypertrophy, and vacuolated macrophages at the injection site and in the lymph node and multiple other tissues. These findings were generally not associated with evidence of necrosis, degeneration, or inflammation. However, 1 HD animal was found to have renal tubular degeneration characterized by basophilia, vacuolation, and single cell necrosis. Vacuolation and degeneration of renal tubular epithelial cells associated with basophilic granulation have been previously reported in animal studies of other phosphorothioate oligonucleotides.

No clear evidence of reproductive or developmental toxicity was seen in mice and rabbit studies, and the ASO was negative for genotoxicity in a standard battery of in vitro and in vivo assays. Justification for a carcinogenicity study waiver included the technical limitations of conducting carcinogenicity studies by the IT route, the pharmacology of the target (production of SMN protein), mechanism of action (modulated splicing of mRNA from the *SMN2* gene), and a lack of nonclinical indications of genotoxic, proliferative, endocrine disruption, or immunosuppressive potential. Additionally, the sponsor stated that a standard rodent carcinogenicity study using parenteral dosing would not provide additional insights into the potential carcinogenic risk of ISIS

396443 since exposure to rat CNS tissues would be low. However, given the systemic exposure demonstrated clinically and concerns about hybridization-independent effects of phosphorothioate oligonucleotides, e.g., proinflammatory or immunostimulatory effects, there is still a case to be made for conducting a standard carcinogenicity study by the parenteral route. For example, the lymphoproliferation and multiorgan lymphohistiocytic infiltrates previously reported with oligonucleotide administration to mice were noted in connection with the malignant lymphoma found in a single LD female in the 13-week SC study. This finding in a single LD animal was not considered drug-related. Nevertheless, the drug-related findings in both this study and the 13-week SC impurity study in mice were attributed in part to cellular activation and cytokine production due to the proinflammatory effects of the oligonucleotide. Rodents appear to be particularly sensitive to this generalized immune stimulation, whereas primates are thought to be less sensitive at equivalent doses. The mixed monocellular infiltrates in liver and other organs that are characteristic of the response in rodents are reportedly absent even after long-term systemic exposure to other phosphorothioate oligonucleotides in monkeys. And, no systemic effects were seen in the IT monkey studies despite higher plasma and tissue concentrations than would be expected clinically (compare Table IVC.11 [p. 43] with mean human plasma C_{max} and AUC₀₋₂₄ values of 829 ng/mL and 2181 ng.h/mL at 12 mg dose in clinical PK study ISIS 396443-CS3A). However, in light of the significant systemic exposures demonstrated in patients receiving IT treatment and the lifespan potential of some patients, a lifetime SC carcinogenicity study in one rodent species is recommended.

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/s/

J EDWARD FISHER
12/20/2016

LOIS M FREED
12/20/2016

MEMORANDUM

DEPARTMENT OF HEALTH & HUMAN SERVICES
Public Health Service
Food and Drug Administration

Division of Neurology Products (HFD-120)
Center for Drug Evaluation and Research

Date: December 20, 2016

From: Lois M. Freed, Ph.D.
Supervisory Pharmacologist

Subject: NDA 209-531 (ISIS 396443; nusinersen; Spinraza)

NDA 209-531 has been submitted by the sponsor (Biogen Idec) for use of ISIS 396443 (nusinersen) as a treatment for spinal muscular atrophy. Submission of most of Module 3, Module 4, the CMC and nonclinical portions of Module 2, and part of Module 1 on August 9, 2016, initiated a rolling submission. The remainder of the information (completion of Modules 1, 2, and 3 and Module 5) was received on September 23, 2016, and the NDA was filed (*Filing Communication, October 25, 2016*). Clinical development was conducted under IND 110011 by ISIS Pharmaceuticals (now Ionis Pharmaceuticals).

The following nonclinical data were submitted to the NDA:

- Pharmacology
- Safety pharmacology (male Sprague-Dawley rat, cynomolgus monkey)
- PK/ADME
- Toxicology
 - 14- (4-week interim sacrifice) and 53-week intrathecal studies in cynomolgus monkey
 - 13-week subcutaneous study in CD-1 mouse
 - 13-week subcutaneous study of impurities in CD-1 mouse
- Reproductive and Developmental Toxicology (subcutaneous)
 - Combined fertility and embryofetal development study in CD-1 mouse
 - Embryofetal development study in New Zealand White rabbit
- Genetic Toxicology
 - Ames assay
 - In vitro chromosomal aberration assay in CHO cells
 - In vivo micronucleus assay (subcutaneous) in ICR mouse

The nonclinical data have been reviewed by Dr. Fisher (*Pharmacology/Toxicology Review and Evaluation, NDA 205-931, December 20, 2016*). Based on his comprehensive review of the data, Dr. Fisher has concluded that the nonclinical data are adequate to support approval, with a post-marketing requirement for a carcinogenicity study in one species.

The following is a brief discussion of the nonclinical data, with focus on the neurotoxicity and the potential adverse effects on learning and memory observed in monkey. Dr. Fisher's review should be consulted for a more detailed description and discussion of all the nonclinical data.

Pharmacology

ISIS 396443 is a fully modified 18-base 2'-O-(2-methoxyethyl) phosphorothioate antisense oligonucleotide designed to increase the amount of full-length *SMN* protein by promoting the inclusion of exon 7 in the *SMN2* mRNA. ISIS 396443 is thought to modulate exon 7 inclusion through displacement of inhibitory heterogeneous nuclear ribonucleoproteins (hnRNPs) from their target site (ISS-N1) in intron 7 of the *SMN2* pre-mRNA.

The sponsor conducted in vitro and in vivo studies to assess the pharmacological activity of ISIS 396443. In vivo studies (using intracerebroventricular and intrathecal [IT] administration) were conducted in transgenic mouse models of SMA because only humans are known to have the *SMN2* gene. Four different mouse models were used, which differed in severity of disease (mild to aggressive). Based on the data from these studies, the sponsor concluded that efficacy was achieved at tissue (brain and spinal cord) concentrations of ~2-10 µg/g. In addition, bolus injections resulted in lower ED₅₀ values compared to infusions, supporting the use of IT bolus injections in clinical trials.

No adverse effects were observed in a 28-day pulmonary and cardiovascular safety study of ISIS 396443 (0, 0.02, 0.06, and 0.2 mg/day by continuous IT infusion) in male Sprague-Dawley rat (8/group). Safety pharmacology parameters were also included in the IT toxicity studies in cynomolgus monkey (discussed below).

PK/ADME

The PK/ADME of ISIS 396443 was assessed in an IT study (ISIS 396443-APK01) in adult cynomolgus monkey (3-4/group/time point). ISIS 396443 was administered either by intravenous (IV) or IT bolus injection at a dose of 1 mg on Days 1, 8, 15, and 22. Animals were sacrificed on Day 29, 85, 183, 253, or 365.

Following IT bolus administration, the $t_{1/2}$ values for ISIS 396443 in CNS tissues ranged from 74.4 days in lumbar spinal cord to 275 days in hippocampus. $T_{1/2}$ values in plasma and csf were 52.2 and 102 days, respectively. Plasma data suggested higher (~2-fold) systemic exposures with IT dosing; this is likely due to initial sampling at 1 hr post dose, which would be expected to result in an underestimate of exposure following IV dosing.

The tissue data (Day 29) demonstrate substantial transfer of ISIS 396443 into the systemic circulation following IT dosing, with concentrations in liver similar between routes; concentrations in kidney cortex were ~2-fold higher following IV dosing.

Plasma and tissue concentration data for ISIS 396443 are summarized in the following sponsor's tables:

Table 4. Pharmacokinetic Plasma Parameters After Multiple 1 mg IT and IV Bolus Injections

PK Parameter	N	IT - Lumbar	N	IV
C _{max} (µg/mL)	17	0.154 ± 0.114	3	0.367 ± 0.0440
T _{max} (hr) ^a	17	4 (1 - 8)	3	1 (1 - 1)
AUC _{0-24hr} (µg*hr/mL)	17	1.53 ± 0.441	3	0.720 ± 0.0662
AUC _{0-168hr} (µg*hr/mL)	17	2.07 ± 0.532	3	1.02 ± 0.163

Values reported as mean ± standard deviation

^a T_{max} values reported as median (range)

Table 5. Summary of ISIS 396443 CNS Tissue Concentrations in Cynomolgus Monkeys After Multiple 1 mg IT Bolus Administrations

Tissue	Day 29		Day 85		Day 183		Day 253		Day 365	
	N	Mean ± SD	N	Mean ± SD	N	Mean ± SD	N	Mean ± SD	N	Mean ± SD
Brain Stem	4	10.2 ± 8.65	3	6.6 ± 0.797	3	0.931 ± 0.402	3	1.86 ± 2.27	4	0.472 ± 0.343
Cerebellum	4	3.74 ± 2.49	3	13.5 ± 6.86	3	0.575 ± 0.437	3	1.35 ± 1.33	4	1.09 ± 0.800
Cervical Spinal Cord	4	19.9 ± 4.69	3	9.58 ± 0.602	3	3.64 ± 2.44	3	3.50 ± 3.25	4	1.86 ± 1.76
Frontal Cortex	4	13.6 ± 4.98	3	7.39 ± 5.36	3	5.00 ± 3.61	3	7.30 ± 8.61	4	1.96 ± 1.10
Hippocampus	4	3.47 ± 3.24	3	1.75 ± 0.745	3	2.54 ± 1.85	3	2.19 ± 2.96	4	1.61 ± 0.953
Lumbar Spinal Cord	4	91.4 ± 37.9	3	37.5 ± 15.9	3	24.2 ± 18.0	3	10.8 ± 9.14	4	4.34 ± 4.78
Putamen	4	0.493 ± 0.352	3	0.658 ± 0.317	3	0.230 ± 0.106	3	0.437 ± 0.392	4	0.306 ± 0.250
Temporal Cortex	4	16.6 ± 5.93	3	15.3 ± 6.7	3	4.91 ± 1.86	3	4.64 ± 4.86	4	2.07 ± 0.948
Thoracic Spinal Cord	4	35.8 ± 9.51	3	15.7 ± 4.62	3	5.93 ± 3.79	3	3.53 ± 2.27	4	1.96 ± 1.20

Values reported as µg/g

SD = Standard Deviation

Table 6. Summary of ISIS 396443 Systemic Tissue Concentrations in Cynomolgus Monkeys After Multiple 1 mg IT or IV Bolus Administrations

Tissue	Route ^a	Day 29		Day 85		Day 183		Day 253		Day 365	
		N	Mean ± SD	N	Mean ± SD	N	Mean ± SD	N	Mean ± SD	N	Mean ± SD
Kidney Cortex	IT-L	4	61.3 ± 15.8	3	14.3 ± 11.6	3	0.962 ± 0.797	3	0.362 ± 0.197	4	0.652 ± 0.363
	IV	3	112 ± 73.5	-	NSC	-	NSC	-	NSC	-	NSC
Liver	IT-L	4	4.74 ± 2.02	3	0.113 ± 0.0556	3	0.0598 ± 0.0425	3	BLQ	4	BLQ
	IV	3	3.47 ± 1.32	-	NSC	-	NSC	-	NSC	-	NSC
Skeletal Muscle	IT-L	4	0.0140 ± 0.0280	3	BLQ	3	BLQ	3	BLQ	4	BLQ

Values reported as µg/g

BLQ = Below limit of quantitation (<0.015 µg/g); SD = Standard deviation; NSC = No sample collected

^a IV = Intravenous bolus; IT-L = Intrathecal lumbar catheter bolus

Metabolites were measured in the IT toxicity studies in monkey. In plasma, ISIS 396443 accounted for >95% of drug-related material, with one metabolite (17-mer, ISIS 52440)

accounting for ~2-3% of drug-related material. In brain cortex and lumbar spinal cord, ISIS 396443 and metabolite ISIS 52440 accounted for ~80 and 10-20% of drug-related material, respectively.

Additional PK/TK data were collected in the toxicity studies (discussed below).

Toxicology

Acute toxicity was assessed only in cynomolgus monkey (3.6-4.2 years of age). Repeat-dose toxicity studies were conducted in CD-1 mouse (two 13-week studies) and juvenile cynomolgus monkey (14-week [with 4-week interim sacrifice] and 53-week studies; 7-8 and 10-11 months of age, respectively).

Mouse

Two 13-week subcutaneous (SC) toxicity studies were conducted in CD-1 mouse. In Study (b) (4)-192015, ISIS 396443 was administered to juvenile CD-1 mice (10/sex/group) at doses of 0, 1, 10, and 50 mg/kg, weekly on PNDs 4-25 and biweekly thereafter through PND95. (The study was not designed to assess potential effects on developmental parameters.) The primary effects were microscopic changes in liver (Kupffer cell hypertrophy at MD and HD), kidney (basophilic granules in proximal convoluted tubules at MD and HD), injection site (vacuolated macrophages at all doses), and lymph node and other tissues (vacuolated macrophages), consistent with accumulation of ISIS 396443 or pro-inflammatory effects. However, renal tubular degeneration (basophilia, vacuolation, and single cell necrosis), associated with tubular basophilia and basophilic granules, was detected in one HDM.

Plasma exposure data were not collected. Mean liver exposures were <10 (LLQ), 44-45, and 214-220 µg/g at the LD, MD, and HD, respectively.

Study 727-083 was conducted in adult CD-1 mouse (6/sex/group) to qualify a series of related impurities using three different formulations (ISIS 396443 Impurity Mixtures 1, 2, and 3). ISIS 39443 mixtures were administered SC weekly at doses of 5 and 25 mg/kg. Results were generally similar to those observed in Study (b) (4)-192015. Although no renal tubular degeneration was observed, the HD was lower than in the previous study and fewer animals were tested; exposure to kidney was not evaluated in either study.

Plasma exposure data were not collected. Across all mixtures, mean liver exposures were ~29-53 and 141-269 µg/g at the LD and HD, respectively.

Monkey

In Study 396443-AS01, ISIS 396443 was administered at acute IT (lumbar) bolus doses of 0, 1, 3, and 7 mg (2-3/sex/group; 5/sex for TK at 3 mg). The only findings of note were clinical signs (sedation and hindlimb weakness) and neurological impairment

(absent cutaneous and proprioceptive reflexes) in one HDM. The neurological impairment persisted, in the absence of clinical signs. Seven days after injection, csf levels were 0.0102 ± 0.00489 , 0.0459 ± 0.0438 , and 0.0641 ± 0.0334 $\mu\text{g/mL}$ at the LD, MD, and HD, respectively. Plasma exposure and tissue concentrations are summarized in the following sponsor's tables:

Table B Preclinical Plasma Pharmacokinetic Parameters in Adult Monkeys After a Single IT bolus (continued)

Dose (mg)	$C_{\max}$ ($\mu\text{g/mL}$)		$T_{\max}$ (hr)		AUC_{0-24} (hr* $\mu\text{g/mL}$)	
	N	Mean $\pm$ SD	N	Median (range)	N	Mean $\pm$ SD
1	6	0.222 ± 0.131	6	5 (1.0 – 10)	6	2.03 ± 0.408
3	13	0.455 ± 0.213	13	4 (2 – 8)	13	4.50 ± 1.83
7	6	2.61 ± 1.93	6	2 (1 – 4)	6	14.6 ± 4.32

Table C ISIS 396443 Tissue Concentrations 7 Days after a Single Intrathecal Bolus in Cynomolgus Monkeys in Study 396443-AS01

Organ	ISIS 396443 ($\mu\text{g/g}$)					
	1 mg		3 mg		7 mg	
	N	Mean $\pm$ SD	N	Mean $\pm$ SD	N	Mean $\pm$ SD
Brain Cerebellum	6	1.43 ± 1.31	6	5.07 ± 3.14	6	8.75 ± 10.7
Brain Cortex	6	2.42 ± 1.37	6	6.98 ± 3.63	6	12.8 ± 12.4
Brain Pons	6	1.46 ± 1.08	6	3.51 ± 1.66	6	4.92 ± 3.22
Cervical Spinal Cord	6	2.72 ± 0.676	6	5.6 ± 2.91	6	10.7 ± 5.43
Liver	6	1.89 ± 0.401	6	9.95 ± 8.48	6	36.0 ± 26.5
Lumbar Spinal Cord	6	8.33 ± 4.89	6	19.2 ± 7.55	6	19.6 ± 14.3
Thoracic Spinal Cord	6	6.07 ± 4.26	6	10.3 ± 5.04	6	18.4 ± 19

The highest tissue concentrations were detected in lumbar spinal cord at the lower doses and in liver at the HD. Hippocampus was not sampled.

In the 14-week toxicity study (396443-AS03), ISIS 396443 was administered at doses of 0, 0.3, 1, and 3 mg by slow IT (lumbar) bolus injection weekly for 4 weeks (interim sacrifice in 3/sex/group on Day 36) weeks, followed in the remaining animals by biweekly dosing for an additional 5 doses (3-4/sex/group sacrificed on Day 106). Two per sex for control and HD groups were sacrificed after a 12-week recovery period (sacrificed on Day 183).

There were no deaths considered drug-related and no adverse effects on clinical pathology parameters, gross pathology, or organ weights were observed. Transient effects (decreased or absent) on lower spinal (cutaneous, sensory foot, and/or tail) reflexes were observed at the HD.

At the 4-week interim sacrifice, microscopic findings in CNS tissues were detected only in one MD male (as listed in the Pathology Report):

- Spinal cord, Infiltrates, Perivascular, Mononuclear Cells (slight)
- Spinal cord, Gray Matter, Microglial Focus/Foci (slight)
- Brain (various regions), Infiltrates, Perivascular, Mononuclear Cell (slight)
- Brain (various regions), Infiltrates, Perivascular, Eosinophils (slight)
- Brain (various regions), Microglial Focus/Foci (slight)

These findings were considered the result of distribution of ISIS 396443 into CNS tissue, "...with a slight focal reaction," but not adverse.

At the 14-week terminal sacrifice, microscopic findings in CNS tissue were detected at the MD (1 mg) and HD (3 mg). Selected findings in main-study and recovery animals are summarized in the following table:

TISSUE	FINDING	MALES		FEMALES	
		MD	HD	MD	HD
MAIN STUDY					
hippocampus (inferior)	vacuolation (neuronal area)	1/3	3/3	0/4	4/4
	neuronal necrosis	0/3	0/3	0/4	1/4
	glial cell necrosis	0/3	1/3	0/4	0/4
spinal cord (lumbar gray matter)	perivascular macrophages	0/3	0/3	0/4	1/4
RECOVERY					
hippocampus (inferior)	vacuolation (neuronal area)		2/2		2/2
	neuronal necrosis		0/2		0/2
	glial cell necrosis		0/2		1/2
spinal cord (lumbar gray matter)	perivascular macrophages		0/2		1/2

According to the study pathologist, necrotic cells were detected in association with vacuolation in several HD animals and were considered adverse.

The toxicokinetic (TK) data are summarized in the following tables; dose levels are given as mg/dose.

Csf levels (mean $\pm$ SD; $\mu\text{g/mL}$) on Day 106, 7 days after the last dose:

CISTERNA MAGNA			LUMBAR		
0.3	1	3	0.3	1	3
0.016 $\pm$ 0.0064	0.022 $\pm$ 0.011	0.139 $\pm$ 0.083	0.015 $\pm$ 0.010	0.011 $\pm$ 0.0085	0.078 $\pm$ 0.073

Plasma exposures (mean $\pm$ SD) on Day 99:

DOSE (mg)	C _{max} ($\mu\text{g/mL}$)	AUC _(0-24 hr) ($\mu\text{g}\cdot\text{hr/mL}$)
0.3	0.12 $\pm$ 0.06	1.05 $\pm$ 0.40
1	0.56 $\pm$ 0.14	3.51 $\pm$ 0.67
3	1.56 $\pm$ 0.26	11.1 $\pm$ 2.26

Tissue concentrations (mean $\pm$ SD; $\mu\text{g/g}$) on Day 106, 7 days after the last dose:

TISSUE	DOSE (mg)		
	0.3	1	3
spinal cord			
cervical	10.6 $\pm$ 6.21	26.8 $\pm$ 6.4	82.8 $\pm$ 27.8
thoracic	20.2 $\pm$ 4.59	38.4 $\pm$ 7.06	101 $\pm$ 46.5
lumbar	31.9 $\pm$ 15.5	78.6 $\pm$ 53.3	169 $\pm$ 54.2
brain			
cortex	15.2 $\pm$ 3.9	35.4 $\pm$ 13.7	166 $\pm$ 45.8
cerebellum	15 $\pm$ 10.2	23.7 $\pm$ 20.1	51.2 $\pm$ 15.9
pons	7 $\pm$ 2.39	13 $\pm$ 3.67	45.9 $\pm$ 18.4
liver	1.88 $\pm$ 0.377	42.5 $\pm$ 40.9	115 $\pm$ 65.1

Plasma and tissue (liver) data demonstrate substantial systemic exposure following IT administration. Hippocampus was not sampled.

In the 53-week toxicity study (396443-AS06), ISIS 396443 was administered by IT (lumbar) bolus injection at doses of 0, 0.3, 1.0, and 4.0 mg weekly for the first 5 doses, followed by a dose every 6 weeks thereafter (5/sex/group; 8 doses). An additional 2/sex for C and HD were followed for a 26-week recovery period.

There were no unscheduled deaths or adverse effects on body weight, ophthalmology, ECG, clinical pathology, or bone parameters (including femur length). As observed in the previous monkey studies, there were transient deficits (decreased or absent) in lower spinal (patellar, grip, and anal) reflexes at the HD.

Learning and memory were tested using the Wisconsin General Test Apparatus, pre-dose, during study Weeks 25-28 and 49-52 (end of dosing period), and at the end of the recovery period (Weeks 22-25). As Dr. Fisher notes, it is difficult to evaluate these data because of the high intra- and interanimal variability and the small number of animals (5-7/sex/group). However, during Weeks 49-52 and Recovery Weeks 22-25, the days required to complete the learning task were increased at the HD in both males and females at both testing periods. These data (in days) are summarized in the table below:

DOSE (mg)	WEEKS 49-52		RECOVERY WEEKS 22-25	
	MALES	FEMALES	MALES	FEMALES
0	6.0 $\pm$ 2.8	5.3 $\pm$ 2.6	5.0	3.5
0.3	7.4 $\pm$ 3.0	5.0 $\pm$ 1.0		
1	7.0 $\pm$ 1.9	4.6 $\pm$ 1.5		
4	9.1 $\pm$ 4.6	7.2 $\pm$ 3.4	10.0	3.0

The sponsor attributed the apparent drug-related adverse effect to prolonged habituation periods (7 in one HDM vs 3-4 in other animals) or performance on the memory test. Regarding the latter, the one HDM (#17340) who took the longest to pass the learning test (18 days) scored 100% on the memory test. The sponsor suggested that the high score on the memory test resulted in difficulty in reversal learning (i.e., learning that a new

stimulus was being rewarded). However, two control males, who also scored 100% on the memory test, passed the learning test quickly (within 3-4 days).

An expanded neurohistopathology evaluation was conducted, with examination of 8 sections from all animals and special stains (GFAP, Fluoro-Jade B), where appropriate, in addition to H&E. ISIS 396443-related microscopic findings were detected at the MD (1.0 mg) and HD (4.0 mg). Selected findings are summarized in the following table:

TISSUE	FINDING	MALE		FEMALES	
		MD	HD	MD	HD
MAIN STUDY					
hippocampal area (neurons)	large vacuoles	2/5	3/5	0/5	2/5
inferior hippocampus (neurons)	large vacuoles	0/5	0/5	0/5	1/5
inferior hippocampus/amygdylod, hippocampal area	vacuoles/flocculent material	0/5	2/5	0/5	1/5
inferior hippocampus/amygdylod, hippocampal area	necrotic cells/cellular debris	1/5	3/5	0/5	0/5
total number of affected animals		2/5	3/5	0/5	2/5
RECOVERY					
hippocampal area (neurons)	large vacuoles		2/2		1/2
inferior hippocampus (neurons)	large vacuoles		0/2		0/2
inferior hippocampus/amygdylod, hippocampal area	vacuoles/flocculent material		1/2		0/2
inferior hippocampus/amygdylod, hippocampal area	necrotic cells/cellular debris		1/2		0/2
total number of affected animals			2/2		1/2

SEX	DOSE (mg)	ANIMAL #	FINDINGS
MAIN STUDY			
M	1	17345	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) necrotic cells/cellular debris
		17346	none
		17349	none
		17352	none
		17358	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons)
	4	17339	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) vacuoles/flocculent material necrotic cells/cellular debris
		17342	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) vacuoles/flocculent material necrotic cells/cellular debris
		17348	none
		17355	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) necrotic cells/cellular debris
		17356	none
F	1	17369	none
		17373	none
		17374	none
		17381	none
		17383	none
	4	17368	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) inferior hippocampus, medial CA3 region large vacuoles (neurons)
		17377	none
		17382	none
		17387	inferior hippocampus/amygdylod hippocampal area large vacuoles vacuoles/flocculent material
		17390	none
RECOVERY			
M	4	17340	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons)
		17361	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons) vacuoles/flocculent material necrotic cells/cellular debris
F	4	17367	inferior hippocampus/amygdylod hippocampal area large vacuoles (neurons)
		17388	none

Although the sponsor suggests that the large vacuoles represent artifacts, the study pathologist did not include large vacuoles in the findings specifically designated in individual histopathology table as “ARTIFACT.” It is presumed that the vacuoles

represent accumulation of ISIS 396443 and would not be considered adverse if not accompanied by other microscopic changes, such as necrotic cells or cellular debris.

Lumbar csf levels (mean $\pm$ SD [range]; ng/mL) on Day 372, 7 days after the last dose:

DOSES (mg)		
0.3	1	4
12.0 $\pm$ 10.6 [4.68-41.1]	23.3 $\pm$ 16.7 [8.78-68.2]	51.8 $\pm$ 37.8 [16.1-162]

Plasma exposures (mean $\pm$ SD) on Day 365:

DOSE (mg)	C _{max} (µg/mL)	AUC _(0-24 hr) (µg*hr/mL)	AUC _(0-168 hr) (µg*hr/mL)	Cl/F (mL/hr)
0.3	0.0859 $\pm$ 0.0514	0.626 $\pm$ 0.197	0.714 $\pm$ 0.270	521 $\pm$ 158
1	0.454 $\pm$ 0.258	2.79 $\pm$ 1.59	3.94 $\pm$ 3.98	417 $\pm$ 125
4	1.52 $\pm$ 0.535	9.59 $\pm$ 1.90	11.7 $\pm$ 2.63	433 $\pm$ 87.2

Tissue concentrations (mean $\pm$ SD; µg/g) on Day 372, 7 days after the last dose:

TISSUE	DOSE (mg)		
	0.3	1	4
spinal cord			
cervical	6.68 $\pm$ 2.78	15.9 $\pm$ 5.14	35.1 $\pm$ 11.0
thoracic	14.0 $\pm$ 9.34	23.0 $\pm$ 10.6	59.8 $\pm$ 11.7
lumbar	21.4 $\pm$ 9.28	53.5 $\pm$ 33.6	99.0 $\pm$ 71.5
brain			
cortex	4.23 $\pm$ 1.89	16.6 $\pm$ 5.97	71.3 $\pm$ 29.3
cerebellum	4.27 $\pm$ 1.39	20.8 $\pm$ 16.0	39.6 $\pm$ 19.7
hippocampus	10.1 $\pm$ 2.81	28.9 $\pm$ 10.3	88.9 $\pm$ 18.0
pons	4.14 $\pm$ 3.06	8.84 $\pm$ 3.97	34.2 $\pm$ 15.5
kidney cortex	23.7 $\pm$ 17.6	42.8 $\pm$ 19.0	239 $\pm$ 176
liver	0.850 $\pm$ 0.430	2.79 $\pm$ 1.64	21.9 $\pm$ 9.16

Half-lives in plasma and tissues following the last dose (>Day 365):

TISSUE	t _{1/2} (days)
plasma	55.8
csf	111
spinal cord	
cervical	195
thoracic	117
lumbar	138
brain	
cortex	n.c.
cerebellum	n.c.
hippocampus	191
pons	174
kidney cortex	34.5
liver	23.4

The study pathologist considered the microscopic findings in the CNS to be drug-related but concluded that they would be "...unlikely to result in any clinical signs or any alteration that would in any way interfere with the animals [sic] ability to function normally in its environment," apparently based on the lack of clinical signs and the nature of the findings. However, it is not uncommon that overt effects of CNS lesions cannot be detected, even where there is consistent evidence of neurotoxicity (e.g., NMDA receptor antagonists) that has clear regulatory impact. In addition, the results of the learning tasks suggest the possibility of a behavioral correlate.

Reproductive and Developmental Toxicology

ISIS 396443 was tested in a combined fertility and embryofetal development study in male and female CD-1 mice and an embryofetal development study in New Zealand White rabbit.

In the mouse study (396443-AS08), ISIS 396443 was administered SC every other day at doses of 0, 3, 10, and 25 mg/kg for 4 (males) or 2 (females) weeks prior to mating, throughout the 2-week mating period, and continuing for a total of 10 weeks (until GD 15 in females). There were no clear effects of ISIS 396443 on fertility parameters, estrus cyclicity, sperm parameters, or embryofetal development. The fecundity index (# pregnant/# confirmed mated) was reduced in a dose-dependent manner (N%92.9, 86.2, and 83.9% at LD, MD, and HD, respectively; however, as Dr. Fisher noted, the fecundity index was lowest in control (70%), making it difficult to interpret the results.

In the rabbit study (396443-AS09), ISIS 396443 was administered SC every other day during GDs 6 through 19 at doses of 0, 6, 12.6, and 25 mg/kg. There were no clear drug-related effects on developmental parameters.

A pre- and postnatal development study is ongoing; submission of the study should be a post-marketing requirement.

Genetic Toxicology

ISIS 396443 was negative in an adequately conducted standard battery of in vitro (Ames and chromosomal aberration in CHO) and in vivo (mouse micronucleus) assays.

Carcinogenicity

No carcinogenicity studies were conducted to support the NDA, which is consistent with the Division's decision to allow the assessment of carcinogenicity to be conducted post-approval. However, the sponsor has submitted a request for waiver of carcinogenicity studies for reasons discussed by Dr. Fisher. I agree with Dr. Fisher's recommendation that, because of the substantial systemic exposure to ISIS 396443 in humans after IT administration, the sponsor should conduct a 2-year carcinogenicity study as a post-marketing requirement.

Conclusions and Recommendations

A less-than standard battery of nonclinical studies (no chronic toxicity study in rodent [to assess systemic exposure], pre- and postnatal development (PPND) study in rodent, or carcinogenicity studies in mouse and rat) was conducted to support approval of nusinersen. Under the IND, the Division agreed that the lack of carcinogenicity studies would not preclude filing or approval of the NDA. The lack of the other studies was not considered a refuse-to-file issue.

The finding of greatest concern is the neurotoxicity observed in the toxicity studies in monkey, which may be associated with adverse effects on learning. The no-effect dose (0.3 mg) for neurotoxicity in the 53-week monkey study was associated with csf levels (12 ng/mL) generally less than that in SMA patients (7-35 ng/mL); therefore, there is no safety margin based on csf levels. However, considering the seriousness of the indication and the apparently substantial evidence of efficacy in the clinical trials in SMA patients, there is no objection to approval of the NDA from a pharmacology/toxicology standpoint. Completion of the ongoing PPND study and a carcinogenicity study in one species (which will provide chronic toxicity data) should be post-marketing requirements. The neurotoxicity findings should be described in labeling.

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